

PHYSICS



Resonance
Educating for better tomorrow

TARGET: JEE (ADVANCED) 2014

MEDIUM : ENGLISH

CRITICAL QUESTIONS BANK

SUBJECT : PHYSICS

MEDIUM : ENGLISH

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QUESTION FORMAT & MARKING CRITERIA

SUBJECT : PHYSICS

A. Questions Format

In the booklet check that it contains all the **185 questions** and corresponding answer choices are legible. Read carefully the Instructions printed at the beginning of each section.

1. **Section 1** contains **57 multiple choice questions**. Each question has Four choices (A), (B), (C) and (D) out of which **only ONE is correct**.
2. **Section 2** contains **32 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.
3. **Section 3** contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **Only ONE is correct**.
4. **Section 4** contains **14 paragraphs** each describing theory, experiment, data etc. **23 questions** related to Fourteen paragraphs with one or two or three questions on each paragraph. Each question of a paragraph has **ONLY ONE correct answer** among the four choices (A), (B), (C) and (D).
5. **Section 5** contains **5 question**. Each question contains statements given in two columns which have to be matched. Statements in **Column I** are labelled as A,B,C and D whereas statements in **Column II** are labelled as p,q,r,s and t. The answers to these questions have to be appropriately bubbled as illustrated in the following example.
6. **Section 6** contains **6 multiple choice questions**. Each questions has matching lists. The codes for the lists have coices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.
7. **Section 7** contains **45 questions**. The answer to each question is a single-digit integer, ranging from 0 to 9 (both inclusive).
8. **Section 8** contains **19 questions**. The answer to each question is a double-digit integer, ranging from 00 to 99 (both inclusive).

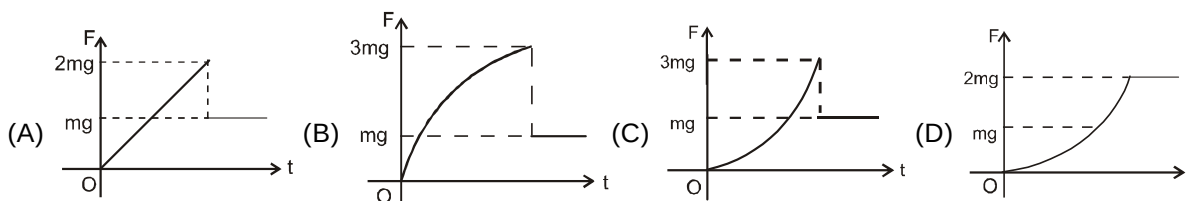
B. Marking Scheme

9. For each question in **Section 1**, you will be awarded **3 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. In all other cases, **minus one (–1) mark** will be awarded.
10. For each question in **Section 2**, you will be awarded **4 marks** if you darken the bubble(s) corresponding to only the correct answer and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
11. For each question in **Section 3**, you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened.
12. **Section 4** contains **5 paragraphs** each describing theory, experiment, data etc. **Ten questions** related to Five paragraphs with two questions on each paragraph. Each question of a paragraph has **ONLY ONE correct answer** among the four choices (A), (B), (C) and (D).
13. For each question in **Section–5**, you will be **awarded 2 marks** for each row in which you have darkened the bubble corresponding to the correct answer. Thus, each question in this section carries a maximum of **8 marks**. There is **no negative marking** for incorrect answer(s) in this section.
14. For each question in **Section 6**, you will be awarded **3 marks** if you darken **all** the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **minus one (–1) mark** will be awarded.
15. For each question in **Section 7**, you will be awarded **4 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
16. For each question in **Section 8**, you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and zero mark if no bubbles are darkened. **No negative marks** will be awarded for incorrect answers in this section.

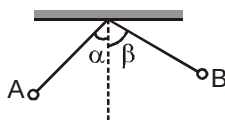
SECTION – 1 : (Only One option correct Type)

This section contains 57 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. An uniform chain of mass m and length ℓ hangs by a thread and touches the surface of a table by its lower end. The thread is cut at time $t = 0$. Which of the following graph best represents the relation between force F exerted by table on the chain with time t . (Assume the fallen part immediately comes to rest after collision with table and do not form heap):



2. Two identical simple pendulums A and B have same point of suspension, having length ℓ each. They are displaced by an angle α and β (α and β are very small and $\beta > \alpha$) and released from rest. Find the time after which B reaches at its initial position for the first time. Assume collision to be elastic and both pendulums move in same plane.



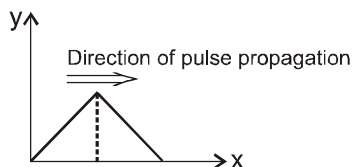
- (A) $\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{\ell}{g}}$ (C) $\frac{\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$ (D) $\frac{2\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

3. A student is performing the experiment of Resonance Column. The diameter of the column tube is 4cm. The frequency of the tuning fork is 512 Hz. The air temperature is 38°C in which the speed of sound is 336 m/s. The zero of the meter scale coincides with the top end of the Resonance Column tube. When the first resonance occurs, the reading of the water level in the column is

- (A) 14.0 cm (B) 15.2 cm (C) 16.4 cm (D) 17.6 cm

4. The wave function of a triangular wave pulse is defined by the relation below at time $t = 0$ sec.

$$y = \begin{cases} mx & \text{for } 0 \leq x \leq \frac{a}{2} \\ -m(x - a) & \text{for } \frac{a}{2} \leq x \leq a \\ 0 & \text{everywhere else} \end{cases}$$

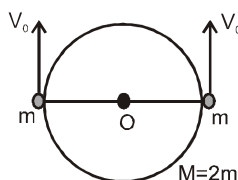


The wave pulse is moving in the $+X$ direction in a string having tension T and mass per unit length μ . The total kinetic energy present with the wave pulse is -

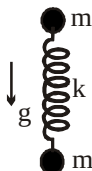
- (A) $\frac{m^2Ta}{2}$ (B) m^2Ta (C) $\frac{3m^2Ta}{2}$ (D) None of these

5. A pendulum bob of mass m is suspended by a massless string and at rest when string is vertical. A constant horizontal force $F = mg$ starts acting on it. The value of maximum tension in string is $T = mg [x\sqrt{2} - 2]$, then the value of x is :
 (A) 1 (B) 2 (C) 3 (D) 0

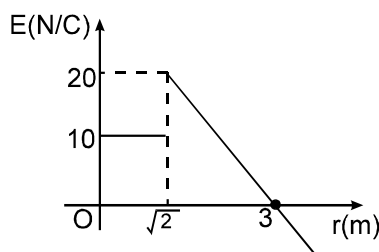
6. An isolated smooth ring of mass $M = 2m$ with two small beads each of mass m is as shown in the figure. Initially both the beads are at diametrically opposite points and have velocity v_0 (for each) in same direction. The speed of the beads just before they collide for the first time is (complete system is placed on a smooth horizontal surface and assume each point of ring is touching the surface)



- (A) v_0 (B) $\frac{2}{\sqrt{3}}v_0$ (C) $\frac{v_0}{2}$ (D) $\frac{\sqrt{3}}{2}v_0$
7. Two large vertical and parallel non conducting plates, have equal & opposite charge density, are at separation of 1 cm and plates are at potential difference of x volt. A proton is released at rest midway between the two plates. It is found to move, at 45° to the vertical. JUST after release. Then x is nearly
 (A) 1×10^{-5} V (B) 1×10^{-7} V (C) 1×10^{-9} V (D) 1×10^{-10} V
8. A mass m is hung on an ideal massless spring. Another equal mass is connected to the other end of the spring. The whole system is at rest. At $t = 0$, m is released and the system falls freely under gravity. Assume that natural length of the spring is L_0 , its initial stretched length is L and the acceleration due to gravity is g . What is distance between masses as function of time.

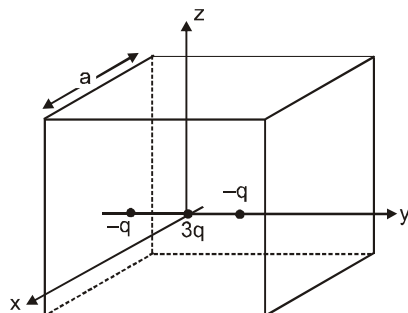


- (A) $L_0 + (L - L_0) \cos \sqrt{\frac{2k}{m}} t$ (B) $L_0 + (L - L_0) \cos \sqrt{\frac{k}{m}} t$
 (C) $L_0 - 2(L + 2L_0) \cos \sqrt{\frac{2k}{m}} t$ (D) $L_0 + (L - L_0) \sin \sqrt{\frac{2k}{m}} t$
9. An electric field 'E' whose direction is radially outward varies as distance from origin 'r' as shown in the graph. E is taken as positive if its direction is away from the origin. Then the work done by electric field on a 2 C charge if it is taken from (1, 1, 0) to (3, 0, 0) is :



- (A) $20(3 - \sqrt{2})$ J (B) -60 J (C) 60 J (D) $20(\sqrt{2} - 3)$ J

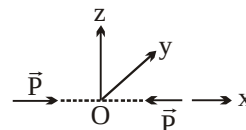
10. A cubical region of side a has its centre at the origin. It encloses three fixed point charges, $-q$ at $(0, -a/4, 0)$, $+3q$ at $(0, 0, 0)$ and $-q$ at $(0, +a/4, 0)$. Choose the incorrect option.



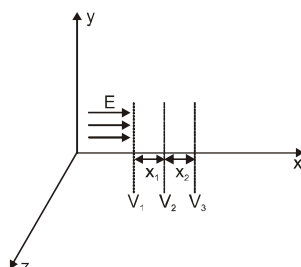
- (A) The net electric flux crossing the plane $x = +a/2$ is equal to the net electric flux crossing the plane $x = -a/2$.
 (B) The net electric flux crossing the plane $y = +a/2$ more than the net electric flux crossing the plane $y = -a/2$.
 (C) The net electric flux crossing the entire region is $\frac{q}{\epsilon_0}$.
 (D) The net electric flux crossing the plane $z = +a/2$ is equal to the net electric flux crossing the plane $z = -a/2$.

11. Two identical dipoles are placed on x -axis at same distance from origin O as shown then

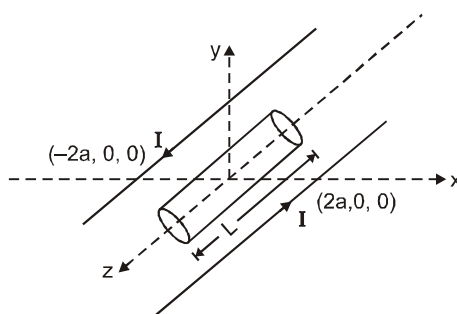
- (A) Electric potential at all points on z -axis is zero
 (B) Electric field at any point with $z > 0$ is directed along $+z$ -axis
 (C) Electric field at origin is zero.
 (D) Electric field at any point with $z < 0$ is directed along negative z -axis.



12. In an electric field shown in figure three equipotential surface are shown. If function of electric field is $E = 2x^2$ V/m, and given that $V_1 - V_2 = V_2 - V_3$ then we have

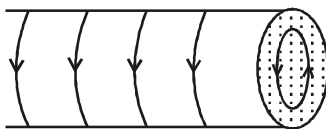


- (A) $x_1 = x_2$ (B) $x_1 > x_2$ (C) $x_2 > x_1$ (D) Data insufficient
13. Two very long wires parallel to the Z -axis (in xz plane) and a distance ' $4a$ ' (along x -axis) apart carry equal currents I in opposite directions as shown in the figure. A cylinder of radius a and length L has its axis on the Z -axis midway between the wires, calculate the net upward magnetic flux through half of the curved cylindrical surface above the x - z plane



- (A) $\frac{\mu_0 IL}{\pi}$ (B) $\frac{\mu_0 IL \ln 3}{\pi}$ (C) $\frac{\mu_0 IL \ln 2}{\pi}$ (D) $\frac{\mu_0 IL}{2}$

14. Magnetic field is uniform and has a magnitude B in the interior of a very long solenoid far from its ends. One of the ends of the solenoid is closed with a thin flat plastic cover. A single small electrical loop of radius R lies on the cover so that its center is on the axis of the solenoid. The electrical current flowing in the loop is I . Then the mechanical tension in the loop's wire is :



- (A) $\frac{BIR}{2}$ (B) BIR (C) $2\pi BIR$ (D) πBIR

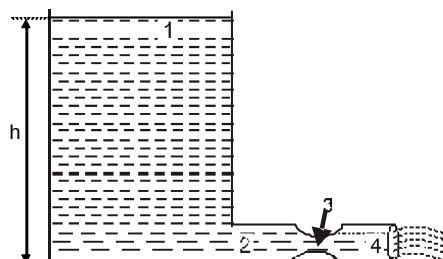
15. The space has electromagnetic field which varies with time whose variation is given as :

$$\vec{B} = \begin{cases} -B_0 \hat{k} & \text{if } 0 \leq t \leq \frac{\pi m}{qB_0} \\ -B_0 \hat{j} & \text{if } \frac{\pi m}{qB_0} \leq t \leq \frac{2\pi m}{qB_0} \\ 0 & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if } 0 \leq t \leq \frac{2\pi m}{qB_0} \\ -E_0 \hat{k} & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases}$$

A charge particle having mass m and positive charge q is given velocity $v_0 \hat{i}$ at origin at $t = 0$ sec. The coordinate of point on xy plane when it again passes through xy plane for the first time is :

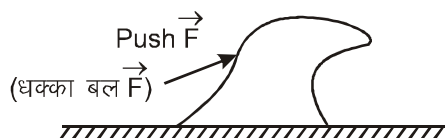
- (A) $\left(\frac{2mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$ (B) $\left(\frac{2mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{mv_0}{qB_0}, 0 \right)$ (C) $\left(\frac{mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$ (D) $\left(\frac{mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{mv_0}{qB_0}, 0 \right)$

16. A large tank is filled with water (density ρ), upto height h . Water is coming out from section (4). Fluid pressure at section (3) at the instant shown is : (Given : Area $A_2 = A_4 = 2A_3$ and A_2, A_3, A_4 are very small as compared to A_1 ; atmospheric pressure = P_0). Assume water to be non-viscous and incompressible.



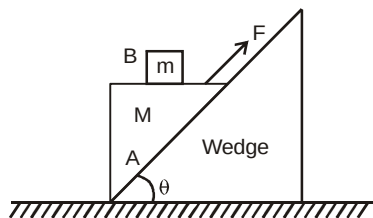
- (A) $P_0 + \rho gh$ (B) $P_0 - \rho gh$ (C) $P_0 - 2\rho gh$ (D) $P_0 - 3\rho gh$

17. A small child tries to move a large rubber toy placed on the ground. The toy does not move but gets deformed under his/her pushing force ($\vec{F}$) which is obliquely upward as shown . Then select correct alternative assuming that deformation is completed :

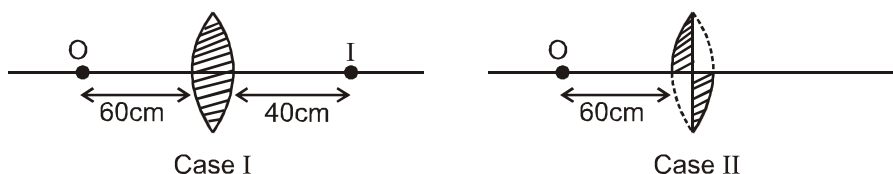


- (A) The resultant of the pushing force ($\vec{F}$), weight of the toy, normal force by the ground on the toy and the frictional force is zero.
 (B) The normal force by the ground is equal and opposite to the weight of the toy.
 (C) The pushing force ($\vec{F}$) of the child is balanced by the equal and opposite frictional force
 (D) The pushing force ($\vec{F}$) of the child is balanced by the total internal force in the toy generated due to deformation

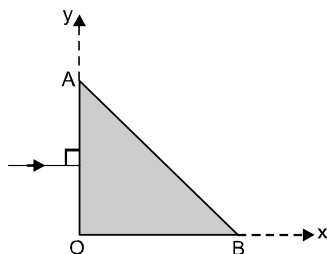
18. Wedge is fixed on horizontal surface. Triangular block A of mass M is pulled upward by applying a constant force F parallel to incline of the wedge as shown in the figure and there is no friction between the wedge and the block A, while coefficient of friction between A and block B of mass m is μ . If there is no relative motion between A and B then frictional force developed between A and B is



- (A) $\left[\frac{F + (m - M)g \sin \theta}{(m + M)} \right] m \cos \theta$ (B) μmg (C) $\left[\frac{F - (m + M)g \sin \theta}{(m + M)} \right] m \cos \theta$ (D) $\mu mg/2$
19. A thin prism of glass is placed in air and water respectively. If $n_g = \frac{3}{2}$ and $n_w = \frac{4}{3}$, then the ratio of deviation produced by the prism for a small angle of incidence when placed in air and water separately is :
- (A) 9 : 8 (B) 4 : 3 (C) 3 : 4 (D) 4 : 1
20. A converging equiconvex thin lens forms real image of a particle as shown in case I. If now lens is cut as shown in case II then select the correct alternative/alternatives :

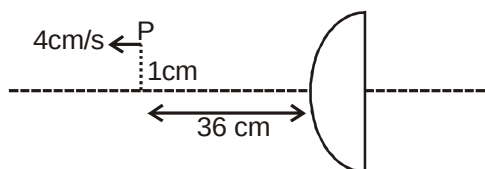


- (A) Image in case II will be at 240 cm from lens.
 (B) Images is erect.
 (C) Image in case II will be at the same location of case I.
 (D) There will be two distinguished images.
21. A triangular medium has varying refracting index $n = n_0 + ax$, where x is the distance (in cm) along x -axis from origin and $n_0 = \frac{4}{3}$. A ray is incident normally on face OA at the mid-point of OA. The range of a so that light does not escape through face AB when it falls first time on the face AB ($OA = 4$ cm, $OB = 3$ cm and $AB = 5$ cm) : (Surrounding medium is air)



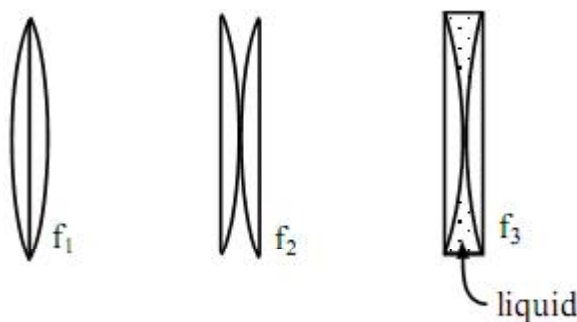
- (A) $a > \frac{1}{9}$ (B) $a > \frac{2}{9}$ (C) $a > \frac{1}{3}$ (D) None of these

22. A very thin plano convex lens of refractive index $\mu = 3/2$ and radius of curvature 15 cm kept fixed as shown :



A point object P starts moving with constant speed 4 cm/s parallel to optical axis from the shown position at $t = 0$. The average speed of image of object from $t = 0$ to $t = 6$ sec is

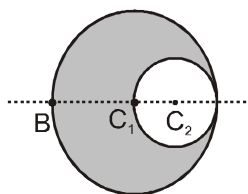
- (A) 20 cm/s
(B) greater than 20 cm/s
(C) less than 20 cm/s
(D) can't say
23. The following figure shows different arrangements of two identical pieces of thin plano-convex lenses. The refractive index of the liquid used is less than that of the glass. The effective focal lengths in the three cases are related as :



- (A) $f_1 = f_2 > f_3$
(B) $f_1 \neq f_2 \neq f_3$
(C) $f_1 = f_2 < f_3$
(D) $f_1 = f_2 = f_3$
24. There is layer of medium of variable refractive index $\mu = \sqrt{2} - \frac{1}{\sqrt{2}} y$ (where $0 \leq y < 1/2$) sandwiched between the layer of glass and air. A beam of light travelling in air at an angle 45° has a width $\Delta\omega$. When the beam enters the layer of glass its width becomes : ($\mu_{\text{glass}} = \sqrt{2}$)

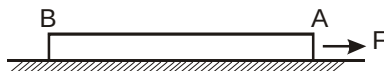
- (A) $\sqrt{\frac{3}{2}} \Delta\omega$
(B) $\sqrt{\frac{2}{3}} \Delta\omega$
(C) $\sqrt{2} \Delta\omega$
(D) $\sqrt{\frac{1}{2}} \Delta\omega$

25. A positively charged sphere of radius r_0 carries a volume charge density ρ (Figure). A spherical cavity of radius $r_0/2$ is then scooped out and left empty, as shown. What is the direction and magnitude of the electric field at point B ?

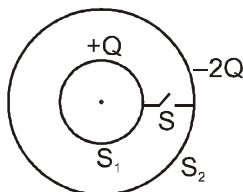


- (A) $\frac{17\rho r_0}{54 \epsilon_0}$ left
(B) $\frac{\rho r_0}{6 \epsilon_0}$ left
(C) $\frac{17\rho r_0}{54 \epsilon_0}$ right
(D) $\frac{\rho r_0}{6 \epsilon_0}$ right
26. A uniform electric field exists in xy plane. The potential of points A (2,2), B(-2, 2) and C(2, 4) are 4V, 16V, 12V, respectively. The electric field is
- (A) $(4\hat{i} + 5\hat{j})\text{V/m}$
(B) $(3\hat{i} + 4\hat{j})\text{V/m}$
(C) $-(3\hat{i} + 4\hat{j})\text{V/m}$
(D) $(3\hat{i} - 4\hat{j})\text{V/m}$

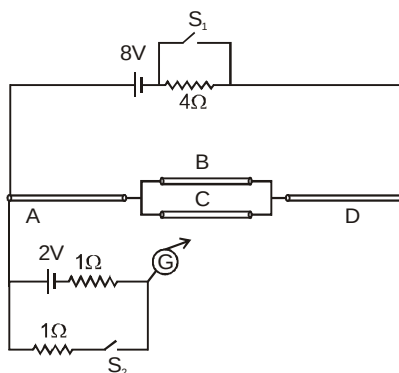
27. A uniform rod is kept at smooth horizontal surface, a constant force is applied on the rod in horizontal direction at end 'A'. Find the ratio of energy stored per unit volume at end A to the energy stored per unit volume in the middle of rod.



- (A) 2 (B) 4 (C) 8 (D) 6
28. We have two concentric conducting sphere S_1 of radius R and S_2 of radius $2R$ and charge $+Q$, $-2Q$ respectively as shown in the figure. Switch S is opened initially. The amount of heat generated after S is closed :



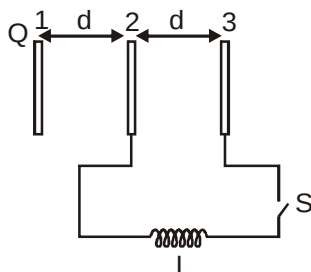
- (A) $\frac{Q^2}{16\pi\epsilon_0 R}$ (B) $\frac{Q^2}{8\pi\epsilon_0 R}$ (C) $\frac{Q^2}{4\pi\epsilon_0 R}$ (D) $\frac{Q^2}{2\pi\epsilon_0 R}$
29. Four wire A, B, C and D each of length $\ell = 10\text{cm}$ and each of area of cross section is 0.1 m^2 are connected in the given circuit. Then, the position of null point is



Given that resistivity

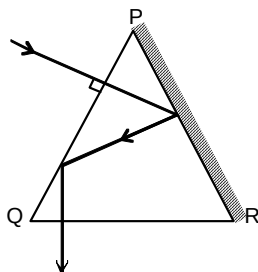
$$\begin{aligned}\rho_A &= 1\ \Omega - \text{m} \\ \rho_B &= 3\ \Omega - \text{m} \\ \rho_C &= 6\ \Omega - \text{m} \\ \rho_D &= 1\ \Omega - \text{m}\end{aligned}$$

- (A) mid point of wire B or wire C when both the switches S_1 and S_2 open.
(B) mid point of wire B when both the switches S_1 and S_2 are closed.
(C) mid point of wire D when both the switches S_1 and S_2 are open.
(D) None of these
30. Three identical large parallel plates are fixed at separation d from each other as shown. The area of each plate is A . Plate 1 is given charge $+Q$, while plates 2 and 3 are neutral and are connected to each other through coil of inductance L and switch S . If resistance of all connecting wires is neglected then the maximum current that will flow through coil after closing switch is (take $C = \epsilon_0 A/d$ and neglect fringe effect)



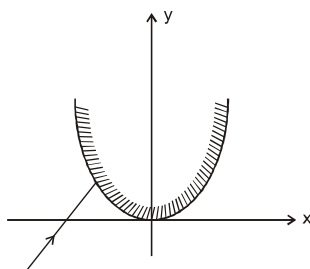
- (A) $\frac{Q_0}{\sqrt{LC}}$ (B) $\frac{Q_0}{2\sqrt{LC}}$ (C) $\frac{3Q_0}{2\sqrt{LC}}$ (D) $\frac{Q_0}{2\sqrt{LC}}$

31. An isosceles glass prism PQR having refractive index μ has one of its faces PR coated with silver. A ray of light is incident normally on the other face PQ as shown in figure ($PQ = PR$). The ray of light upon reflection from PR then undergoes TIR at face PQ and finally emerges through the base of the prism perpendicularly. Then angles of prism are :



- (A) $40^\circ, 70^\circ, 70^\circ$ (B) $50^\circ, 65^\circ, 65^\circ$ (C) $36^\circ, 72^\circ, 72^\circ$ (D) data insufficient
32. You are given a parabolic mirror whose inner surface is silvered so that its outer surface behaves like a mirror.

The equation of the curve formed by its intersection with x-y plane is given by $y = \frac{x^2}{4}$. A ray travelling in x-y plane along line $y = x + 3$, hits the mirror in second quadrant and gets reflected. The unit vector in the direction of reflected ray is :



- (A) $\frac{1}{\sqrt{2}}(-\hat{i} - \hat{j})$ (B) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ (C) $\frac{1}{\sqrt{2}}(-\hat{i} + \hat{j})$ (D) None of these
33. The refractive index in space changes with y, whose function is given as :

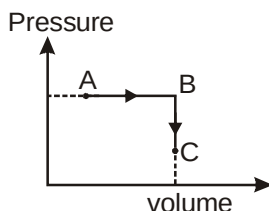
$$\mu(y) = \begin{cases} 1 & \text{if } y \leq 0 \\ ky^2 & \text{if } 0 < y < 2y_0 \\ \frac{\ell}{y^2} & \text{if } 2y_0 \leq y \leq 4y_0 \end{cases}$$

where k and ℓ are positive constant (with proper dimensions). A light ray is incident at point A as shown in figure. When the ray reaches to point B, it becomes parallel to xz-plane. Choose the correct statement.

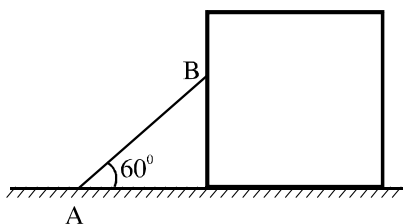
- (A) α is independent of k (B) α is independent of ℓ
 (C) α is independent of both k and ℓ (D) None of these
34. A trinary star system has time period $T = 3$ year, while the distance between its components is 2 astronomical unit. If mass of the sun is represented by M_s , then the total mass of this multiple star system will be :
- Note:**
 (1) System of three star, orbiting around centre of mass of system, is called trinary star system, for simplicity assume all three components to be identical.
 (2) 1 astronomical unit = distance between earth and sun.
 (3) 1 year = time period of earth to complete one full revolution around sun

- (A) $3 M_s$ (B) $2 M_s$ (C) $\frac{7}{9} M_s$ (D) $\frac{8}{9} M_s$

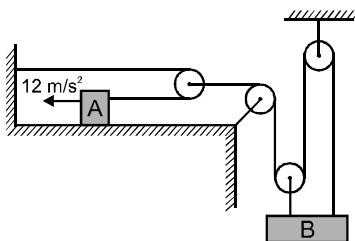
35. A planet is at an average distance d from the sun, and its average surface temperature is constant and equal to T . Assume that the planet receives energy only from the sun, and loses energy only through radiation from its surface. Neglect atmospheric effects. If $T \propto d^{-n}$, the value of n is (Power of sun assumed to be constant)
- (A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
36. A hot black body losses heat at the rate of $30 \text{ Jm}^{-2}\text{s}^{-1}$ and its most intense radiation corresponds to $15,000 \text{ \AA}$. When the temperature of this body is further increased then its most intense radiation corresponds to $7500 \text{ \AA}$, then find the rate of loss of heat i . (Surrounding temperature is half of the initial temperature of black body) :
- (A) $480 \text{ Jm}^{-2}\text{s}^{-1}$ (B) $\frac{30}{17} \text{ Jm}^{-2}\text{s}^{-1}$ (C) $510 \text{ Jm}^{-2}\text{s}^{-1}$ (D) $495 \text{ Jm}^{-2}\text{s}^{-1}$
37. Two moles of ideal helium gas are in a rubber balloon at 30°C . The balloon is fully expandable and can be assumed to require no energy in its expansion. The temperature of the gas in the balloon is slowly changed to 35°C . The amount of heat required in raising the temperature is nearly (take $R = 8.3 \text{ J/mol. K}$)
- (A) 62 J (B) 104 J (C) 124 J (D) 208 J
38. In the process ABC for an ideal mono-atomic gas, the temperature at states A and C are equal. The heat released in the process BC is Q . The work done in the process A to B is equal to (AB is isobaric and BC is isochoric process) :



- (A) Q (B) $\frac{Q}{2}$ (C) $\frac{2Q}{3}$ (D) $\frac{3Q}{2}$
39. Rod AB is placed against a block which is moving towards right with a speed of 1m/s . If at an instant when the rod makes an angle 60° with the horizontal and end A is sliding towards left with a speed of 1m/s . Then the speed of the point of contact 'B' of the rod is :

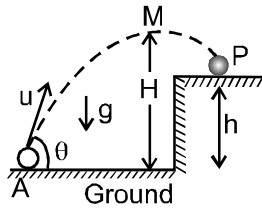


- (A) 1m/s (B) $\frac{\sqrt{3}}{2} \text{ m/s}$ (C) $\frac{1}{2} \text{ m/s}$ (D) $\frac{\sqrt{7}}{\sqrt{3}} \text{ m/s}$
40. Assuming only translational motion of blocks A and B. Find the acceleration of B.



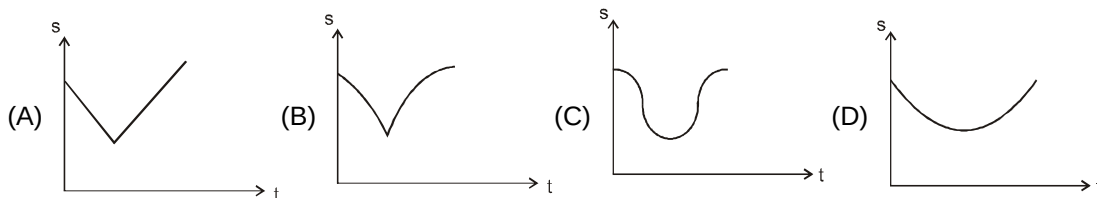
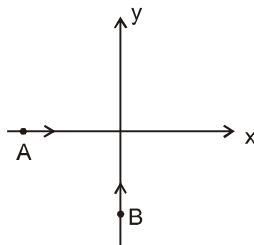
- (A) 6 m/s^2 (B) 2 m/s^2 (C) 4 m/s^2 (D) None of these

41. A ball is projected from ground with initial velocity u at an angle θ (with horizontal) from point A on ground. It strikes a point 'P' at height 'h' from ground level. Provided that the maximum height attained by the ball is H , the angle made by velocity vector of the ball with horizontal at point P at the moment of striking is:

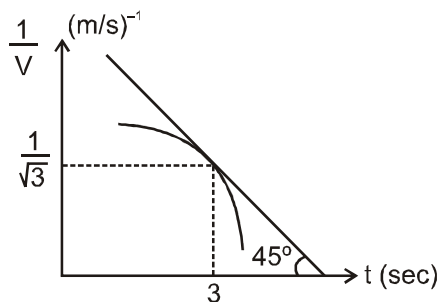


- (A) $\tan^{-1}\left(\frac{\sqrt{2gh}}{u\cos\theta}\right)$ (B) $\tan^{-1}\left(\frac{\sqrt{2gh}}{u\sin\theta}\right)$ (C) $\tan^{-1}\left(\frac{\sqrt{2g(H-h)}}{u}\right)$ (D) $\tan^{-1}\left(\frac{\sqrt{2g(H-h)}}{u\cos\theta}\right)$

42. Particle A and B are moving with constant velocities along positive x and positive y axis respectively, as shown in figure. The graph of separation between them with time is represented by (Assume that they will not collide):

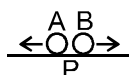


43. The graph shows the variation of $\frac{1}{V}$ (where V is the velocity of the particle) with respect to time. Then find the value of acceleration at $t = 3$ sec.

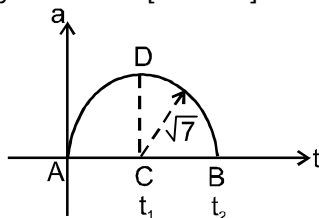


- (A) 3 m/s^2 (B) 5 m/s^2 (C) 1 m/s^2 (D) None of these
44. Which of the following statements is true ?
 (A) Initial velocity of the particle is zero.
 (B) Between $t = 3.5$ to $t = 4.5$ sec, the acceleration of particle is variable.
 (C) Between $t = 7$ to $t = 100$ sec, the acceleration-time graph will be a straight line passing through origin.
 (D) None of the statements is correct.

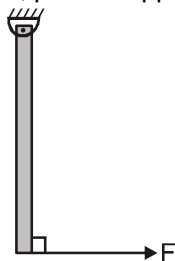
45. If initially, the particle was at $x = 0$, then the position of particle at $t = 2$ seconds will be :
 (A) 10 m (B) $\frac{32}{3}$ m (C) $\frac{40}{3}$ m (D) 14 m
46. The speed of a motor boat in still water is 20 km/hr and river flow is 5 km/hr. A float is dropped from the boat when it starts moving upstream. After moving 1.5 km the boat returns back. The boat will catch the float after (from initial instant).
 (A) 6 min (B) 12 min (C) 10 min (D) 15 min
47. The speed of a particle moving along straight line becomes half after every next second. The initial speed is v_0 . The total distance travelled by the particle will be -
 (A) v_0 (B) $2v_0$ (C) ∞ (D) None
48. Two small balls A and B start from rest from same point P on ground with accelerations a_1 and a_2 respectively in opposite directions as shown. If ball A maintains acceleration a_1 for 5 seconds and B maintains acceleration a_2 for 3 seconds (after which their accelerations become zero) then find magnitude of velocity of A as seen by B at the end of 4 seconds :



- (A) $4a_1 + 4a_2$ (B) $4a_1 - 3a_2$ (C) $4a_1 + 3a_2$ (D) $-4a_1 + 3a_2$
49. The acceleration time graph of the motion of a particle from A to B is semicircle with radius of $\sqrt{7}$ unit. Find the velocity of particle at t_2 if at t_1 , velocity is 3m/sec. [$\pi = 22/7$].

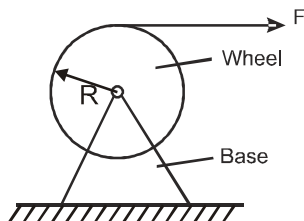


- (A) $\frac{15}{2}$ m/sec (B) 8 m/sec (C) $\frac{17}{2}$ m/sec (D) None of these
50. A rod of length ℓ is given two velocities v_1 and v_2 in opposite directions at its two ends at right angles to the length of the rod. The distance of the instantaneous axis of rotation from the end which is given velocity v_1 is:
 (A) $\frac{v_2 \ell}{v_1 + v_2}$ (B) $\left(\frac{v_1 + v_2}{v_1} \right) \ell$ (C) $\left(\frac{v_1 \ell}{v_1 + v_2} \right)$ (D) $\left(\frac{v_1 + v_2}{v_2} \right) \ell$
51. A uniform ring of mass 'm' and radius 'R' is projected horizontally with velocity v_0 on a rough horizontal floor, so that it starts off with a purely sliding motion and it acquires a purely rolling motion after a distance d. If the coefficient of friction between the ground and ring is μ , then work done by friction in the process is (Ring does not loose contact with surface)
 (A) $-\mu mgd$ (B) $-\frac{1}{4}mv_0^2$ (C) μmgd (D) $-\frac{1}{8}mv_0^2$
52. A rod of mass m and length ℓ rests on a smooth horizontal ground and is hinged at one of its end. At the other end, a horizontal force F is applied whose magnitude is constant and the direction is always perpendicular to the rod. When the rod rotates by 90° angle, power supplied by this force at that instant is :



- (A) $\sqrt{\frac{3F^3 \pi \ell}{m}}$ (B) $\sqrt{\frac{3F^3 \pi \ell}{2m}}$ (C) $\sqrt{\frac{3F^3 \ell}{2m}}$ (D) $\sqrt{\frac{3F^3 \ell}{m}}$

53. A wheel (disc of mass m , radius R) is mounted on a base of mass M . The whole system is placed on rough horizontal surface. The coefficient of friction between the base & the surface is μ . Initially the system is at rest. A time varying horizontal force $F = 2t$ is applied tangentially on the wheel, where t is time in second & force F is in newton. The kinetic energy of the disc when the system is about to slide on ground is :



- (A) $\frac{1}{m} \left(\frac{\mu (M) g}{2} \right)^4$ (B) $\frac{1}{m} \left(\frac{\mu (m + M) g}{2} \right)^4$ (C) $\frac{1}{m} \left(\frac{\mu (m + M) g}{2} \right)^2$ (D) $\frac{1}{m} \left(\frac{\mu (M) g}{2} \right)^2$

54. Large number (N) of coherent waves superimpose at a point in the medium represented by-

$$y_1 = a \sin (kx - \omega t)$$

$$y_2 = a \sin (kx - \omega t + \phi)$$

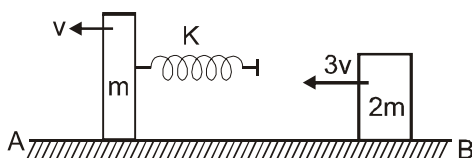
$$y_3 = a \sin (kx - \omega t + 2\phi)$$

.....

$$y_N = a \sin(kx - \omega t + N\phi)$$

where ϕ is very small. Then amplitude of resultant wave is-

- (A) zero (B) Na (C) $(a/\phi) \sin(N\phi/2)$ (D) $2(a/\phi) \sin(N\phi/2)$
55. In experiment of resonance tube for determination of speed of sound, the intensity of sound at resonance will decrease if : (Assume 100% reflection in each case)
- (A) Water is replaced by mercury
 (B) Water is replaced by oil
 (C) Area of cross-section of the tube is increased
 (D) Area of cross-section of the tube is decreased.
56. 5 kg ice at -40°C is mixed with 4.5 kg water at 40°C . Then final amount of water in the mixture will be : [Specific heat of ice = $0.5 \text{ cal/gm } ^\circ\text{C}$, Specific heat of water = $1 \text{ cal/gm } ^\circ\text{C}$, Latent heat of fusion of ice = 80 cal/gm]
- (A) 5.5 kg (B) 7 kg (C) 3.5 kg (D) none of these
57. AB is a long frictionless horizontal surface. One end of an ideal spring of spring constant K is attached to a block of mass m , which is being moved left with constant velocity v , and the another end is free. Another block of mass $2m$ is given a velocity $3v$ towards the spring. Work done by external agent in moving m with constant velocity v in long time will be :

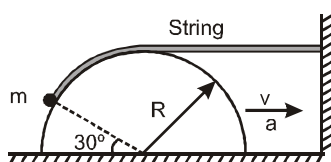


- (A) $-5mv^2$ (B) $-8mv^2$ (C) $8mv^3$ (D) None of these

SECTION – 2 : (One or more options correct Type)

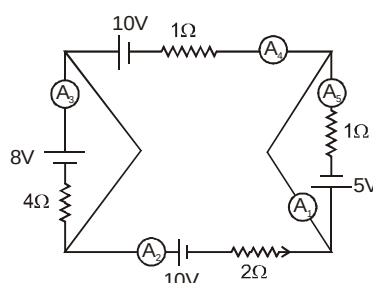
This section contains 32 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE** are correct.

58. An ideal spring has natural length 40 cm and spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and other end of the spring is attached to ceiling. The block released from the position, where the spring has length 45 cm. ($g = 10 \text{ ms}^{-2}$)
 (A) the block will perform SHM of amplitude 5 cm.
 (B) the block will have maximum velocity $30\sqrt{5}$ cm/sec
 (C) the block will have maximum acceleration 15 m/s^2
 (D) the minimum potential energy of the spring will be zero
59. The particle displacement of a travelling longitudinal wave is represented by $S = f(x, t)$. The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true ?
 (A) $\left| \frac{\partial S}{\partial x} \right|_C = \left| \frac{\partial S}{\partial x} \right|_R$
 (B) $\left| \frac{\partial S}{\partial t} \right|_C = \left| \frac{\partial S}{\partial t} \right|_R = 0$
 (C) $(\text{pressure})_C - (\text{pressure})_R = 2 \left| \frac{\partial S}{\partial x} \right|_C \cdot (\text{bulk modulus of air})$
 (D) Particles of air are stationary mid-way between 'C' and 'R'.
60. A point object of mass m is slipping down on a smooth hemispherical body of mass M and radius R . The point object is tied to a wall by an ideal string as shown. At a certain instant shown in figure, speed of the hemisphere is v and its acceleration is a . Then speed v_p and acceleration a_p of the point object is. (Assume all the surfaces in contact are frictionless).



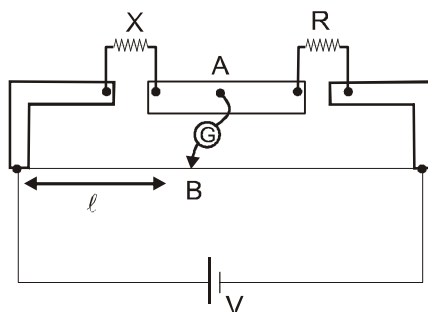
- (A) $v_p = v \sin 60^\circ$ (B) $v_p = v$ (C) $a_p = a$ (D) $a_p = \sqrt{\left(\frac{a\sqrt{3}}{2} + \frac{v^2}{R}\right)^2 + \left(\frac{a}{2}\right)^2}$

61. In the given circuit all ammeters are ideal. Then choose correct statement.

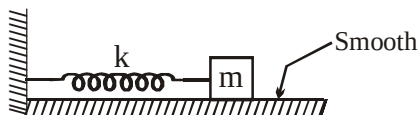


- (A) Reading of $A_2 =$ Reading of A_5 . (B) Reading of A_2 and Reading of A_4 both are zero.
 (C) Reading of A_1 is 5 amp (D) Reading of $A_1 = \frac{5}{2}$ times reading of A_3 .

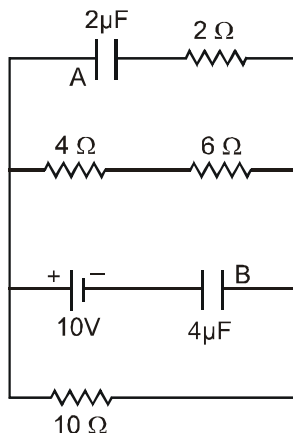
62. A meter bridge is used to predict the value of unknown resistance X , it is observed that current is flowing from A to B initially (when jockey is fixed at a point B). If the temperature of X is increased, then select possible option/s :



- (A) current reduces in magnitude
(B) current remain constant
(C) current become zero
(D) current flows in opposite direction
63. Figure shows an ideal spring block system, force constant of spring is k which has been compressed by an amount x_0 . If x is instantaneous deflection of spring from its natural length, mark the correct option(s).

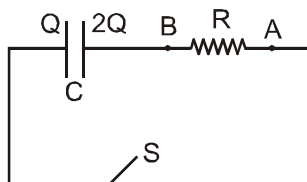


- (A) Instantaneous power developed by spring is $P = kx \sqrt{\frac{k}{2m}(x_0^2 - x^2)}$
- (B) Maximum power of spring is $\frac{k}{2} \sqrt{\frac{k}{m}} x_0^2$
- (C) Maximum power occurs at $x = \frac{x_0}{\sqrt{2}}$
- (D) Maximum power occurs at $x = \frac{x_0}{2}$
64. There is a circuit diagram in which capacitors, resistors & battery are given. Then choose the correct option (s) (Assume initially all the capacitors are uncharged)



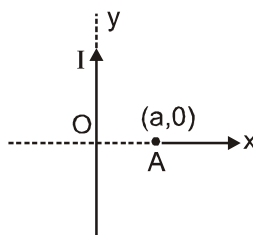
- (A) At steady state charge on capacitor A is zero
(B) At steady state charge on capacitor B is zero
(C) At steady state charge on capacitor B is $40\mu\text{C}$
(D) Electric current in 2Ω resistor is 5A

65. Select correct alternative/s If both plates of parallel plate capacitor are given positive charge as shown in figure and at $t = 0$ switch is closed :



- (A) current in resistance 'R' will flow from A to B
 (B) electric field intensity between plate of capacitor will decrease continuously
 (C) current in the circuit at t is $\frac{Q}{2RC} e^{-t/RC}$
 (D) total heat developed in the circuit long after switch is closed is $\frac{Q^2}{4C}$

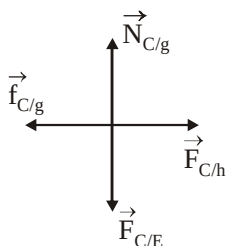
66. An infinitely long wire carrying a current I in vertically upward direction (y -direction) is placed near the surface of earth as shown in figure. Neglect the earth's magnetic field :



At some time t a particle is at point $A(a, 0)$ and moving towards origin with velocity V . If the acceleration of particle at point A is $2g$ in downward direction then :

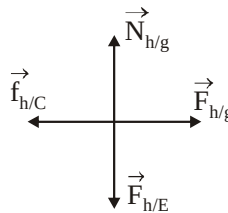
- (A) particle is negatively charged
 (B) particle is positively charged
 (C) magnitude of charge on the particle is $\left(\frac{\pi amg}{\mu_0 VI} \right)$
 (D) magnitude of charge on the particle is $\left(\frac{2\pi amg}{\mu_0 VI} \right)$

67. Consider a cart being pulled by a horse with constant velocity. The horse exerts force $\vec{F}_{C/h}$ on the cart. (The subscript indicate the force on the cart due to horse.). The first subscript denotes the body on which force acts and second due to which it acts. (N = Normal reaction)



Free body diagram of cart

C $\rightarrow$ Cart
 g $\rightarrow$ ground
 h $\rightarrow$ horse
 E $\rightarrow$ Earth

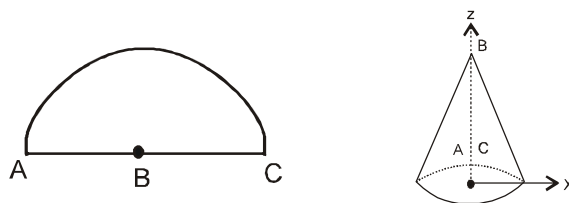


Free body diagram of horse

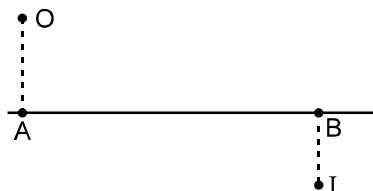
Choose the correct statement(s) :

- (A) $\vec{f}_{C/g}$, $\vec{N}_{C/g}$, $\vec{N}_{h/g}$ are external forces on a system consisting of horse and cart.
 (B) $\vec{F}_{h/g} + \vec{f}_{C/g} = 0$.
 (C) $\vec{N}_{C/g}$ and $\vec{F}_{C/E}$ are action reaction pairs.
 (D) $\vec{F}_{C/h}$ and $\vec{f}_{h/C}$ are action reaction pairs.

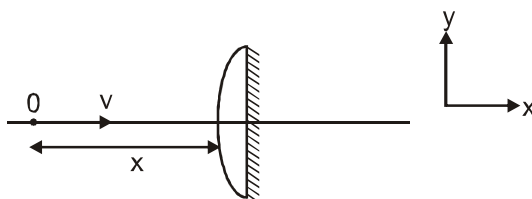
68. A semi-circular silver sheet of radius R and mass m is folded to form a cone with its base in x - y plane as shown in figure. If a block of mass m is sliding on its inclined surface with constant velocity, and a ray in x - z plane along $-\hat{j}$ is incident on it then which of the following is / are correct.



- (A) $\mu_k = \sqrt{3}$
 (B) $\mu_k = \frac{1}{\sqrt{3}}$
 (C) unit vector along the reflected ray will be $\frac{\hat{i}}{2} + \frac{\sqrt{3}}{2}\hat{k}$
 (D) unit vector along the reflected ray will be $\frac{\hat{i}}{2} + \frac{\sqrt{3}}{2}\hat{j}$
69. A luminous point object is placed at O, whose image is formed at I as shown in figure. Line AB is the optical axis. Which of the following statement is/are correct ?

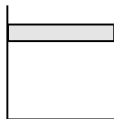


- (A) If a lens is used to obtain the image, then it must be a converging lens and its optical centre will be the intersection point of line AB and OI.
 (B) If a lens is used to obtain the image, then it must be a diverging lens and its optical centre will be the intersection point of line AB and OI.
 (C) If a mirror is used to obtain the image then the mirror must be concave and object and image subtend equal angles at the pole of the mirror.
 (D) I is a real Image.
70. Point A (0, 1) and B (12, 5) are object-image pair (one of the point acts as object and the other point as image) x -axis is the principal axis of the mirror. Then this object image pair is:
 (A) due to a convex mirror of focal length 2.5 cm
 (B) due to a concave mirror having its pole at (2, 0)
 (C) real virtual pair
 (D) data is insufficient for (A) and (B)
71. When object 'O' moves towards a fixed lens mirror combination, select correct choice / choices :



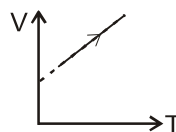
- (A) Image moves towards negative x -axis
 (B) Speed of image and object may be same
 (C) Image may move faster than object
 (D) Image may come closer to arrangement

72. An ideal gas has molar heat capacity at constant pressure $C_p = 5R/2$. The gas is kept in a cylindrical vessel fitted with a piston which is free to move. Mass of the frictionless piston is 9 kg. Initial volume of the gas is 0.0027 m^3 and cross-section area of the piston is 0.09 m^2 . The initial temperature of the gas is 300 K. Atmospheric pressure $P_0 = 1.05 \times 10^5 \text{ N/m}^2$. An amount of $2.5 \times 10^4 \text{ J}$ of heat energy is supplied to the gas such that piston move without acceleration, then

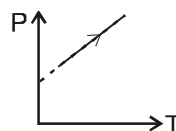


- (A) Initial pressure of the gas is $1.06 \times 10^5 \text{ N/m}^2$ (B) Final temperature of the gas is 1000 K
(C) Final pressure of the gas is $1.06 \times 10^5 \text{ N/m}^2$ (D) Work done by gas is $9.94 \times 10^3 \text{ J}$

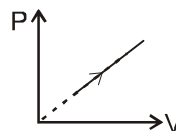
73. S_1 : The process is represents isobaric process.



- S_2 : The process represents isochoric process.

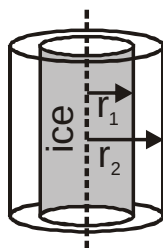


- S_3 : The process represents isothermal process.



- (A) F F F (B) T T F (C) T T T (D) F F F

74. A 100 cm long cylindrical flask with inner and outer radius $r_1 = 2 \text{ cm}$ and $r_2 = 4 \text{ cm}$ respectively, is completely filled with ice at 0°C as shown in the figure. The constant temperature outside (surrounding) the flask is 40°C . Assume heat exchange occurs only through the curved surface of the flask. (Thermal conductivity of the flask is $0.693 \text{ W.m}^\circ\text{C}$, $L_{\text{ice}} = 80 \text{ cal/gm}$ and $1 \text{ cal} = 4200 \text{ J}$).



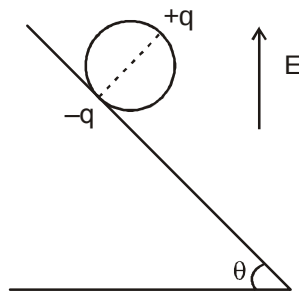
- (A) Rate of heat flow from surrounding to the flask is $80\pi \text{ J/s}$.

- (B) The rate at which ice melts is $\frac{\pi}{4200} \text{ Kg/s}$.

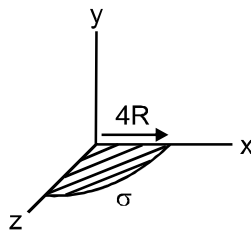
- (C) The rate at which ice melts is $100\pi \text{ Kg/s}$.

- (D) Rate of heat flow from surrounding to the flask is $40\pi \text{ J/s}$.

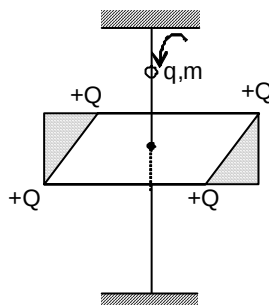
75. Two point charges $+q$ and $-q$ are fixed on diametrically opposite point on the uniform ring, such that $-q$ is at bottom and is in contact with a perfect insulator incline plane. Total mass is equal to m . It remains in equilibrium on the rough inclined plane in the presence of uniform vertical electric field $\vec{E}$. Then



- (A) the value of friction force is $mg \sin \theta$. (B) the value of friction force is less than $mg \sin \theta$.
 (C) the value of electric field is $\frac{mg}{2q}$. (D) the value of electric field is $\frac{mg \tan \theta}{2q}$.
76. Quarter non-conducting disc of radius $4R$ having uniform surface charge density σ is placed in xz -plane then which of the following is the correct :

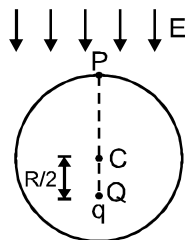


- (A) electric potential at $(0, 3R, 0)$ is $\frac{\sigma R}{4 \epsilon_0}$
 (B) electric potential at $(0, 0, 0)$ is $\frac{\sigma R}{2 \epsilon_0}$
 (C) electric field at $(0, 3R, 0)$ is symmetric with x and z axis
 (D) electric field intensity at $(-4R, 0, -4R)$ is equally inclined with x and z axis
77. Figure shows four charges fixed on the vertex of a square in horizontal plane. A charged bead is constrained to move along a wire which passes through centre of square & perpendicular to its plane. Mark the **correct** statement(s).

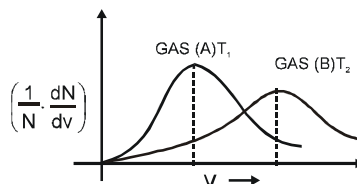


- (A) In gravity free space a positively charged bead cannot be in stable equilibrium at any position.
 (B) In gravity free space a negatively charged bead cannot be in stable equilibrium at any position.
 (C) In presence of gravity a negatively charged bead cannot be in stable equilibrium at any position above the plane of fixed charges.
 (D) In presence of gravity a positively charged bead cannot be in stable equilibrium position above the plane of fixed charges.

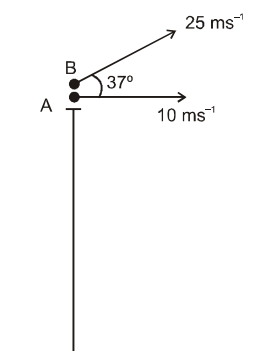
78. A positive charge q is placed inside a neutral hollow conducting sphere of radius R , as shown in figure. Whole system is placed in uniform external vertical electric field pointing downward (line PCQ is also vertical) then select the correct statement(s) about electric field at point P. Point P is a point of the material inside the conductor.



- (A) Electric field due to outer surface of sphere at point P is zero
- (B) Magnitude of electric field due to inner surface of sphere at point P is $\frac{q}{9\pi\epsilon_0 R^2}$
- (C) Magnitude of electric field at point P, due to hollow sphere is less than E
- (D) Direction of resultant electric field at point P due to charge q and hollow sphere is vertically upward
79. According to Maxwell's distribution of molecular speeds, for the below graph drawn for two different samples of gases A and B at temperature T_1 and T_2 respectively, which of the following statements is/are INCORRECT :

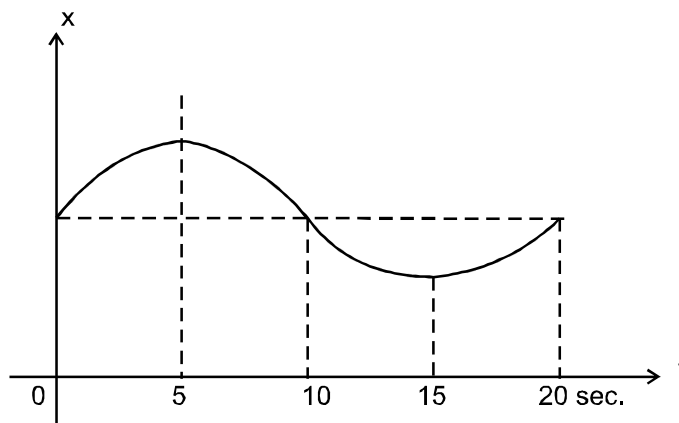


- (A) If $T_1 = T_2$, then molecular mass of gas B (M_B) is greater than molecular mass of gas A (M_A).
- (B) If molecular mass of gas A (M_A) is equal to molecular mass of gas B (M_B), then $T_1 > T_2$.
- (C) If $T_1 < T_2$, then molecular mass of gas B (M_B) is necessarily less than molecular mass of gas A (M_A).
- (D) If gas A is O_2 and gas B is N_2 , then considering them to be ideal gases, T_1 is necessarily less than T_2 .
80. Particle A and B are projected at $t = 0$ from a high tower then which of following statement is/are true about their motion when both particles are in air :

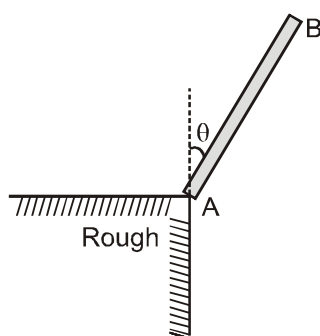


- (A) their relative velocity is non zero constant
- (B) separation between two particle increase then decrease.
- (C) distance between two particle at $t = 2$ sec is $10\sqrt{17}$ m
- (D) acceleration of particle with respect to each other is zero.

81. In a projectile motion from a point on horizontal surface to another point on the same surface (Take $\vec{a}$ = acceleration and $\vec{v}$ = instantaneous velocity)
- (A) $\vec{a} \cdot \vec{v} = 0$ at maximum height (B) $\vec{a} \cdot \vec{v} = 0$ only if angle of projection is 90°
 (C) $\vec{a} \times \vec{v} = \text{const.}$ every where in air (D) None of these
82. At what angle should a body be projected with a velocity 24 ms^{-1} just to pass over the obstacle 14 m high at a distance of 24 m . [Take $g = 10 \text{ ms}^{-2}$]
- (A) $\tan \theta = 1$ (B) $\tan \theta = 2$ (C) $\tan \theta = 3$ (D) $\tan \theta = 19/5$
83. Position time graph for a particle moving on straight line is shown in figure. Assume same slope of $x-t$ graph at $t = 0$ and $t = 20 \text{ s}$. Select correct alternative/s :

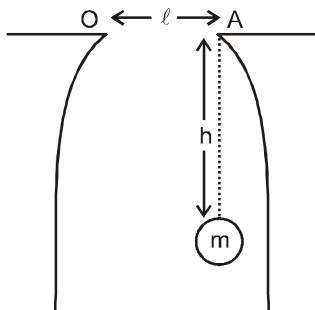


- (A) Average velocity of particle from $t = 0$ to $t = 20 \text{ sec.}$ is zero.
 (B) Acceleration of particle from $t = 10 \text{ sec.}$ to $t = 15 \text{ sec.}$ is positive.
 (C) Average acceleration of particle from $t = 0$ to $t = 20$ is zero.
 (D) Average velocity & instantaneous velocity becomes same in magnitude and direction more than once from $t = 0$ to $t = 20 \text{ sec.}$
84. A uniform rod of mass m and length ℓ is released from rest in the vertical position on a rough (surface is sufficiently rough to prevent just sliding) square corner A, shown in figure, then choose correct options :

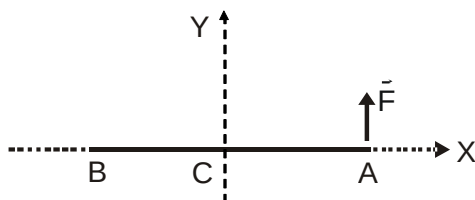


- (A) If the rod just begins to slip when $\theta = 37^\circ$ with vertical, then the coefficient of static friction μ between the rod and the corner is 0.3 .
 (B) If the end of the rod is notched so that it cannot slip, then the angle θ at which contact between the rod and the corner ceases is 53° .
 (C) If the rod just begins to slip when $\theta = 37^\circ$, then the coefficient of static friction μ between the rod and the corner is $3/4$.
 (D) If the end of the rod is notched so that it cannot slip, then the angle θ at which contact between the rod and the corner ceases is 37° .

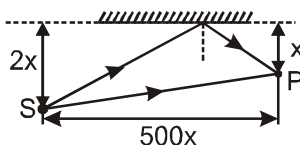
85. A stone is dropped from point A, which is at a horizontal distance ℓ from point 'O'. When the stone has fallen a depth 'h' :



- (A) Angular momentum of the stone about the point 'O' is $m\ell\sqrt{2gh}$
- (B) Angular velocity of the stone about the point 'O' is $\frac{\sqrt{2gh}}{\ell}$
- (C) Angular velocity of the stone about the point 'O' is $\frac{\ell}{\ell^2 + h^2} \sqrt{2gh}$
- (D) Net torque on the stone about the point 'O' is $(mg)(\sqrt{\ell^2 + h^2})$
86. AB is a uniform rod of mass 1 kg and length 1m. Rod is placed on a smooth horizontal table in xy plane along X-axis. Initially centre of mass C of the rod is at origin as shown. A constant force $\vec{F} = 1\hat{j}$ Newton starts acting on the rod at end A. Choose correct options :

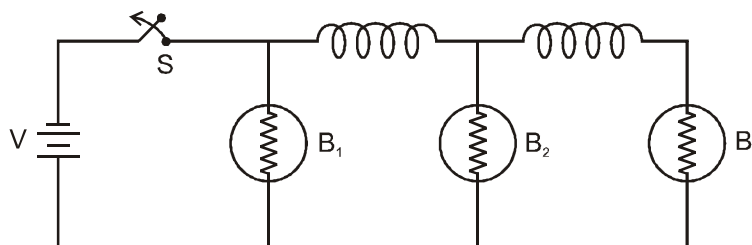


- (A) C will move on Y-axis
- (B) after a finite time rod will be parallel to Y-axis momentarily.
- (C) rod undergoes oscillatory motion about point A after some time.
- (D) angular velocity of rod will be proportional to $\sqrt{\sin\theta}$ where θ is angle made by rod with X-axis, $(0 \leq \theta < \pi/2)$
87. The point S is monochromatic source of light emitting light of wavelength λ . At the point P at a distance x from the mirror as shown in the figure, interference takes place between two light rays one directly coming from source S and another after reflection from the mirror such that a maxima is formed. The value of x may be :

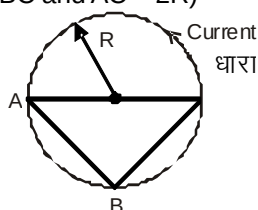


- (A) 120λ (B) 125λ (C) 62.5λ (D) 187.5λ

88. The circuit shown in figure consisting of three identical lamps and two coils is connected to a direct current source. The ohmic resistance of the inducting coils is negligible. The switch is closed for a long time and then opened. Which of the following statements is/are correct for the instant immediately after opening the switch?



- (A) All the lamps are turned off
(B) Brightness of B_2 & B_3 remains unchanged
(C) Brightness of B_1 suddenly decreases.
(D) Brightness of B_1 suddenly increases
89. Inside a very long solenoid of radius R , consider the triangular circuit ABC as shown in figure. Axis of solenoid is perpendicular to the plane of paper. If the magnetic field inside the solenoid changes at the rate dB/dt , then the magnitude of induced emf ($AB = BC$ and $AC = 2R$)



- (A) in the triangular circuit ABC is $R^2 \left| \frac{dB}{dt} \right|$
(B) in the triangular circuit ABC is $\frac{R^2}{2} \left| \frac{dB}{dt} \right|$
(C) between the ends of wire AB is $\frac{R^2}{2} \left| \frac{dB}{dt} \right|$, if AC and BC were removed from the circuit
(D) between the ends of wire BC is $\frac{R^2}{2} \left| \frac{dB}{dt} \right|$, if AC and AB were removed from the circuit,

SECTION - 3 (True & False Statement Type)

This Section Contains 3 questions. Each questions contains 4 statements S_1, S_2, S_3 & S_4 . Each statement is either true (T) or false (F). Each questions has 4 choices (A), (B), (C) and (D) each of which contains whether S_1, S_2, S_3 & S_4 are true or false. Exactly one choice contains the correct order of truthness or falseness of S_1, S_2, S_3 & S_4 respectively and is the correct choice.

90. S_1 : When a concave mirror is held under water, its focal length increases.
 S_2 : When a convex lens is held under water, its focal length increases.
 S_3 : If electric field is zero at a point, the electric potential must also be zero at that point.
(A) T T T (B) F F F (C) F T F (D) T F T
91. S_1 : Two particles are in projectile motion at small height under gravity. The path of one particle as seen by another particle will be a straight line.
 S_2 : If $\frac{dv}{dx} = 0$ at $x = x_1$ and $\frac{d^2v}{dx^2} < 0$ at $x = x_1$ then v has maximum value at $x = x_1$.
 S_3 : In order to cross a river of uniform width (flow is also uniform) in shortest time, swimmer must swim in perpendicular direction of river flow.
 S_4 : If acceleration of particle is positive then its speed increase.
State, in order, whether S_1, S_2, S_3, S_4 are true or false
(A) FFTF (B) FTTF (C) TFTF (D) FTFF

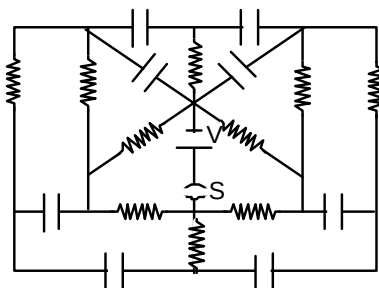
92. S_1 : If pressure of a gas is increased keeping the temperature constant the speed of sound wave in the gas increases.
- S_2 : In a travelling sinusoidal sound wave the phase difference between displacement wave and pressure wave is $\frac{\pi}{2}$.
- S_3 : The apparent frequency is v_1 , when a source approaches a stationary observer with speed u and is v_2 when the observer approaches the same stationary source with the same speed u . Then $v_2 < v_1$, if $u < v$ where v is the speed of sound.
- (A) T F T (B) F T T (C) F F T (D) F T F

SECTION – 4 : (Paragraph Type)

This section contains 10 **paragraphs** each describing theory, experiment, data etc. **23 questions** relate to two paragraphs with **one or two or three questions** on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 93

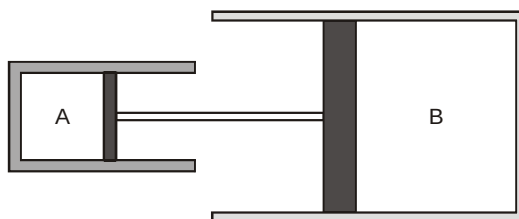
A system of capacitors and resistors are connected to an ideal battery of emf V as shown in figure. Initially all the capacitors are uncharged and switch is open. Now switch S is closed at $t = 0$. (Capacitance of each capacitor is ' C ' and resistance of each resistor is R and all of the symbols have their usual meaning)



93. Current through battery just after switch is closed.
- (A) $\frac{7}{13} \frac{V}{R}$ (B) $\frac{17}{13} \frac{V}{R}$ (C) $\frac{2V}{R}$ (D) None of these

Paragraph for Questions 94 and 95

Two fixed and horizontal cylinders A and B having pistons (both massless) of cross sectional area 100 cm^2 and 200 cm^2 respectively, are connected by massless rod. The piston can move freely without friction. The cylinder A contains 100 gms of an ideal gas ($\gamma = 1.5$) at pressure 10^5 N/m^2 and temperature T_0 . The cylinder B contains identical gas at same temperature T_0 but has different mass. The piston are held at the state such that volume of gas in cylinder A and cylinder B are same and is equal to 10^{-2} m^3 . The walls and piston of cylinder A are thermally insulated where as gas in cylinder B is maintained at constant temperature T_0 . The whole system is in vacuum. Now the pistons are slowly released and they move towards left and mechanical equilibrium is reached at the state when the volume of gas in cylinder A becomes $25 \times 10^{-4} \text{ m}^3$.



94. The mass of gas in cylinder B is
 (A) 200 gms (B) 600 gms (C) 500 gms (D) 1 kg
95. The change in internal energy of gas in cylinder A is :
 (A) 2000 J (B) 1000 J (C) 500 J (D) 3000 J

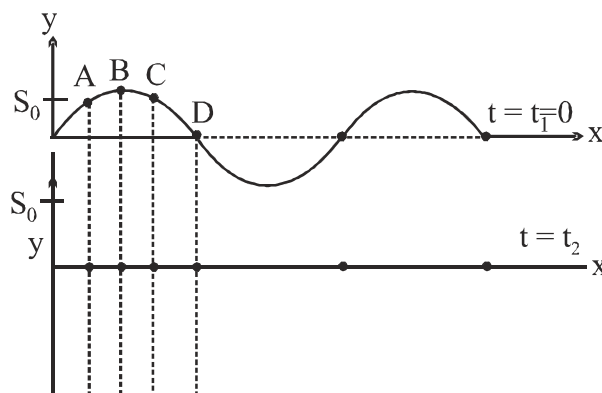
Paragraph for Question Nos. 96 to 97

A regular hexagon of side a stands vertically with one side on the ground and a particle is projected from ground so as to graze its four upper vertices and return back to ground.

96. The angle of projection of particle required is :
- (A) $\tan^{-1}(\sqrt{3})$ (B) $\tan^{-1}\left(\sqrt{\frac{28}{3}}\right)$ (C) $\tan^{-1}\left(\sqrt{\frac{29}{2}}\right)$ (D) $\tan^{-1}(\sqrt{6})$
97. The range of the particle on the ground is :
- (A) $3a$ (B) $\sqrt{5}a$ (C) $\sqrt{7}a$ (D) $\sqrt{6}a$

Paragraph for Question Nos. 98 to 99

The graphs show the standing wave on a string at two successive instants of time t_1, t_2 . A, B, C are points on the string (so is the maximum displacement amplitude of the standing wave)



98. Energy of which of the following elements is flowing towards right through the given point at t_1 .
- (A) A (B) B (C) C (D) All of these
99. Near which of the following point has maximum net energy at time t_2 :
- (A) A (B) B (C) C (D) D

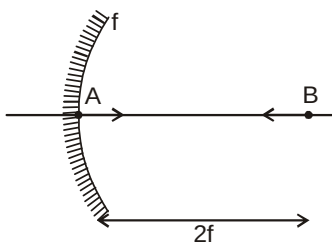
Paragraph for Question Nos. 100 to 101

A particle is projected from a point P (0, 0, 600 m) lying in horizontal X – Z plane with initial velocity of $\vec{u} = (80\hat{i} + 100\hat{j} - 60\hat{k})$ m/s at $t = 0$. It hits a smooth vertical wall in X - Y plane with y-axis in vertically upward direction. If co-efficient of restitution $e = \frac{1}{2}$, ($g = 10$ m/s²)

100. The ball hits the wall at time t when its velocity is making an angle θ with the horizontal then
- (A) $t = 10$ s and $\theta = 0^\circ$ (B) $t = 10$ s and $\theta = 37^\circ$
 (C) $t = 20$ s and $\theta = 90^\circ$ (D) $t = 5$ s and $\theta = 45^\circ$
101. Position of the point of impact with the horizontal X-Z plane after collision with the wall is
- (A) (800, 0, 300) m (B) (1600, 0, 300) m
 (C) (800, 0, 150) m (D) (1600, 0, 600) m

Paragraph for Question Nos. 102 to 103

At time $t = 0$, two point objects A and B respectively are at pole and centre of curvature of a fixed concave mirror of focal length f . The velocity vectors of A and B are always $\vec{v}_A = u\hat{i}$ and $\vec{v}_B = -u\hat{i}$ respectively, where $\hat{i}$ is unit vector along principal axis directed from pole towards focus and u is a positive constant.



102. The distance between images of A and B will be $4f$ at time $t =$

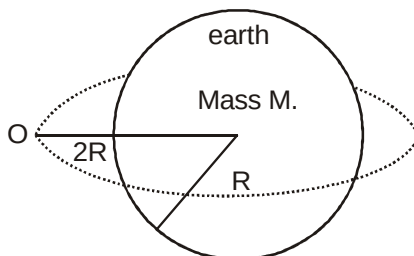
(A) $\frac{f}{\sqrt{2}u}$ (B) $\frac{\sqrt{2}f}{u}$ (C) $\frac{f}{2u}$ (D) $\frac{f}{4u}$

103. The magnitude of relative velocity of image of A and image of B at $t = 0$ is :

(A) u (B) $2u$ (C) $\frac{u}{2}$ (D) $\sqrt{2}u$

Paragraph for Question Nos. 104 to 105

A satellite is moving around earth (mass M , Radius R) in circular orbit of radius $2R$ in equatorial plane.



104. Find the area on the earth, this satellite covers for communication purpose in its one complete revolution w.r.t earth.

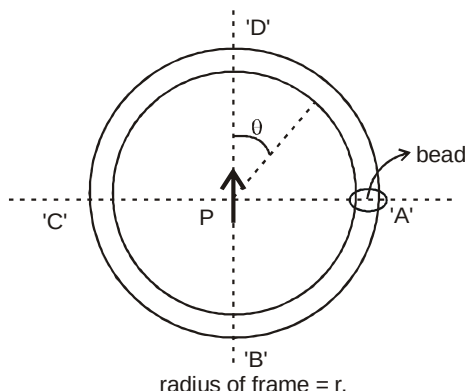
(A) $2\pi R^2$ (B) $\sqrt{3}\pi R^2$ (C) $2\sqrt{3}\pi R^2$ (D) $\frac{4}{\sqrt{3}}\pi R^2$

105. A particle is projected from this satellite radially outward relative to satellite with speed u . Find the minimum value of u so that particle escape from the gravitational force of earth. (Mass of earth $\gg \gg$ mass of satellite $\gg \gg$ mass of particle)

(A) $\sqrt{\frac{GM}{R}}$ (B) $\sqrt{\frac{GM}{2R}}$ (C) $\sqrt{\frac{2GM}{R}}$ (D) $2\sqrt{\frac{GM}{R}}$

Paragraph for Question Nos. 106 to 107

A small charged bead (mass = m , charge = Q) can slide on a circular frictionless, fixed insulating wire frame. A point like dipole (dipole moment P) is fixed at the centre of circle. Bead is released from rest from given position 'A'. (Neglect gravity)



106. Speed of bead as a function of θ is

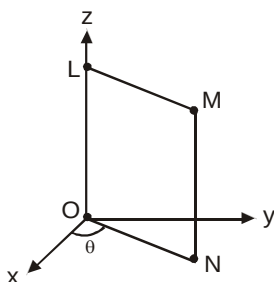
(A) $\sqrt{-\frac{QP \sin \theta}{2\pi\epsilon_0 m r^2}}$ (B) $\sqrt{-\frac{QP \cos \theta}{2\pi\epsilon_0 m r^2}}$ (C) $\sqrt{\frac{QP \sin \theta}{2\pi\epsilon_0 m r^2}}$ (D) $\sqrt{\frac{QP \cos \theta}{2\pi\epsilon_0 m r^2}}$

107. Choose **incorrect** statements.

- (A) Speed of bead is maximum at point 'B'.
 (B) Normal force on the bead at point 'A' is zero
 (C) Tangential acceleration of bead at point 'A' is zero
 (D) Motion of bead is periodic.

Paragraph for Question Nos. 108 to 109

The electric field intensity at all points in space is given by $\vec{E} = \sqrt{3} \hat{i} - \hat{j}$ volts/metre. A square frame LMNO of side 1 metre is shown in figure. The point N lies in x-y plane. The initial angle between line ON and x-axis is $\theta = 60^\circ$



108. The magnitude of electric flux through area enclosed in square frame LMNO is -

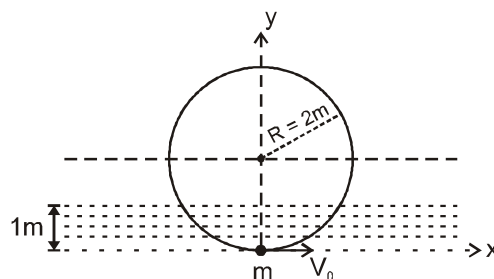
- (A) 0 volt metre (B) 1 volt metre (C) 2 volt metre (D) 4 volt metre

109. The square frame LMNO is now rotated about z-axis by an angle 30° , such that θ either increases or decreases. Then pick up the correct statement .

- (A) The magnitude of electric flux increases from initial value as θ is increased.
 (B) The magnitude of electric flux increases from initial value as θ is decreased.
 (C) The magnitude of electric flux may increase or decrease from initial value as θ is changed.
 (D) The magnitude of electric flux will decrease from initial value as θ is changed.

Paragraph for Question Nos. 110 to 112

A smooth and vertical circular wire frame of radius 2m is fixed inside water as shown. A small bead of specific gravity 0.5 is threaded on the wire and is kept at the origin. If the bead is imparted velocity v_0 towards positive x axis, it moves on the wire frame then neglecting effect of viscosity

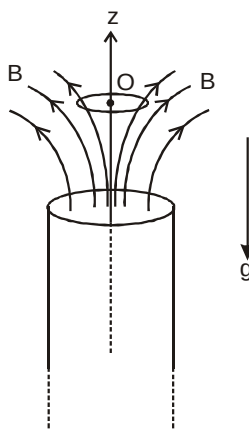


110. Minimum value of V_0 so that it completes vertical circle will be :
 (A) $5\sqrt{2}$ m/s (B) $2\sqrt{20}$ m/s (C) $2\sqrt{10}$ m/s (D) $\sqrt{70}$ m/s
111. Inside water, as bead moves up, with time its speed.
 (A) Increases at constant rate (B) Increases at decreasing rate
 (C) Increases with increasing rate (D) moves with constant acceleration
112. If bead is given minimum velocity V_0 at origin as calculated in first question then acceleration 'a' of bead just after it comes out of water is :
 (A) $\sqrt{5}g$ (B) $g < a < 2g$ (C) $2g < a < 3g$ (D) $3g < a < 4g$

Paragraph for Questions 113 and 115

A thin superconducting (zero resistance) ring is held above a vertical long solenoid, as shown in the figure. The axis of symmetry of the ring is same as that of the solenoid. The cylindrically symmetric magnetic field around the ring can be described approximately in terms of the vertical and radial component of the magnetic field vector as $B_z = B_0(1 - \alpha z)$ and $B_r = B_0\beta r$, where B_0 , α and β are positive constants, and z & r are vertical and radial position coordinates, respectively. Initially plane of the ring is horizontal, has no current flowing in it. When released, it starts to move downwards with its axis still vertical. Initial coordinates of the centre of the ring 'O' is $z = 0$ and $r = 0$.

In the given diagram point O is on the axis and slightly above the solenoid having vertical and radial position coordinates as (0, 0). Ring has mass m , radius r_0 and self inductance L . Assume the acceleration due to gravity as g .



113. Find the magnitude of current in the ring at a vertical position z .
 (A) $\frac{1}{L}B_0\alpha\pi r_0^2z$ (B) $\frac{1}{L}B_0\alpha\beta\pi r_0^2z$ (C) $\frac{1}{2L}B_0\beta\pi r_0^2z$ (D) None of these
114. Find the vertical coordinates z for equilibrium position of the ring.
 (A) $\frac{mgL}{2B_0^2\alpha\beta\pi^2r_0^4}$ (B) $-\frac{mgL}{2B_0^2\alpha\beta\pi^2r_0^4}$ (C) zero (D) None of these
115. Find the time period of SHM (for small displacement along z -axis) of the ring.
 (A) $\frac{1}{B_0r_0^2}\sqrt{\frac{2mL}{\alpha\beta}}$ (B) $\frac{1}{2B_0r_0^2}\sqrt{\frac{2mL}{\alpha\beta}}$ (C) $\frac{1}{B_0r_0^2}\sqrt{\frac{mL}{3\alpha\beta}}$ (D) ring will not perform SHM

SECTION – 5 : (Matrix - Match Type)

This section contains 4 questions. Each question contains statements given in two columns which have to be matched. Statements in **Column I** are labelled as A,B,C and D whereas statements in **Column II** are labelled as p,q,r, s and t. Any given statement in Column-I can have correct matching with ONE OR MORE statement(s) in Column-II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

If the correct matches are A-p, s and t ; B-q and r ; C-p and q ; and D-s and t ; then the correct darkening of bubbles will look like the following :

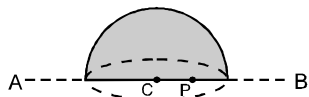
	p	q	r	s	t
A	☒	☒	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

116. In each figure of column-I, a spherical uniform charge distribution is given. in column-II, direction of electric field is represented by arrows. Match the direction of electric field due to charge distribution at given point in figures of column-I.

Column-I

Column-II

- (A) A solid hemisphere, uniformly positive charged direction of resultant electric field at point P must be (point P lies on diameter AB as shown



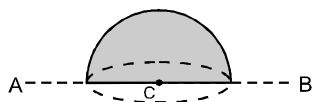
- (B) A solid uniformly positive charged hemisphere with a hemispherical cavity and C_1 and C_2 are the centres of hemispheres, direction of resultant electric field at point C_2 can be



- (C) A solid uniformly positive charged hemisphere with a hemispherical cavity and C_1 and C_2 are the centres of hemispheres, $C_1P = C_2P = \frac{R}{4}$ where R is radius of hemisphere and radius of cavity is $\frac{R}{2}$. Direction of resultant electric field at point P must be



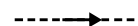
- (D) A solid hemisphere, uniformly charged, direction of resultant electric field at centre C can be



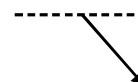
(p)



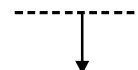
(q)



(r)



(s)



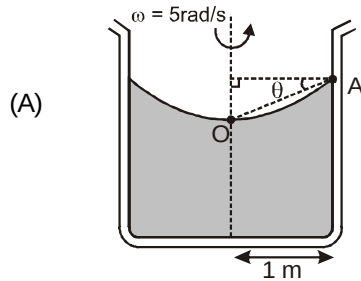
(t)



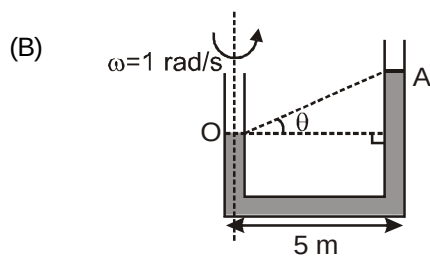
117. Match the difference in level and angle (θ) of line connecting O to A in Column I with the respective options in Column II. (Liquid is ideal in all cases and $g = 10 \text{ m/s}^2$)

Column I

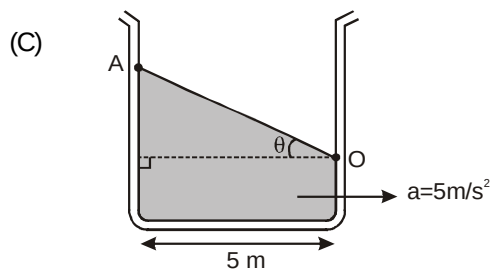
Column II



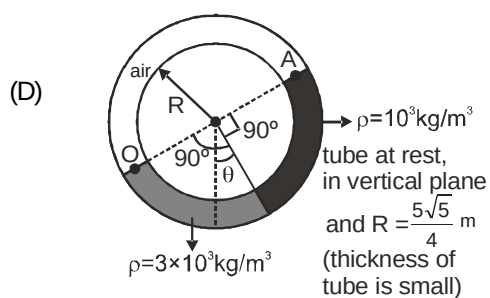
(p) The difference in the vertical height between O and A is 1.25 m



(q) The difference in the vertical height between O and A is 2.5 m



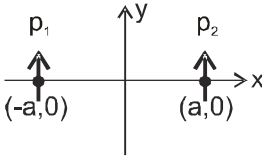
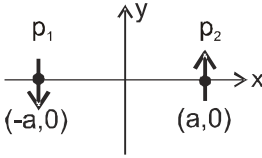
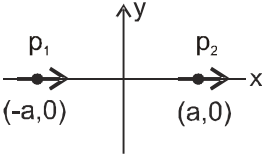
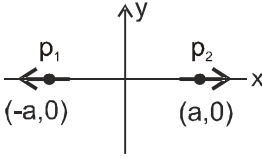
(r) $\theta = \tan^{-1} \left(\frac{1}{2} \right)$



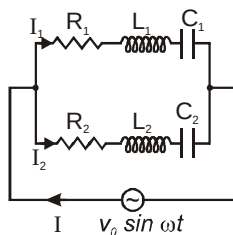
(s) $\theta = \tan^{-1} \left(\frac{1}{4} \right)$

(t) $\theta = \tan^{-1} \left(\frac{5}{4} \right)$

118. In each situation of column-I, two electric dipoles having dipole moments $\vec{p}_1$ and $\vec{p}_2$ of same magnitude (that is, $p_1 = p_2$) are placed on x-axis symmetrically about origin in different orientations as shown. In column-II certain inferences are drawn for these two dipoles. Then match the different orientations of dipoles in column-I with the corresponding results in column-II and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the OMR.

Column-I	Column-II
(A)  $(\vec{p}_1 \text{ and } \vec{p}_2 \text{ are perpendicular to x-axis as shown})$	(p) The torque on one dipole due to other is zero.
(B)  $(\vec{p}_1 \text{ and } \vec{p}_2 \text{ are perpendicular to x-axis as shown})$	(q) The potential energy of one dipole in electric field of other dipole is negative
(C)  $(\vec{p}_1 \text{ and } \vec{p}_2 \text{ are parallel to x-axis as shown})$	(r) There is one straight line in x-y plane (not at infinity) which is equipotential
(D)  $(\vec{p}_1 \text{ and } \vec{p}_2 \text{ are parallel to x-axis as shown})$	(s) Electric field at origin is zero. (t) The force one dipole due to other is non zero

119. An alternating L-C-R circuit is shown in figure, then match the column : (Here symbols have their usual meaning)

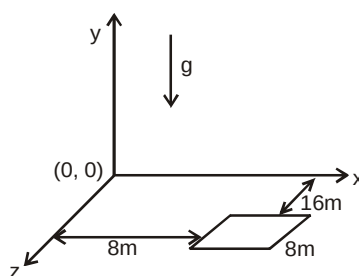


Column I	Column II
(A) If $\omega L_1 - \frac{1}{\omega C_1} = R_1$ and $\omega L_2 - \frac{1}{\omega C_2} = R_2$	(p) I_1 and I_2 are in same phase
(B) If $\omega L_1 - \frac{1}{\omega C_1} = R_1$ and $\frac{1}{\omega C_2} - \omega L_2 = R_2$	(q) $I = I_1 + I_2$; (I, I_1, I_2 rms value of currents)
(C) If capacitor C_1 and inductor L_2 are removed from circuit and $\omega L_1 = R_1$, $\frac{1}{\omega C_2} = R_2$	(r) Magnitude of phase difference between I_1 and I_2 is $\frac{\pi}{2}$
(D) If capacitors C_1 and C_2 are both removed from circuit and $\omega L_2 = R_2$, $\omega L_1 = R_1$	(s) $I = \sqrt{I_1^2 + I_2^2}$; (I, I_1, I_2 rms value of currents) (t) I_1 is lagging and I_2 is leading from source voltage

SECTION – 6 : (Matching list Type)

This section contains 2 multiple choice questions. Each question has matching lists. The codes for the lists have choice (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

120. A square platform of side length 8 m is situated in x–z plane such that it is at 16 m from the x–axis and 8 m from the z-axis as shown in figure. A particle is projected with velocity $\vec{v} = (v_2\hat{i} + 25\hat{j})$ m/s relative to wind from origin and at the same instant the platform starts with acceleration $\vec{a} = (2\hat{i} + 2.5\hat{j})$ m/s². Wind is blowing with velocity $v_1\hat{k}$. ($g = 10$ m/s²)



List I

List II

- | | | |
|-----|--|--------|
| (P) | Least possible values of v_2 (in m/s) so that particle hits the platform or edge of platform is | (1) 4 |
| (Q) | Least possible value of v_1 (in m/s) so that particle hits the platform or edge of platform is | (2) 6 |
| (R) | If t is the time (in second) after particle hits the platform then $2t$ is equal to | (3) 8 |
| (D) | Value of displacement with respect to ground (in m) of the particle in y–direction, when v_2 has its minimum possible value is (till particle hits the platform or edge of platform) | (4) 20 |

Codes :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 4 | 3 | 1 |
| (B) | 2 | 1 | 3 | 4 |
| (C) | 2 | 3 | 4 | 1 |
| (D) | 2 | 1 | 4 | 3 |

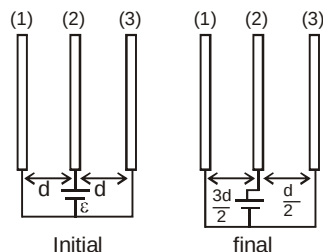
121. Match List I with List II and select the correct answer using the codes given below the lists :

List I

List II

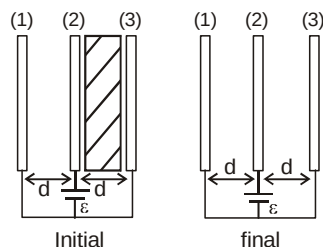
(P) Three large conducting parallel plates each having same area are connected to a battery. Now middle plate shift towards third plate very slowly keeping 1st and 3rd plate fixed. Initial and final situation Shown in figure. If E_1 is the energy stored between the plates initially and E_2 finally. Then

(1) $\frac{E_1}{E_2} > 1$



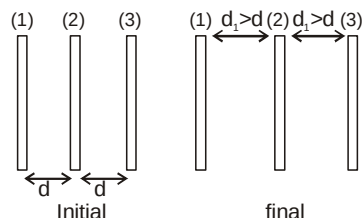
(Q) Three large conducting parallel plates each having same area are connected to battery and space between 2nd and 3rd plates is filled with dielectric. Now dielectric is removed very slowly. If E_1 is the energy stored between the plates initially and E_2 finally. Then

(2) $\frac{E_1}{E_2} < 1$



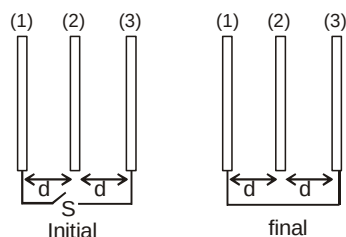
(R) Three large conducting parallel plates each having same area and having charge $-Q$, $2Q$ and $-Q$ respectively initially. Now first plate shift towards left and 3rd towards right keeping 2nd fixed. If F_1 is the force between plate-1 and 3 initially and F_2 is the force between them finally. Then

(3) $\frac{F_1}{F_2} < 1$



(S) Three large conducting parallel plates each having same area, charge on 1st plates $-Q$, on 2nd plate Q and third is uncharged. Now switch is closed, If F_1 is the force between plate-1 and 3 initially and F_2 is the force between them finally. Then

(4) $\frac{F_1}{F_2} = 1$



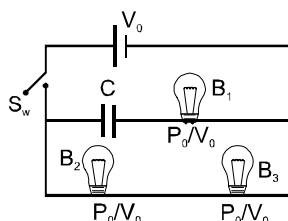
Codes :

	P	Q	R	S
(A)	2	1	3	4
(B)	1	2	4	4
(C)	2	1	4	3
(D)	1	2	4	3

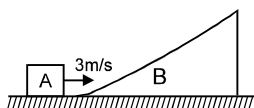
SECTION – 7 : (Integer value correct Type)

This section contains 45 questions. The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

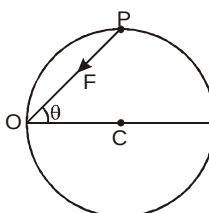
122. A uncharged capacitor is connected in circuit as shown in figure. Power ratings of bulbs are given in diagram. At $t = 0$ switch is closed then calculate ratio of power consumption in the circuit $t = 0$ and $t = \infty$. (Internal resistance of battery and connecting wires is negligible)



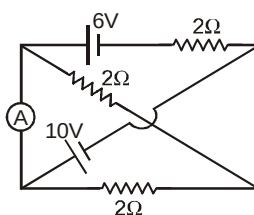
123. In the figure shown A is of mass 1 kg and B of mass 2 kg. A moves with velocity 3 m/s and rises on B which is initially at rest. All the surfaces are smooth. By the time A reaches the highest point on B, find the work done by A on B in joule



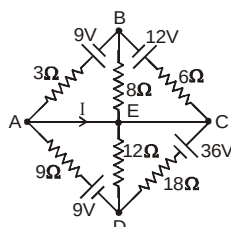
124. A particle P is moving on a circle under the action of only one force, which always acts towards a fixed point O lying on the circumference. Find ratio of $\frac{d^2\theta}{dt^2}$ to $\left(\frac{d\theta}{dt}\right)^2$ at the moment when $\theta = 45^\circ$. (C is centre of circle)



125. An ideal ammeter is connected in a circuit as shown in circuit diagram. What will be the reading of ammeter (in S.I. units).

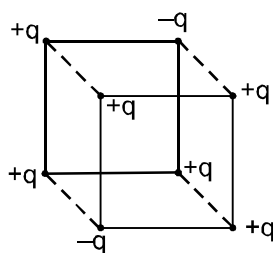


126. In the shown circuit all cells are ideal. If current in branch BE is x ampere and current in branch AE is y ampere then value of $x + y$ (in ampere) is

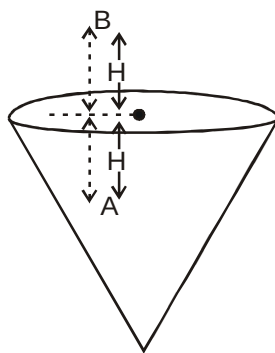


127. A positive charge $+q_1$ is located to the left of a negative charge $-q_2$. On a line passing through the two charges, there are two places where the total potential is zero. The reference is assumed to be at infinity. The first place is between the charges and is 4.00 cm to the left of the negative charge. The second place is 7.00 cm to the right of the negative charge. If $q_2 = -12/11 \mu\text{C}$, what is the value of charge q_1 in μC .
128. A particle is uncharged and is thrown vertically upward from ground level with a speed of $5\sqrt{5} \text{ m/s}$. As a result, it attains a maximum height h . The particle is then given a positive charge $+q$ and reaches the same maximum height h when thrown vertically upward with a speed of 13 m/s. Finally, the particle is given a negative charge $-q$. Ignoring air resistance, determine the speed (in m/s) with which the negatively charged particle must be thrown vertically upward, so that it attains exactly the same maximum height h . (Assume uniform electric field and $g = 10\text{m/s}^2$)

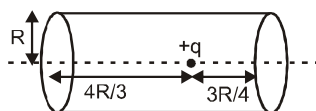
129. The side of the cube is ' ℓ ' and point charges are kept at each corner as shown in diagram. Interaction electrostatic potential energy of all the charges is $\frac{4kq^2}{\sqrt{N} \ell}$, then N is :



130. A point charge Q is kept at point 'A' the flux through the inclined surface of cone is $\frac{3Q}{5\epsilon_0}$. Now another charge $-Q$ is also placed at point B. If net flux through the inclined surface is $\frac{nQ}{5\epsilon_0}$. Find n .

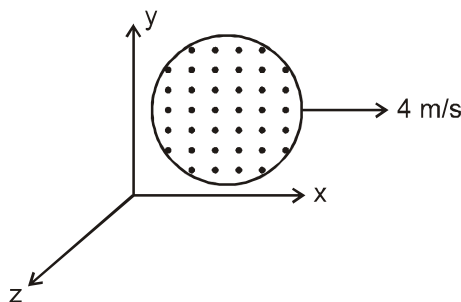


131. A point charge $+q$ is placed on the axis of a closed cylinder of radius R and height $\frac{25R}{12}$ as shown. If electric flux coming out from the curved surface of cylinder is $\frac{xq}{10 \epsilon_0}$, then calculate x .

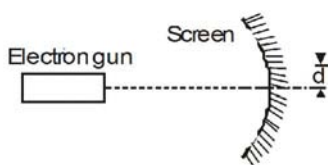


132. A thin conducting wire in the shape of a 'figure of eight' is situated with its circular loops in two planes making an angle of 120° with each other if the current in the loop is I and the radius is R , the magnetic induction at a point of intersection P of the two axes of the loops is $\frac{N\mu_0 I}{48R}$. Find N .

133. A uniform magnetic field $\vec{B} = 0.25\hat{k}$ T exists in a circular region of radius $R = 5$ m. A loop of radius $R = 5$ m lying in $x - y$ plane encloses the magnetic field at $t = 0$ and then pulled at uniform velocity $\vec{v} = 4\hat{i}$ m/s. Find the emf induced (in volts) in the loop at $t = 2$ sec.

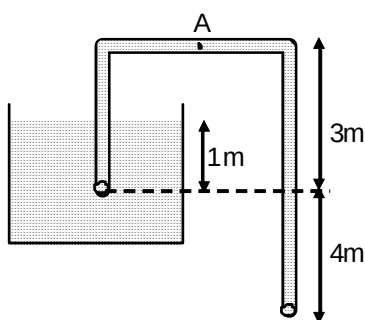


134. Estimate the approximate maximum deflection d of the electron beam (in mm) near the centre of a TV screen due to the Earth's magnetic field of 5.0×10^{-5} T. Assume that the screen is 20 cm from the electron gun which produces electrons of speed $v = 3.0 \times 10^7$ ms $^{-1}$ as shown in figure. You may assume that d is so small that the magnetic force always acts in the same direction

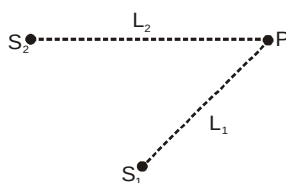


135. A liquid of specific gravity 0.5 is filled in a large container shown in figure. Area of cross section of tube is negligible as compare to the area of cross section of container then the magnitude of gauge pressure at point

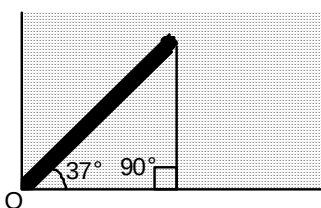
A is given by $\frac{x}{20}$ atm then x is :



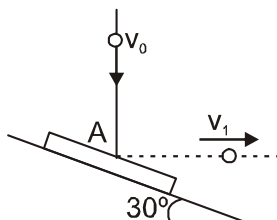
136. Two sound sources shown in the figure vibrate in phase. By moving S_1 along PS_1 consecutive minima are heard when $L_1 - L_2$ has values, 20 cm, 60 cm and 100 cm. If the frequency of sound source is $\frac{1700}{n}$ Hz. Then find the value of n : [Speed of sound is 340 m/s]



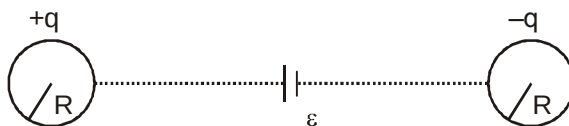
137. A point source of power 50π watts is producing sound waves. The velocity of sound is 330 m/s, density of air is 1.0 kg m^{-3} . Then (using $\pi = 22/7$) at $r = \sqrt{330}$ m from the point source then the pressure amplitude is $x \text{ Nm}^{-2}$. Find the value of x .
138. The work done in increasing the size of a soap film from $10 \text{ cm} \times 6 \text{ cm}$ to $10 \text{ cm} \times 11 \text{ cm}$ is $3 \times 10^{-4} \text{ J}$. If surface tension (in Nm^{-1}) of the film is $N \times 10^{-2}$. Then calculate N .
139. A string fixed at both ends is vibrating in the lowest mode of vibration for which a point at quarter of its length from one end is a point of maximum displacement. The frequency of vibration in this mode is 100 Hz the frequency emitted when it vibrates in the next mode such that this point is again a point of maximum displacement is $\eta \times 100 \text{ Hz}$. Find the value of η .
140. A cylindrical rod of uniform cross section, is attached at O in a water tank. The linear mass density of rod is $\lambda_0 x$, where x is distance of the element of the rod, from end O as shown in figure. If the tension in string is given by $\frac{10^4}{P} \text{ N}$ then P is (Length of rod 1m, radius of area of cross section is $\frac{1}{\sqrt{\pi}} \text{ m}$, $\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ ms}^{-2}$, $\lambda_0 = 10^3$ in S.I. unit)



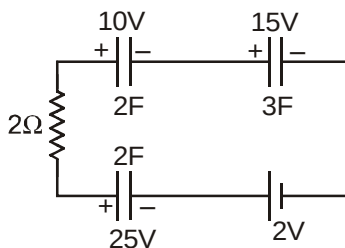
141. A steel ball falling vertically strikes a fixed rigid plate A with velocity v_0 and rebounds horizontally as shown. Assuming surface to be same and the effect of gravity on motion of ball to be neglected. If the coefficient of the restitution for the ball is 'e' then the value of $\frac{1}{e}$ is :



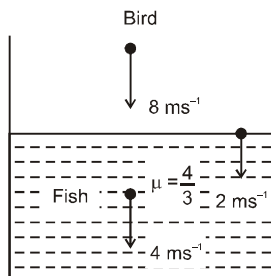
142. Two metallic spheres each of radius R separated by a large distance and connected with a battery of emf ε as shown. In the electric equilibrium the charge on the spheres are $+q$ and $-q$. If $K = \frac{q}{2\pi\epsilon_0 R \varepsilon}$, find the value of K .



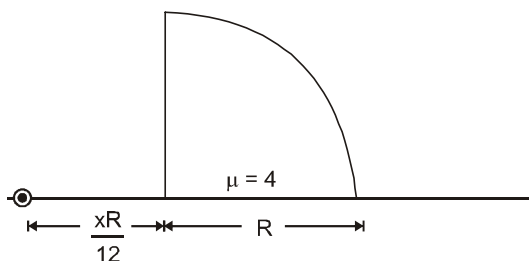
143. In the figure shown find the electric power (in watts) supplied by the ideal cell for the values shown.



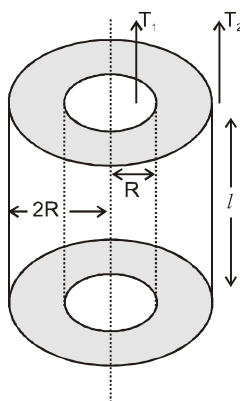
144. Speed of bird with respect to fish in ms^{-1} at given instant, if water surface is moving downward 2 m/sec at this time:



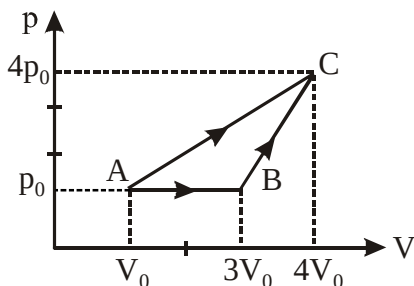
145. Find the value of x so that image of object formed at infinite from a quarter cylinder of refractive index $\mu = 4$.



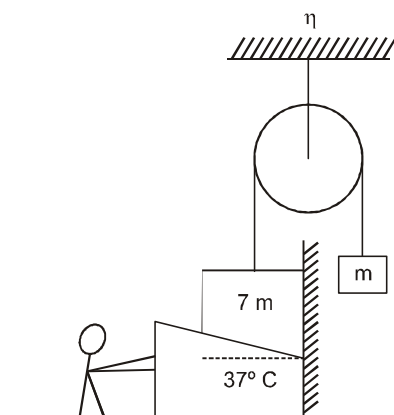
146. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that ($T_1 > T_2$) as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and $2R$ respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{\ell}{2}$ inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk . If rate of radially outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2 , then find the value of n .



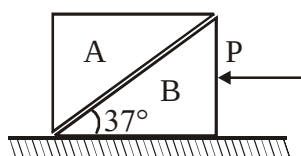
147. A certain quantity of ideal gas takes up 56 J of heat in the process AB and 360 J in the process AC . What is the number of degrees of freedom of the gas.



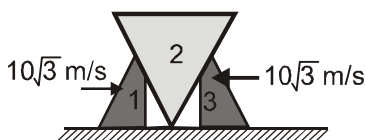
148. All the surfaces are smooth, pulley is ideal and frictionless. If force applied by person to keep the system in equilibrium is $\frac{\eta mg}{2}$ then calculate η :



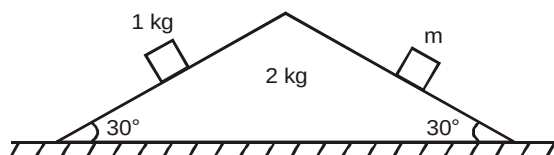
149. Blocks A and B each have same mass $m = 1$ kg. The largest horizontal force P which can be applied to B so that A will not slip up on B is $5X$ N. Neglect any friction. Find the value of X .



150. One equilateral wedge and two other wedges are arranged on a rough horizontal surface as shown in figure. Now wedge 1 and 3 are moved horizontally with velocity $10\sqrt{3}$ m/s as shown. If velocity of wedge-2 is $5x$ ms⁻¹ then x is :



151. In the given figure all surfaces are smooth. The system is released from rest. Find the value of m in kg for which 1kg block does not slide on wedge :



152. In a certain hypothetical radioactive decay process, species A decays into species B and species B decays into species C according to the given reactions.

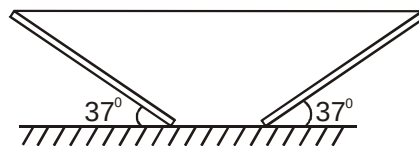


The decay constant for species A is $\lambda_1 = 1$ sec⁻¹ and that for species B is $\lambda_2 = 100$ sec⁻¹. Initially 10^4 moles of species of A were present, while there were none of B and C. It was found that species B reaches its maximum number at a time $t_0 = 2 \ln(10)$ sec. Calculate the value of maximum number of moles of B.

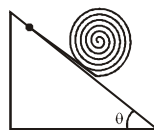
153. A particle projected from ground at some angle with verticle has displacement (w.r.t. point of projection) $\vec{S}_1 = 6\hat{i} + 8\hat{j}$ and $(8\hat{i} + 6\hat{j})$ at some moment t_1 and t_2 respectively. If the range of the projectile is

$\frac{37x}{7}$ m then calculate x .

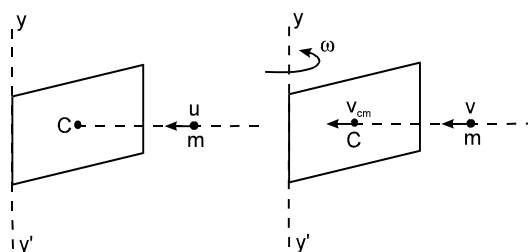
154. A particle moves in straight line with constant acceleration 'a'. The displacements of particle from origin at $t = t_1$, $t = t_2$ and $t = t_3$ are s_1 , s_2 and s_3 respectively. If t_1 , t_2 , t_3 are in A.P. with common difference d and displacements are in G.P. Find the value of n (any number) if $a = \frac{(\sqrt{s_3} - \sqrt{s_1})^2}{nd^2}$
155. A juggler tosses a ball up in the air with initial speed u . At the instant it reaches its maximum height h , he tosses up a second ball with the same initial speed. If the two balls will collide at a height $\frac{xh}{4}$ then x is :
156. A particle with instantaneous velocity $\vec{V} = (2\hat{i} + 3\hat{j} + x\hat{k}) \text{ m/s}$ and instantaneous acceleration $\vec{a} = 3\hat{i} + 4.5\hat{j} - 6\hat{k} \text{ m/s}^2$. If path of the particle is a straight line then magnitude of x is :
157. A solid billiard ball of radius 'R' and mass 'm' initially at rest is given a sharp impulse by a cue, held horizontally at a distance $\frac{R}{2}$ above the centre. Just after the impulse, the velocity of centre of mass of the ball is $v = 10 \text{ m/s}$. The coefficient of friction between ball and table is $\mu = \frac{1}{2}$. The ball starts rolling without slipping t seconds after impulse is given. Find value of $\frac{1}{t}$ (in sec^{-1})
158. Two uniform identical rods of same mass are tied together with the help of a string and balanced as shown in the figure. The minimum coefficient of friction for which the system will remain in equilibrium in the position is $\frac{2}{x}$. The value of 'x' is



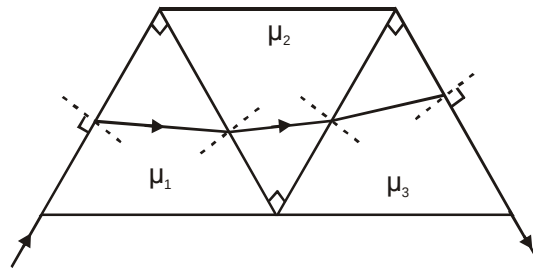
159. A length $L = 15\text{m}$ of flexible tape is tightly wound. It is then allowed to unwind as it rolls down a fixed incline that makes an angle $\theta = 30^\circ$ with the horizontal, the upper end of the tape being fixed. Find the time taken (in second) by the tape to unwind completely. Neglect radius at any time w.r.t. the length of the rope.



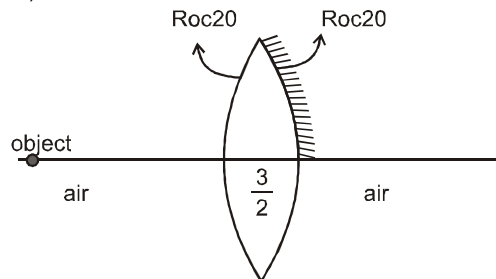
160. A uniform square plate of mass $m = 100 \text{ gm}$ and side $a = 24 \text{ cm}$ can rotate about a smooth vertical axis passing through one edge. It is initially at rest. A particle of mass $m = 100 \text{ gm}$ is moving horizontally and perpendicular to the plane of the plate with velocity $u = 70 \text{ cm/s}$. It collides with the plate elastically at the centre of the plate. Find the angular velocity (in rad/s) of the plate just after collision ?



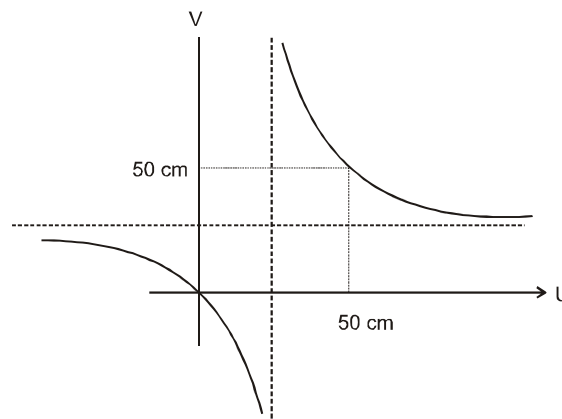
161. A ray of light enters at grazing angle of incidence into an assembly of three isosceles right-angled prisms having refractive indices $\mu_1 = \sqrt{2}$, $\mu_2 = \sqrt{x}$ and $\mu_3 = \sqrt{3}$. If finally emergent light ray also emerges at grazing angle then calculate x :



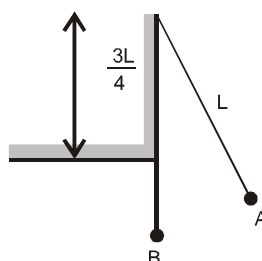
162. Distance between two images formed by upper and lower part of the point object placed at 30 cm from given lens is $(60 + x)$ cm, then the value of x is :



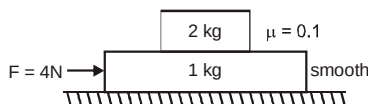
163. UV graph for convex mirror shown in figure. If the magnification of a real object placed at 25 cm from mirror is $\left(\frac{4}{x}\right)$ then x is :



164. A pendulum has period T for small oscillations. An obstacle is placed directly beneath the pivot, so that only the lowest one quarter of the string can follow the pendulum bob when it swings to the left of its resting position as shown in the figure. The pendulum is released from rest at a certain point A. The time taken by it to return to that point is $\frac{nT}{4}$. Find the value of n ?



165. 2 kg block is kept on 1 kg block as shown. Both blocks are initially at rest. The friction between 1 kg block and fixed surface is absent and the coefficient of friction between 2 kg block and 1 kg block is $\mu = 0.1$. A constant horizontal force $F = 4 \text{ N}$ is applied on 1 kg block. If the work done by the friction on 1 kg block in 2 s is $-XJ$, then find the value of X .

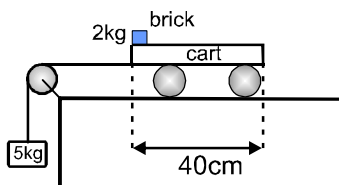


166. Wind entering in a wind mill with a velocity of 20 m/sec , facing area of the windmill is 10 m^2 and density of air is 1.2 kg/m^3 . If wind energy is converted into electrical energy with 33.3% efficiency, if electrical power produced by the wind mill $4N \text{ KW}$ then N is :

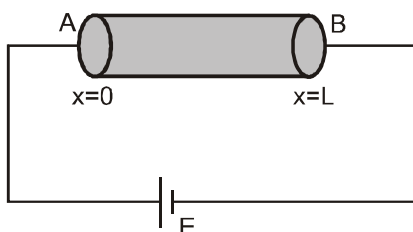
SECTION – 8 : (Integer value correct Type)

This section contains 19 **questions**. The answer to each question is a **Two digit integer**, ranging from 00 to 99 (both inclusive).

167. A cart of mass 3 kg is pulled by a 5 kg object as shown in figure. Length of cart is 40 cm moves on frictionless table. There is a small brick of mass 2 kg on the cart which falls at 0.8 sec. after the start of motion. If the coefficient of friction between cart and brick is $\frac{x}{100}$. Then value of x is : ($g = 10 \text{ m/s}^2$)

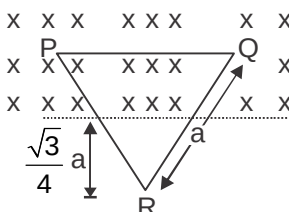


168. AB is a solid cylinder of radius a_0 and length L . Resistivity of the material varies with x -coordinate from end A as $\rho = \frac{\rho_0 x}{L}$. An ideal battery of emf E is connected from end A and B as shown.

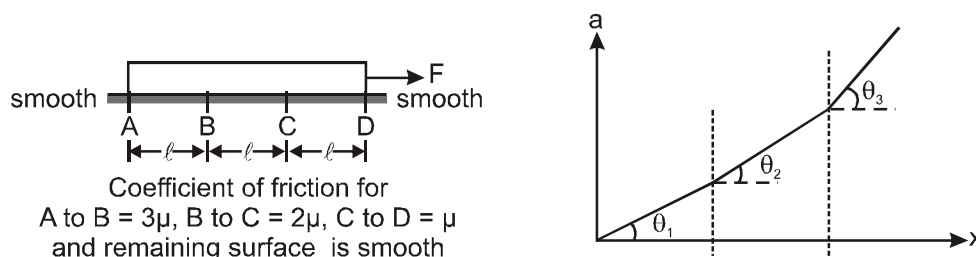


If power gradient (i.e. $\frac{dP}{dx}$, from end A) is $\frac{N\pi a_0^2 E^2}{\rho_0 L^3} x$, then find N

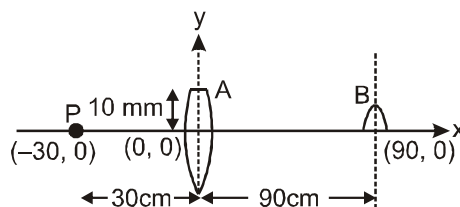
169. An equilateral triangular frame PQR of mass 2 kg and side 1 m is at rest under the influence of horizontal magnetic field 10 T and gravitational field as shown in the figure. The plane of the frame is vertical find the magnitude of current (in ampere) in the frame so that the frame remains at rest [Neglect EMF induced due to motion of the loop] ($g = 10 \text{ m/s}^2$)



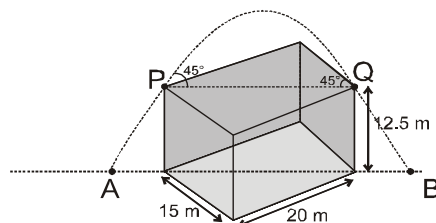
170. A $1\mu\text{C}$ charge is projected opposite to the magnetic field of 10T in opposite direction of the field with 10^6 ms^{-1} then force experienced by charge out given time (in newton).
171. In the diagram, the variation of coefficient of friction of the surface is shown. The external force F applied on board is equal to $2\mu\text{Mg}$. Now the board of mass M is gently pushed so that it starts moving. The variation of acceleration a of board with respect to displacement x has been shown. Find the value of $\left(\frac{\tan\theta_1}{\tan\theta_3}\right)^3$.



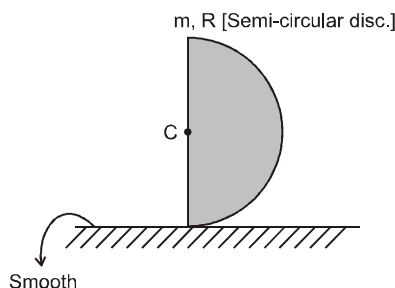
172. A thin convex lens having focal length 20 cm is cut into two parts (A and B) 10 mm above the principle axis. The lower part 'A' is placed with its optical centre at origin and upper part 'B' at $(90\text{ cm}, 0)$ as shown in the figure. A point object 'P' is placed at $(-30\text{ cm}, 0)$. If the distance of final image from x-axis is $N\text{ mm}$, then find the value of N . (Assuming paraxial ray approximation to remain valid)



173. Electrons in a hydrogen like atom ($Z = 3$) make transitions from the fourth excited state to the third excited state and from the third excited state to the second excited state. The resulting radiations are incident on a metal plate to eject photoelectrons. The stopping potential for photoelectrons ejected by shorter wavelength is 3.95 eV . If the stopping potential for the photoelectrons ejected by the longer wavelength is $N \times 10^{-2}\text{ V}$, then find N . (Take energy of the hydrogen atom in ground state as -13.6 eV)
174. Light of intensity I is incident on a fixed plane surface at an angle 30° with the surface. If 50% light is reflected and remaining light is absorbed then radiation pressure on the plate is $\frac{3I}{nC}$, then find n (speed of light is C):
175. A commander fires a shell at certain angle of projection from 'A' which clears the building (cuboid) of dimensions $20 \times 15 \times 12.5\text{ m}^3$ in it's diagonal plane. Shell just clears the roof diagonally at points P and Q and falls on the other side of the building at B, and makes 45° angle with horizontal at P and Q as shown in the figure. If the range AB of the shell is $N \times \sqrt{3}$ metre then find N . ($g = 10\text{ m/s}^2$)

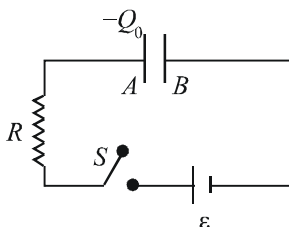


176. If angular acceleration of the uniform disc just after release from rest position shown in figure is $\frac{8\pi g}{NR}$, then calculate N. (C is centre of semi-circular disc.) ($\pi^2 = 10$)

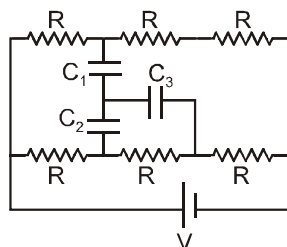


177. A closed organ pipe of length 99.4 cm is vibrating in its first overtone and is always in resonance with a tuning fork having variable frequency $f = (300 - 2t)$ Hz, where t is time in second. The rate by which radius of organ pipe changes when its radius is 1 cm is $\frac{1}{N}$ m/s then find N. (speed of sound in organ pipe = 320 m/s)
178. A constant tangential force of 0.02 N is acting on a large wooden plate of area 10 m^2 floating on the surface of river and plate moves with the constant speed 2 m/s on the river surface. The river is 1 m deep and the water in contact with the river bed is stationary. Assuming constant speed gradient, coefficient of viscosity of water is 10^{-n} poise. Find the value of n.
179. The figure shows an RC circuit with a parallel plate capacitor. Before switching on the circuit, plate A of the capacitor has a charge $-Q_0$ while plate B has no net charge. Now, at $t = 0$, the circuit is switched on. How much time (in second) will elapse before the net charge on plate A becomes zero.

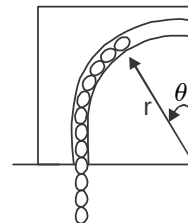
(Given $C = 1 \mu\text{F}$, $Q_0 = 1 \text{ mC}$, $\varepsilon = 1000 \text{ V}$ and $R = \frac{2 \times 10^6}{\ln 3} \Omega$)



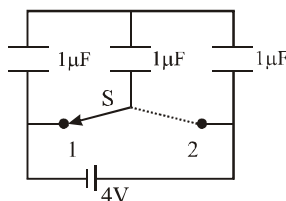
180. In the shown circuit, all three capacitor are identical and have capacitance $C \mu\text{F}$ each. Each resistor has resistance of $R \Omega$. An ideal cell of emf V volts is connected as shown. If the magnitude of potential difference across capacitor C_3 in steady state is $\frac{a}{b} V$ then value of $a \times b$ is :



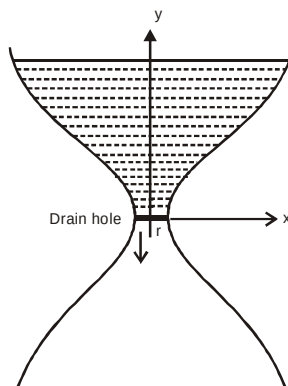
181. The flexible bicycle - type chain of length $5\pi^2$ m and mass per unit length ρ is released from rest with angle $\theta = 0^\circ$ in the smooth circular channel of radius $r = 10\pi$ m and falls through the hole in the supporting surface shown in figure. If v is the velocity of chain in m/s as the last link leaves the slot is : (take $\pi^2 = 10$).



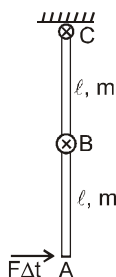
182. If switch S is connected with point-1. initially. Net heat produced in the circuit after switch S is connected to point 2 is $\frac{x}{12} \mu\text{J}$, then x is :



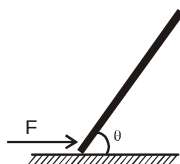
183. The shape of an ancient water clock jar is such that the water level descends at a constant rate at all times. The water level falls by λ m/s and the shape of the jar is given by $y \propto x^n$. Find the value of n if the radius r of the drain hole can be assumed to be very small.



184. Two uniform rods each of mass $m (=1\text{kg})$ and length $\ell (=1\text{m})$ are hinged at B and pivoted at C. If a horizontal impulse $F\Delta t (=7 \text{ N.s})$ is applied to the end A of the lower rod then find the angular velocity ω_2 (in rad/sec) of the upper rod just after the impulse. (Δt is very small and rods can rotate freely w.r.t hinge and pivots)



185. A force F is applied to a uniform, thin rod of mass 4 kg and length $\ell = 50 \text{ cm}$. The rod has pure translational motion in the vertical plane along a smooth, horizontal surface as shown. If $F = 60 \text{ N}$, the angle θ for translation of rod in the given orientation comes to be $\theta = \tan^{-1}\left(\frac{2}{X}\right)$. Find X.



CRITICAL QUESTION BANK

ANSWER KEY

PHYSICS

- | | | | | | | | |
|--|--|-----------|---|-----------|------------|-----------|--|
| 1. (C) | 2. (B) | 3. (B) | 4. (A) | 5. (C) | 6. (D) | 7. (C) | |
| 8. (A) | 9. (A) | 10. (B) | 11. (C) | 12. (B) | 13. (B) | 14. (A) | |
| 15. (A) | 16. (D) | 17. (A) | 18. (C) | 19. (D) | 20. (A) | 21. (B) | |
| 22. (B) | 23. (C) | 24. (A) | 25. (A) | 26. (D) | 27. (B) | 28. (A) | |
| 29. (A) | 30. (D) | 31. (C) | 32. (A) | 33. (A) | 34. (D) | 35. (C) | |
| 36. (C) | 37. (D) | 38. (C) | 39. (D) | 40. (B) | 41. (D) | 42. (D) | |
| 43. (A) | 44. (D) | 45. (B) | 46. (B) | 47. (B) | 48. (C) | 49. (C) | |
| 50. (C) | 51. (B) | 52. (A) | 53. (B) | 54. (D) | 55. (C) | 56. (A) | |
| 57. (B) | 58. (BCD) | 59. (ACD) | 60. (BD) | 61. (BCD) | 62. (ACD) | 63. (BC) | |
| 64. (AC) | 65. (BC) | 66. (BD) | 67. (ABD) | 68. (AC) | 69. (ACD) | 70. (ABC) | |
| 71. (ABCD) | 72. (ACD) | 73. (AD) | 74. (AB) | 75. (AC) | 76. (ABCD) | 77. (AC) | |
| 78. (BD) | 79. (ABCD) | 80. (AD) | 81. (AC) | 82. (AD) | 83. (ABC) | 84. (AB) | |
| 85. (AC) | 86. (ABCD) | 87. (CD) | 88. (BD) | 89. (ACD) | 90. (C) | 91. (C) | |
| 92. (B) | 93. (C) | 94. (D) | 95. (A) | 96. (B) | 97. (C) | 98. (A) | |
| 99. (B) | 100. (A) | 101. (B) | 102. (C) | 103. (B) | 104. (C) | 105. (B) | |
| 106. (B) | 107. (C) | 108. (C) | 109. (D) | 110. (C) | 111. (C) | 112. (D) | |
| 113. (A) | 114. (B) | 115. (A) | 116. (A) – r ; (B) – r ; (C) – r ; (D) – s, t | | | | |
| 117. (A) – p, t ; (B) – p, s ; (C) – q, r ; (D) – q, r | 118. (A) – p, r, t ; (B) – p, q, s, t ; (C) – p, q, r, t ; (D) – p, s, t | | | | | | |
| 119. (B) – r, s, t ; (C) – r, s, t ; (D) – p, q | 120. (B) | 121. (C) | 122. 3 | 123. 1 | | | |
| 124. 2 | 125. 7 | 126. 1 | 127. 4 | 128. 9 | 129. 3 | 130. 1 | |
| 131. 7 | 132. 3 | 133. 6 | 134. 6 | 135. 7 | 136. 2 | 137. 5 | |
| 138. 3 | 139. 3 | 140. 6 | 141. 3 | 142. 1 | 143. 2 | 144. 6 | |
| 145. 1 | 146. 4 | 147. 5 | 148. 9 | 149. 3 | 150. 6 | 151. 6 | |
| 152. 2 | 153. 2 | 154. 1 | 155. 3 | 156. 4 | 157. 7 | 158. 3 | |
| 159. 3 | 160. 5 | 161. 3 | 162. 6 | 163. 8 | 164. 3 | 165. 8 | |
| 166. 4 | 167. 40 | 168. 04 | 169. 04 | 170. 00 | 171. 27 | 172. 30 | |
| 173. 75 | 174. 08 | 175. 25 | 176. 30 | 177. 72 | 178. 02 | 179. 02 | |
| 180. 18 | 181. 30 | 182. 64 | 183. 04 | 184. 06 | 185. 03 | | |

SOLUTIONS

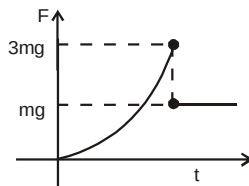
1. $F_{\text{net}} = F_{\text{dynamic}} + F_{\text{static}}$
At any time t when x length is on the table then :
 $F_{\text{static}} = x\lambda g$

$$F_{\text{dynamic}} = V_{\text{rel}} \left(\frac{\lambda dx}{dt} \right) = \sqrt{2gx} \times \lambda \sqrt{2gx} = 2gx\lambda$$

$$F_{\text{net}} = 3 \times \left(\frac{1}{2}gt^2 \right) \lambda g$$

$$\Rightarrow F_{\text{net}} = 3\lambda g \cdot \frac{1}{2}gt^2 = \frac{3}{2}\lambda g^2 t^2 = \frac{3}{2} \frac{mg^2 t^2}{\ell}$$

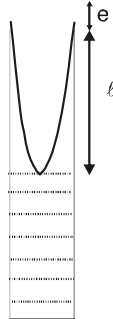
At $t = \sqrt{2\ell/g}$ all the chain will be on table so



$F_{\text{net}} = 0 + mg$ (since At this position $F_{\text{dyn}} = 0$)
So the correct graph is represented by option (3)

2. After first collision B reaches up to angular displacement α and after second collision 'B' reaches to its initial position.

$$\text{So, total time taken is } t = \frac{T}{4} + \frac{T}{4} + \frac{T}{4} + \frac{T}{4} = T$$



3.

First resonance occurs at fundamental frequency

$$f = \frac{V}{4(\ell + e)} \Rightarrow \ell + e = \frac{V}{4f} \text{ (where } e = 0.6 \times 2 = 1.2 \text{ cm)}$$

$$\ell + e = \frac{336}{4 \times 512} = 0.164 \text{ m}$$

$$\ell = 16.4 - 1.2 = 15.2 \text{ cm}$$

4. Kinetic energy present with the wave pulse :

$$\int_0^{a/2} \frac{1}{2} \mu dx \left(\frac{\partial y}{\partial t} \right)^2 + \int_{a/2}^a \frac{1}{2} \mu dx \left(\frac{\partial y}{\partial t} \right)^2$$

$$\left(\frac{\partial y}{\partial t} \right) = \frac{mdx}{dt} \text{ for } 0 \leq x \leq \frac{a}{2} \text{ where } dx/dt = \sqrt{\frac{T}{\mu}}$$

$$\left(\frac{\partial y}{\partial t} \right) = - \frac{mdx}{dt} \text{ for } \frac{a}{2} \leq x \leq a$$

5. Tension in string is maximum when Torque on bob about 'O' is zero.

When bob is at A tension is maximum

$$T - Mg \sqrt{2} = \frac{mV^2}{\ell} \quad \dots\dots\dots (1)$$

$$\text{and } \frac{1}{2} mV^2 = mg \frac{\ell}{\sqrt{2}} - mg\ell \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$= \sqrt{2} mg\ell - mg\ell \quad \dots\dots\dots (2)$$

$$T = mg \sqrt{2} + 2 \sqrt{2} mg\ell - 2mg = mg [3\sqrt{2} - 2]$$

$\therefore X = 3$ **Ans. 3**

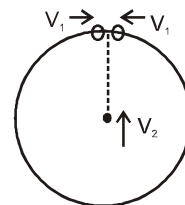
6. Just before the collision, situation is as shown
Let the beads have velocity V_1 w.r.t. the ring and the ring has velocity V_2 .

Then, by momentum conservation

$$2mv_0 = (M + 2m)V_2$$

$$\Rightarrow V_2 = \frac{V_0}{2} \quad \dots\dots\dots (1)$$

By mechanical energy conservation



$$\frac{1}{2} (2m)V_0^2 = \frac{1}{2} 2mv^2 + \frac{1}{2} 2m \left(\frac{V_0}{2} \right)^2$$

where v = velocity of beads w.r.t. ground

$$\text{So, } V = \frac{\sqrt{3}}{2} V_0$$

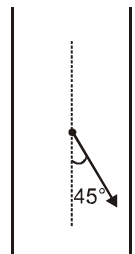
7. $qE = mg$

$$X = Ed$$

$$= \left(\frac{mg}{q} \right) d$$

$$X = \frac{1.6 \times 10^{-27} \times 10}{1.6 \times 10^{-19}} \times 1 \times 10^{-2}$$

$$X = 10^{-9} \text{ Volt}$$



8. In CM frame both the masses execute SHM with

$$\omega = \sqrt{\frac{k}{\mu}} = \sqrt{\frac{2k}{m}} \text{ SHM}$$

Initially particles are at extreme

$$\text{distance} = L_0 + (L - L_0) \cos \sqrt{\frac{2k}{m}} t$$

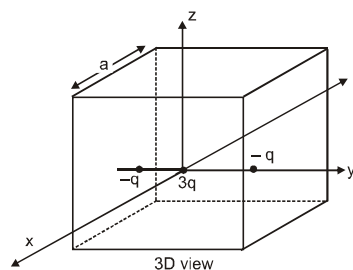
9. Work done on 2 C charge = $\int q \vec{E} \cdot d\vec{r} = q \int_1^3 E dr$

$$[\because r \text{ for } (1, 1, 0) = \sqrt{2} \text{ \& } r \text{ for } (3, 0, 0) = 3]$$

$$= 2 \times \text{area of E-r graph from } r = \sqrt{2} \text{ m to } r = 3$$

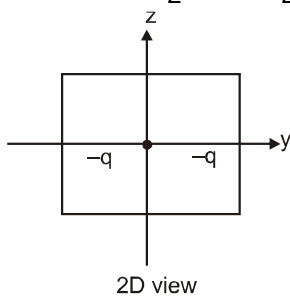
$$= 2 \times \left[\frac{1}{2} (3 - \sqrt{2}) 20 \right] = 20 (3 - \sqrt{2}) \text{ J.}$$

10.



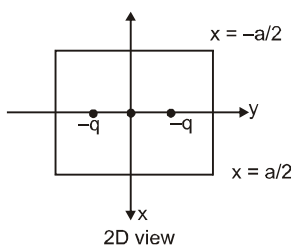
$$\text{Net flux through } x = \frac{a}{2} \text{ \& } x = -\frac{a}{2} \text{ will be equal}$$

$$\text{Net flux through } y = \frac{a}{2} \text{ \& } y = -\frac{a}{2} \text{ will be equal}$$



$$\phi_{\text{net}} = \frac{q_{\text{in}}}{\epsilon_0} = \left(\frac{-q + 3q - q}{\epsilon_0} \right) = \frac{q}{\epsilon_0}$$

$$\text{Net flux through } Z = +\frac{a}{2} \text{ \& } x = +\frac{a}{2} \text{ will be equal.}$$



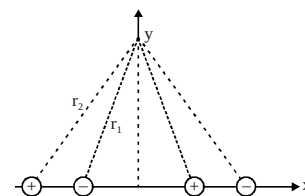
Ans. A, C & D

11. Potential foHko

$$r_1 < r_2$$

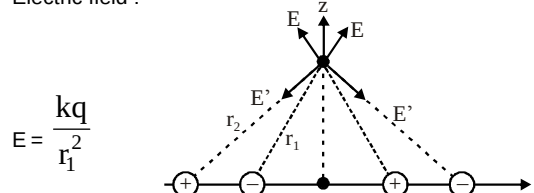
$$V_+ = \frac{2kq}{r_1}$$

$$V_- = \frac{-2kq}{r_2}$$



$$V_{\text{net}} = V_+ + V_- = 2kq \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \neq 0$$

Electric field :



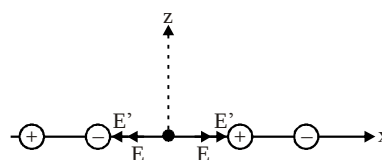
$$E = \frac{kq}{r_1^2}$$

$$E' = \frac{kq}{r_2^2}$$

$E' < E$, hence electric field for $z > 0$ is directed along +ve z-axis.

$\therefore$ Similarly for $z < 0$, electric field is directed along -ve z-axis.

Hence electric field at origin is zero.



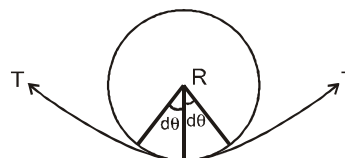
12. Rate of change of potential is increasing

$$\therefore v_1 - v_2/x_1 < v_2 - v_3/x_2$$

$$\therefore x_1 > x_2$$

$$13. \int_a^{3a} \frac{\mu_0 IL}{2\pi} \left[\frac{dx}{x} + \frac{dx}{4a-x} \right] = \frac{\mu_0 IL \ln 3}{\pi}$$

$$14. B \text{ at end} = \frac{1}{2} B \text{ at interior} = \frac{1}{2} B$$



$$IdL \left(\frac{B}{2} \right) = 2T \sin \theta$$

$$dL = R(2d\theta)$$

$$I R \cdot 2d\theta \frac{B}{2} = 2T d\theta$$

$$T = \frac{BIR}{2}$$

$$15. \frac{2mv_0}{qB_0} = \frac{1}{2} \left(\frac{qE_0}{m} \right) t^2$$

$$\frac{4m^2v_0}{q^2E_0B_0} = t^2$$

$$x = v_0 \times t \Rightarrow x = v_0 \frac{2m}{q} \sqrt{\frac{v_0}{E_0B_0}}$$

$$16. \text{Velocity of efflux at section (4) is } v = \sqrt{2gh}$$

Applying Bernoulli's equation between section (3) and (4)

$$P_3 + \frac{1}{2} \rho v_3^2 = P_4 + \frac{1}{2} \rho v_4^2$$

$$\Rightarrow P_3 + \frac{1}{2} \rho (2\sqrt{2gh})^2 = P_0 + \frac{1}{2} \rho (\sqrt{2gh})^2$$

$$\Rightarrow P_3 = P_0 - 3\rho gh.$$

17. Since toy is not accelerating so net external force on toy is zero. So (A)

$$18. a = \left[\frac{F - (m+M)g \sin \theta}{(m+M)} \right]$$

$$\text{So, } f = ma \cos \theta$$

$$= \left[\frac{F - (m+M)g \sin \theta}{(m+M)} \right] m \cos \theta$$

$$19. \delta_a = \left(\frac{3}{2} - 1 \right) \times A = \frac{A}{2}$$

$$\delta_w = \left(\frac{3/2}{4/3} - 1 \right) A = \frac{A}{8}$$

$$\frac{\delta_{\text{air}}}{\delta_{\text{water}}} = \frac{4}{1}$$

$$20. \frac{1}{40} - \frac{1}{-60} = \frac{1}{f} \quad \dots(1)$$

$$\frac{1}{v} - \frac{1}{-60} = \frac{1}{2f} \quad \dots(2)$$

$$\frac{1}{v} + \frac{1}{60} = \frac{1}{2} \cdot \left(\frac{1}{40} + \frac{1}{60} \right)$$

$$\Rightarrow \frac{1}{v} + \frac{1}{60} = \frac{1}{2} \cdot \frac{100}{40 \times 60}$$

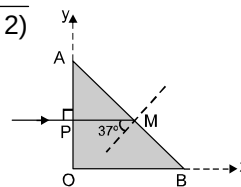
$$\Rightarrow -\frac{1}{v} = \frac{1}{48} - \frac{1}{60}$$

$$\Rightarrow v = 240 \text{ cm}$$

$$21. \text{Clearly, } PM = \frac{3}{2} \text{ cm}$$

$$37^\circ > \sin^{-1} \frac{1}{n_0 + a(3/2)}$$

$$\frac{3}{5} > \frac{1}{n_0 + \frac{3a}{2}}$$



$$3n_0 + \frac{9a}{2} > 5$$

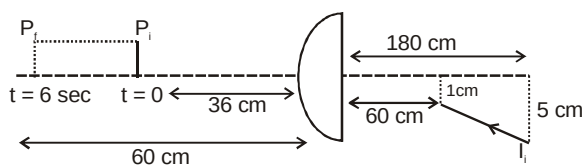
$$\frac{9a}{2} > 1 \quad ; \quad a > \frac{2}{9}$$

$$22. \frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{15} \right)$$

$$f = 30 \text{ cm.}$$

$$\frac{1}{v} + \frac{1}{36} = \frac{1}{30} \quad \frac{1}{v} = \frac{1}{30} - \frac{1}{36} = \frac{1}{180}$$

$$v = 180 \text{ cm}$$



Ans. (B)

23. Let the focal length of each piece be f

$$\text{Then } \frac{1}{f_1} = \frac{1}{f} + \frac{1}{f}$$

$$\frac{1}{f_2} = \frac{1}{f} + \frac{1}{f}$$

$$\Rightarrow f_1 = f_2$$

For the third arrangement the liquid forms a concave lens which has a diverging effect. So $f_3 > f_1 = f_2$

$$24. \frac{\Delta\omega}{\Delta\omega'} = \frac{\cos \theta_{\text{glass}}}{\cos \theta_{\text{air}}} \text{ putting the values we have } \Delta\omega' = \sqrt{\frac{3}{2}} \Delta\omega$$

25. Electric field on surface of a uniformly charged sphere is given

$$\text{by } \frac{Q}{4\pi\epsilon_0 R^3} = \frac{\rho R}{3\epsilon_0}$$

Electric field at outside point is given by

$$E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{\rho R^3}{3\epsilon_0 r^2}$$

$$|\bar{E}_r| = \frac{\rho r_0}{3\epsilon_0} - \frac{\rho \left(\frac{r_0}{2} \right)^3}{3\epsilon_0 \left(\frac{3r_0}{2} \right)^2} = \frac{17\rho r_0}{54\epsilon_0} \text{ left}$$

26. $\vec{E} = E_x \hat{i} + E_y \hat{j}$, $\Delta V = -E_x \Delta x - E_y \Delta y$

for A and B

$$16 - 4 = -E_x (-2 - 2) - E_y (2 - 2)$$

$$E_x = 3 \text{ V/m}$$

for B and C

$$12 - 16 = -E_x \{2 - (-2)\} - E_y (4 - 2)$$

$$E_y = -4 \text{ V/m}$$

$$\therefore \vec{E} = (3\hat{i} - 4\hat{j}) \text{ V/m}$$

27. $U = \frac{1}{2} \frac{(\text{stress})^2}{Y}$

$$\text{Stress} = \frac{\text{Tension}}{\text{area}}$$

$$\frac{U_A}{U_{\text{middle}}} = \frac{T_A^2}{(T_{\text{middle}})^2} = \frac{F^2}{(F/2)^2} = 4$$

28. $U_i = \left(\frac{1}{2} \frac{Q^2}{8\pi\epsilon_0 R} + \frac{1}{2} \frac{Q^2}{8\pi\epsilon_0 R} \right)$

$$U_i = \frac{1}{2} \frac{Q^2}{8\pi\epsilon_0 R}$$

$$\therefore \Delta \text{Heat} = \frac{1}{16} \frac{Q^2}{\pi\epsilon_0 R}$$

29. Both open.

$$R_A = 1\Omega, R_B = 3\Omega, R_C = 6\Omega, R_D = 1\Omega$$

Current in upper circuit when S_1 is open is $I_1 = 1\text{A}$

Null point is mid point of B or C when S_2 and S_1 both open

When both closed, current in upper circuit

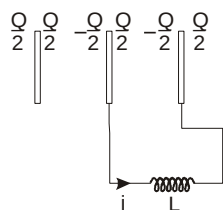
$$I_1 = 2\text{A}$$

Current in lower circuit $I_2 = 1\text{A}$

$$\Delta V = 2 - I_2 (1) = 1\text{V}$$

null point is mid point of A.

30. $\frac{\left(\frac{Q}{2}\right)^2}{2C} = \frac{1}{2} Li_0^2$



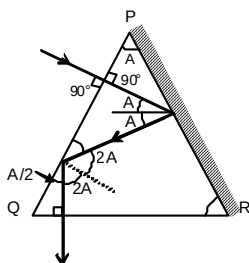
$$\Rightarrow i_0 = \frac{Q}{2\sqrt{LC}}$$

31. $\frac{A}{2} + 2A = 90^\circ$

$$\frac{5A}{2} = 90^\circ$$

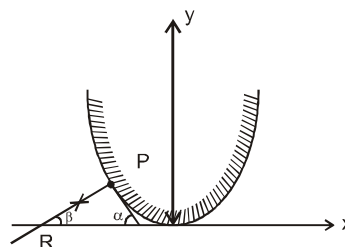
$$A = 36^\circ$$

$$\therefore \text{Angles } (36^\circ, 72^\circ, 72^\circ)$$



32. Point of intersection of two curves. $y = \frac{x^2}{4}$ and $y = x + 3$ is, $(-2, 1)$

$$\frac{dy}{dx} = \frac{x}{2} \text{ i.e. } \left(\frac{dy}{dx} \right)_{x=-2} = \frac{-2}{2} = -1$$



$$\tan \beta = \text{slope of incident ray RP} = 1 \text{ or } \beta = 45^\circ$$

$$\text{and } \tan \alpha = \left| \left(\frac{dy}{dx} \right)_{x=-2} \right| = 1 \Rightarrow \alpha = 45^\circ$$

From geometry we can show that $\angle i = \angle r = 0^\circ$ i.e., the ray is incident normally.

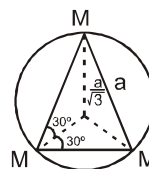
$$\text{Hence, the desired unit vector : } A = \frac{1}{\sqrt{2}} (-\hat{i} - \hat{j})$$

33. $\sin \alpha = \frac{\ell}{4y_0^2}$, $\alpha = \sin^{-1} \left(\frac{\ell}{4y_0^2} \right)$

34. For earth and sun system

$$T_0^2 = \frac{4\pi^2}{GM_s} R^3 \quad \dots\dots\dots(i)$$

For trinary star system



$$2 \frac{Gm^2}{a^2} \frac{\sqrt{3}}{2} = m\omega^2 \frac{a}{\sqrt{3}} \Rightarrow \omega^2 = \frac{3Gm}{a^3}$$

$$T^2 = \frac{4\pi^2}{\omega^2} = \frac{4\pi^2}{3Gm} a^3 \quad \dots\dots\dots(ii)$$

$$\text{Given, } T = 3T_0 \quad a = 2R$$

$$9T_0^2 = \frac{4\pi^2}{3Gm} 8R^3$$

$$\Rightarrow 9 \frac{4\pi^2}{GM_s} R^3 = \frac{8}{3} \frac{4\pi^2}{Gm} R^3$$

$$\Rightarrow m = \frac{8M_s}{27}$$

$$\text{Total mass} = 3m = \frac{8}{9} M_s$$

35. Let P = power radiated by the sun, R = radius of planet.

$$\text{Power received by planet} = \frac{P}{4\pi d^2} \times \pi R^2.$$

$$\text{Power radiated by planet} = (4\pi R^2) \sigma T^4.$$

$$\text{For thermal equilibrium, } \frac{P}{4\pi d^2} \times \pi R^2 = 4\pi R^2 \sigma T^4.$$

$$\text{or } T^4 \propto \frac{1}{d^2}$$

$$\text{or } T \propto \frac{1}{d^{1/2}} \quad \text{or } T \propto d^{-1/2}.$$

$\therefore$ Ans. (C)

36. Wein's displacement law is,

$$\lambda_m T = b$$

So, when λ_m becomes half

$\therefore$ T doubles

$$u_i = \sigma(T^4 - T_0^4)$$

$$u_i = \sigma \left(T^4 - \left(\frac{T}{2} \right)^4 \right)$$

$$u_f = \sigma \left((2T)^4 - \left(\frac{T}{2} \right)^4 \right)$$

$$\frac{u_i}{u_f} = \frac{1 - \frac{1}{16}}{16 - \frac{1}{16}} = \frac{15}{255}$$

$$u_f = \frac{255}{15} \times 30$$

$$u_f = 510 \text{ Jm}^{-2}\text{s}^{-1}.$$

$$37. \Rightarrow \Delta Q = nC_p \Delta T = \frac{2 \times 5 \times 8.31 \times 5}{2}$$

$$\Delta Q = 207.75 \text{ J}$$

$$38. T_A = T_C$$

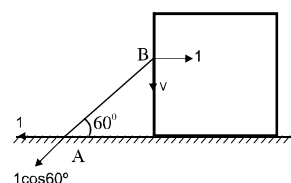
$$\Delta U_{BC} = -Q$$

$$\Delta U_{AB} = Q$$

$$\Rightarrow n \frac{3}{2} R \Delta T = Q$$

$$W_{AB} = nR \Delta T = 2 \frac{Q}{3}$$

- 39.

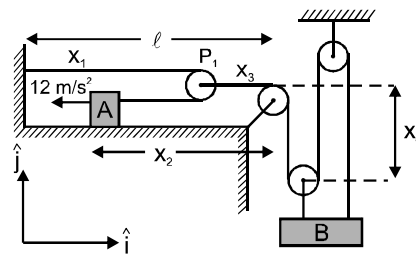


$$1 \cos 60^\circ = v \cos 30^\circ - 1 \cos 60^\circ$$

$$v = \frac{2}{\sqrt{3}}$$

$$v_{\text{net}} = \sqrt{\frac{7}{3}}.$$

$$40. a_B = 2 \text{ m/s}^2 \quad (\uparrow)$$



$$(\ell - x_3) + (x_2 - x_3) = k_1$$

$$\& \quad x_3 + 3x_4 = k_2$$

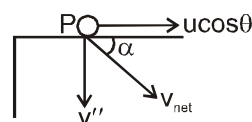
$$a_2 - 2a_3 = 0 \Rightarrow a_3 = \frac{a_2}{2} = \frac{12}{2} = 6 \text{ m/s}^2$$

[$a_3 \Rightarrow$ acceleration of P_1 pulley]

$$\& \quad -a_3 + 3a_4 = 0 \Rightarrow a_4 = \frac{a_3}{3} = \frac{6}{3} = 2 \text{ m/s}^2.$$

[$a_4 \Rightarrow$ acceleration of block B]

41. M is the highest point of ball's motion.
Horizontal component of velocity remains the same i.e. $u \cos \theta$.



At point P, let α is the angle formed by V_{net} (net velocity) with horizontal

$$\therefore \tan \alpha = \frac{v''}{u \cos \theta}$$

where v'' = vertical velocity component at point P.

Taking vertically downward as positive and motion from M to P
 $v' = 0 \Rightarrow$ vertically downward velocity at highest point.

$$\begin{array}{c} \uparrow \\ \text{M} \\ \downarrow \\ \text{P} \end{array} \quad v' \downarrow = 0$$

$$\therefore (v'')^2 = (v')^2 + 2g(H-h) \quad (3\text{rd equation of motion})$$

$$\therefore (v'')^2 = (0)^2 + 2g(H-h)$$

$$\therefore (v'') = \sqrt{2g(H-h)}.$$

$$\therefore \tan \alpha = \frac{v''}{u \cos \theta} = \frac{\sqrt{2g(H-h)}}{u \cos \theta}$$

$$\text{or } \alpha = \tan^{-1} \left[\frac{\sqrt{2g(H-h)}}{u \cos \theta} \right].$$

42. 1st sep. will decrease then sep will increase

For slope :- 1st there will be some V_{app} , then V_{app} will decrease slowly and become zero, after V_{sep} will increase gradually. Hence slope will decrease till it become zero afterward it will increase.

53. When the system is about to slide, friction will be equal to the limiting value. Since $a_{cm} = 0$

$$2t_0 = \mu(m + M)g$$

$$t_0 = \frac{\mu(m+M)g}{2}$$

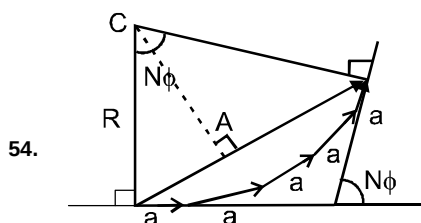
$$\frac{d\omega}{dt} = \frac{2tR}{mR^2} = \frac{4t}{mR}$$

$$\int d\omega = \int \frac{4tdt}{mR}$$

$$\omega = \frac{4}{mR} \cdot \frac{t_0^2}{2} = \frac{2t_0^2}{mR}$$

$$\therefore KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \frac{m R^2}{2} \frac{4 t_0^4}{m^2 R^2}$$

$$= \frac{1}{m} t_0^4 = \frac{1}{m} \left(\frac{\mu (m + M) g}{2} \right)^4$$



$$2R \sin \left(\frac{N\phi}{2} \right) = A$$

$$N\phi = \frac{Na}{R} \Rightarrow A = 2 \left(\frac{a}{\phi} \right) \sin \left(\frac{N\phi}{2} \right)$$

- 55.** Area of cross-section of the tube is increased

- 56.** Heat released by 4.5 kg of water when its temperature falls from 40°C to 0°C is.

$$Q_1 = ms\Delta\theta = (4.5) (1) (40 - 0) = 180 \text{ Kcal}$$

when 5 kg ice at -40°C comes to a temperature of 0°C , it takes an energy

$$Q_2 = ms\Delta\theta = (5) (0.5) (40) = 100 \text{ Kcal}$$

The remaining heat

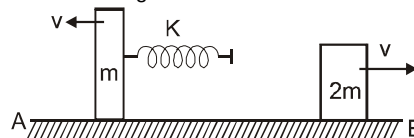
$Q = Q_1 - Q_2 = 80 \text{ Kcal}$ will melt mass m of the ice, where

$$m = \frac{80}{80} = 1 \text{ kg}$$

So, the temperature of the mixture will be 0°C .

mass of water in it is $4.5 + 1 = 5.5$ kg and mass of ice is $5 - 1 = 4$ kg.

- 57.** Situation after long time



$$\text{Work done} = \Delta K = \frac{1}{2} (2m)(v)^2 - \frac{1}{2} (2m) (3v)^2 = -8mv^2$$

- 58.** Extension is 5cm and equilibrium is at the extension of 2 cm so
Amplitude of SHM is 3cm

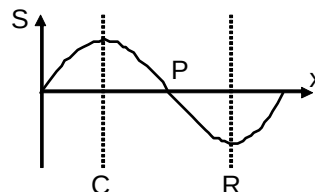
Also $\omega = \sqrt{\frac{k}{m}} = 10\sqrt{5} \text{ rad/s}$

$$v_{\max} = \omega A = 30\sqrt{5} \text{ ms}^{-1}$$

$$a_{\max} = \omega^2 A = 15 \text{ ms}^{-2}$$

59. $P_C - P_B = 2BAK$ $S = A \sin (kx - \omega t)$

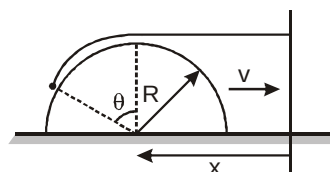
$$\left(\frac{\partial \mathbf{S}}{\partial \mathbf{x}}\right)_c = \mathbf{AK}$$



60. $x + R\theta = \text{constant}$

$$\frac{dx}{dt} + \frac{Rd\theta}{dt} = 0$$

$$\therefore \omega = \frac{d\theta}{dt} = \frac{v}{R}$$



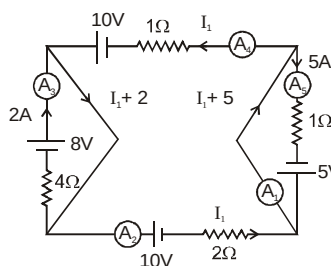
$$\alpha = d\omega/dt = a/R$$

At $\theta = 60^\circ$,

$$\therefore v_p = v \text{ and } a_p = \sqrt{\left(\frac{a\sqrt{3}}{2} + \frac{v^2}{R}\right)^2 + \left(\frac{a}{2}\right)^2}$$

- 61.** Using KVL

$$-I_1 + 10 - 10 - 2I_1 = 0$$



$$I_1 = 0$$

∴ **Ans.** (B), (C), (D).

62. If $x(1 - \ell) < R\ell$ then $v_A > v_B$ if x is increased then current may decrease, become zero or may increase.

63. (A) $-\frac{1}{2}k(x^2 - x_0^2) = \frac{1}{2}mv^2 - 0$

$$\therefore v = \sqrt{\frac{k}{m}(x_0^2 - x^2)}$$

$$\therefore P = F \cdot v = kx \sqrt{\frac{k}{m}(x_0^2 - x^2)}$$

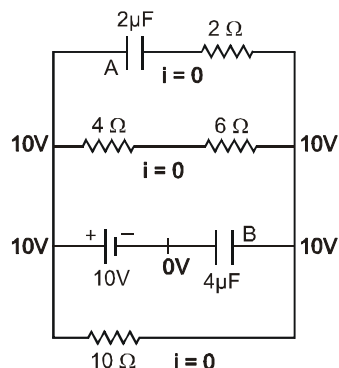
(B) $P = k \sqrt{\frac{k}{m}(x_0^2 x^2 - x^4)}$

(C) $y = x_0^2 x^2 - x^4$

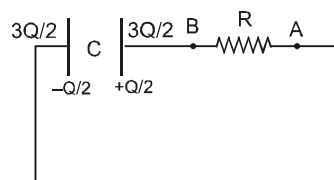
$$\frac{dy}{dx} = 0 \Rightarrow x = \frac{x_0}{\sqrt{2}}$$

$$\therefore P_{\max} \text{ is at } x = \frac{x_0}{\sqrt{2}}$$

64.



65.



$$Q(t) = \frac{Q}{2}(1 - e^{-t/RC})$$

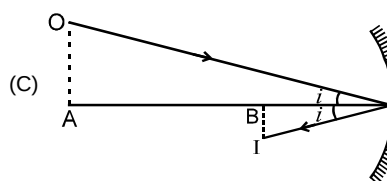
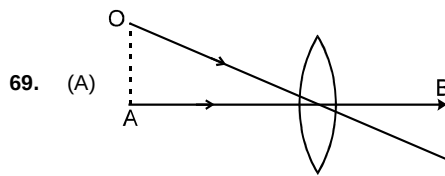
$$i(t) = \frac{Q}{2RC} e^{-t/RC} \quad (\text{from B to A})$$

$$\Delta \text{ heat} = \frac{(Q/2)^2}{2C}$$

66. $B = \frac{\mu_0 I}{2\pi a}$ by the direction of force particle is positively charge.
Now

$$mg + \frac{\mu_0 I}{2\pi a} \times q \times v = 2mg$$

$$\frac{\mu_0 I}{2\pi a} qv = mg \Rightarrow q = \frac{2\pi mga}{\mu_0 IV}$$



(D) Image is inverted $\Rightarrow$ It should be real

70. due to a convex mirror of focal length 2.5 cm
due to a concave mirror having its pole at (2, 0)
real virtual pair

71. The above lens mirror combination is an equivalent converging mirror.

72. $PV = nRT$ (1)
Mass of piston is 9kg & area of piston is 0.09 m^2

$$\text{So } P = P_0 + \frac{9g}{0.09}$$

$$P = 1.06 \times 10^5 \text{ N/m}^2$$

As mass of piston is not changed & Piston is frictionless So process is isobaric $P_f = P_i = 1.06 \times 10^5 \text{ N/m}^2$

$$P_f = P_i = 1.06 \times 10^5 \text{ N/m}^2$$

$$dQ = n C_p dT \quad \dots\dots\dots(2)$$

$$\text{or } P_2 V_2 - P_1 V_1 = n R dT \quad \dots\dots\dots(3)$$

$$(3) \div (2) \quad \frac{P(V_2 - V_1)}{dQ} = \frac{2}{5}$$

$$\Rightarrow P(V_2 - V_1) = \frac{2}{5} \times 2.5 \times 10^4$$

$$W = P\Delta V = 10^4 \text{ J} \quad \& \quad \text{by equation (1)}$$

$$n = \frac{1.06 \times 10^5 \times 0.0027}{R \times 300}$$

$$n = \frac{1.06 \times 270}{2500} = 0.11448$$

or by $dQ = n C_p dT$

$$\therefore T_2 - T_1 = \frac{2.5 \times 10^4}{0.11448 \times \frac{5R}{2}}$$

on solving $T_2 = 10782.18 \text{ K}$

73. S_1 : Isobaric process must pass through origin.
 S_2 : Isochoric process must pass through origin.
 S_3 : Straight line on PV does not represent isothermal process.

$$74. H = \frac{KAd\theta}{dr} = K(2\pi rL) \frac{d\theta}{dr}$$

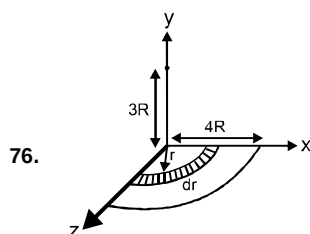
$$\therefore H \int_{r_1}^{r_2} \frac{dr}{r} = 2\pi KL(\theta_2 - \theta_1) = 80\pi$$

$$\Rightarrow \frac{dm}{dt} L = 80\pi$$

$$\Rightarrow \frac{dm}{dt} = \frac{8\pi}{L} = \frac{80\pi}{80 \times 4200} = \frac{\pi}{4200} \text{ Kg/second.}$$

75. $f = mg \sin\theta$
 and
 $PE \sin\theta = mg \sin\theta \cdot R$
 $q \cdot 2R \cdot E \sin\theta = mg \sin\theta \cdot R$

$$E = \frac{mg}{2q} \quad \therefore \text{Ans. (A) \& (C)}$$



76.

$$(A) v = \int_0^{4R} \frac{K \frac{\pi}{2} r dr \sigma}{\sqrt{r^2 + (3R)^2}} = \frac{\sigma R}{4 \epsilon_0}$$

$$(B) v = \int_0^{4R} \frac{K \frac{\pi}{2} r dr \sigma}{r} = \frac{\sigma R}{2 \epsilon_0}$$

77. In space (gravity not zero) net force on +q is vertically upward. So that equilibrium can be achieved in the space. If charge is -q then net force on it is vertically downward so equilibrium not achieved.
78. External electric field and induced charge on outer surface of the sphere will produce a net zero field inside it. Net field at P must be vertically upward.
79. At constant temperature, decrease in molecular mass causes flattening of the graph. For same molecular mass of gas, increase in temperature causes flattening of the graph.

$$(V_{mp})_{T_1} < (V_{mp})_{T_2}$$

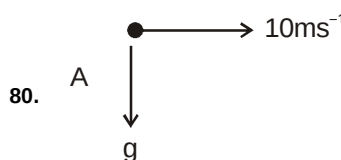
$$\therefore \frac{T_1}{M_A} < \frac{T_2}{M_B} \quad \therefore \frac{T_2}{T_1} > \frac{M_B}{M_A}$$

$$\text{But, } \frac{M_B}{M_A} \text{ can be less than or greater than 1}$$

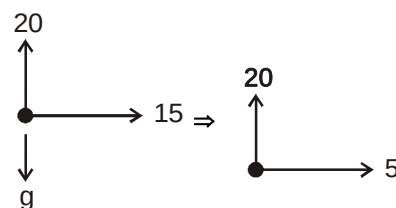
$$\text{Similarly, } \frac{M_A}{M_B} > \frac{T_1}{T_2}$$

If gas A is O_2 and gas B is N_2 , then $M_A > M_B$.
 $(V_{mp})_{T_1} < (V_{mp})_{T_2}$

$$\therefore \frac{T_1}{M_A} < \frac{T_2}{M_B} \quad \therefore \frac{T_2}{T_1} > \frac{M_B}{M_A}$$



80.

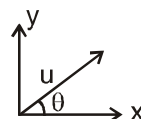


$$\text{Distance} = 5\sqrt{13} \times 2 = 10\sqrt{13}$$

$$81. \vec{v} = u \cos\theta \hat{i} + u \sin\theta \hat{j} - gt \hat{j}$$

$$\vec{a} = -g \hat{j}$$

$$\vec{a} \times \vec{v} = u \cos\theta g \hat{k} = \text{constant}$$



At highest point

$$\vec{a} = -g \hat{j}$$

$$\vec{v} = u \cos\theta \hat{i}$$

$$\Rightarrow \vec{a} \cdot \vec{v} = 0$$

$$82. x = 24 = u \cos\theta \cdot t$$

$$\Rightarrow t = \frac{24}{u \cos\theta} = \frac{1}{\cos\theta}$$

$$y = 14 = u \sin\theta t - \frac{1}{2} g t^2$$

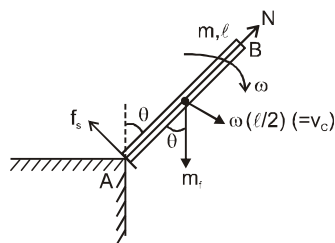
$$\Rightarrow 14 = \frac{u \sin\theta}{\cos\theta} - \frac{5}{\cos^2\theta}$$

$$\Rightarrow 14 = u \tan\theta - 5 \sec^2\theta$$

$$\Rightarrow 5 \tan^2\theta - 24 \tan\theta + 19 = 0$$

$$\Rightarrow \tan\theta = 1, 19/5. \text{ Ans}$$

83. Average velocity of particle from $t = 0$ to $t = 20$ sec. is zero.
Acceleration of particle from $t = 10$ sec. to $t = 15$ sec. is positive.
Average acceleration of particle from $t = 0$ to $t = 20$ is zero.
- 84.



$$mg \ell/2 = mg (\ell/2) \cos \theta + \frac{1}{2} \left(\frac{m \ell^2}{3} \right) \omega^2 \quad \dots(1)$$

$$mg \cos \theta - N = m \omega^2 (\ell/2) \quad \dots(2)$$

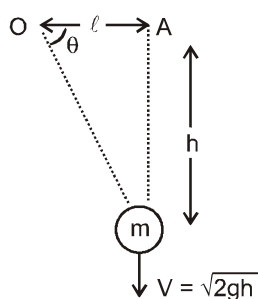
$$(mg \sin \theta) \ell/2 = \frac{m \ell^2}{3} \alpha \quad \dots(3)$$

$$mg \sin \theta - f_s = m (\ell/2) \alpha \quad \dots(4)$$

$$f_s \leq \mu N \quad \dots(5)$$

85. (A) $L = mvr_{\perp} = m(\sqrt{2gh})(\ell)$

(B) & (C) $\tan \theta = \frac{y}{\ell}$



$$\sec^2 \theta \frac{d\theta}{dt} = \frac{1}{\ell} \left(\frac{dy}{dt} \right)$$

$$\omega = \frac{d\theta}{dt} = \frac{1}{\ell \sec^2 \theta} (\sqrt{2gh})$$

$$\omega = \frac{\sqrt{2gh}}{\ell (1 + \tan^2 \theta)} = \frac{\sqrt{2gh}}{\ell \left(1 + \left(\frac{h}{\ell} \right)^2 \right)} = \frac{\ell}{\ell^2 + h^2} \sqrt{2gh}$$

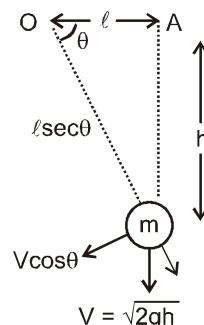
Alternate Solution :

Wrong solution :

$$\omega = \frac{V}{r_{\perp}} = \frac{\sqrt{2gh}}{\ell}$$

Correct solution :

$$\omega = \frac{V_{\perp}}{r} = \frac{\sqrt{2gh} \cos \theta}{\ell \sec \theta}$$



$$\omega = \frac{\sqrt{2gh}}{\ell} \cos^2 \theta = \frac{\sqrt{2gh}}{\ell} \left(\frac{\ell}{\sqrt{\ell^2 + h^2}} \right)^2$$

$$\Rightarrow \omega = \sqrt{2gh} \left(\frac{\ell}{\ell^2 + h^2} \right)$$

(D) $\tau = Fr_{\perp}$
 $\tau = (mg) (\ell)$

86. $T_c = I_c \alpha$

$$\left(\frac{1}{2} \cos \theta \right) 1 = \frac{1(1)^2}{12} \frac{d^2 \theta}{dt^2}$$

$$\int_0^{\omega} \omega d\omega = 6 \int_0^{\theta} \cos \theta d\theta$$

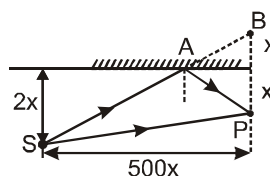
$$\frac{\omega^2}{2} = 6 \sin \theta$$

$$\omega \propto \sqrt{\sin \theta}$$

$T_c = \frac{1}{2} \cos \theta$ positive so θ increases as t increases. So rod will be parallel to y -axis in finite time. C.O.M. moves on y -axis

[if necessary use $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = 2.6$]

87. $\Delta x = (SA + AP) - SP$
 $= (SA + AB) - SP$
 $= SB - SP$
 $= \sqrt{(500x)^2 + (3x)^2} - \sqrt{(500x)^2 + x^2}$



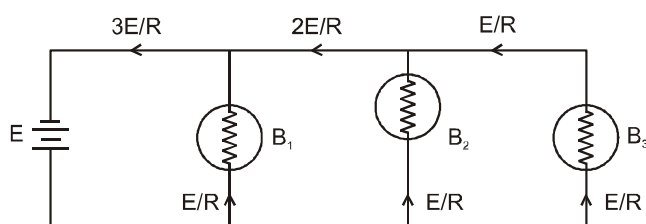
$$= 500x \left[\sqrt{1 + \frac{9}{500^2}} - \sqrt{1 + \frac{1}{500^2}} \right]$$

$$= 500x \left[\left(1 + \frac{9}{2 \times 500^2} \right) - \left(1 + \frac{1}{2 \times 500^2} \right) \right]$$

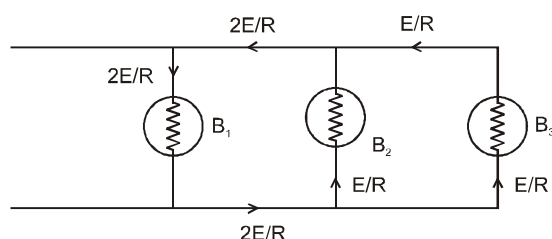
$$= \frac{500x}{2 \times 500^2} [8] = \frac{\lambda}{2}, \frac{3\lambda}{2} \dots$$

$$x = 62.5\lambda, 187.5\lambda \dots$$

88. Due to self induced emf current through B_2 and B_3 remains unchanged but current through B_1 increased.



Just Before

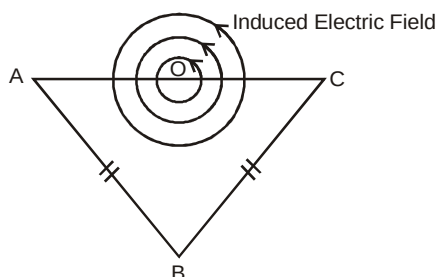


89. Faraday's law induced e.m.f. = - rate of change of magnetic flux
Inside the solenoid, flux = $B \times \text{Area}$
where B is the magnetic flux density

$$\text{Induced e.m.f.} = - \frac{d}{dt} (B \times \text{Area}) = - \text{Area} \times \frac{dB}{dt} = -R^2 \frac{dB}{dt}$$

since area of triangle ABC is R^2 .

(b) the important point is that a changing magnetic field induces a circulating electric field (from which we derive the induced e.m.f.). in other words, there are no radial components of electric field (radiating outwards or inwards) so that no e.m.f. is actually induced in the length AC of the circuit.



Hence the induced e.m.f. is shared out between AB and BC.

$$\text{So for AB, induced e.m.f.} = - \frac{1}{2} R^2 \frac{dB}{dt}$$

90. S_1 : The focal length of a concave mirror depends only on its radius of curvature.

$$S_2: \frac{1}{f} = (n_{\text{rel}} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$n_{\text{rel}} = \frac{n_{\text{rel}}}{n_{\text{surrounding}}}$$

$$n_{\text{surrounding}} \uparrow \quad n_{\text{rel}} \downarrow \quad f \uparrow$$

$$S_3: \text{Since } E = \frac{-dv}{dr} : \text{if } E = 0$$

$V = \text{constant}$ not necessarily equal to zero

91. $S_1 \rightarrow \text{True}$ Both are falling under gravity.
 $S_2 \rightarrow \text{False}$ Function has maxima at $x = x_1$, not necessary maximum value also.

$$S_3 \rightarrow \text{True} \quad T = \frac{W}{V} \text{ to minimize } T, V \text{ will be maximum.}$$

i.e. whole effort of swimmer must towards opposite bank.

92. S_1 : When pressure is increase density also increases in same proportion
 S_2 : True

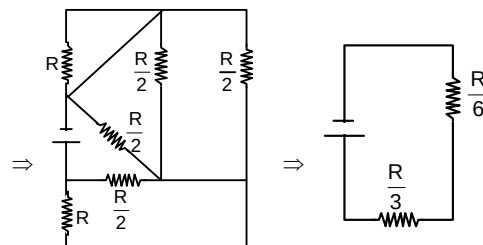
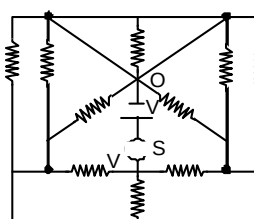
$$S_3: v_1 = \frac{v}{v-u} v$$

$$v_2 = \left[\frac{u+v}{v} \right] v$$

$$v_1 - v_2 = v \left[\frac{v}{v-u} - \frac{u+v}{v} \right]$$

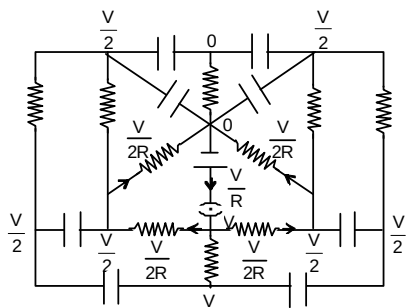
$$= v \left[\frac{v^2 - v^2 + u^2}{(v-u)v} \right] = \frac{u^2 v}{(v-u)v} > 0$$

93. At $t = 0$ All the capacitors are short circuited. So current at $t = 0$ can be calculated as below.



$$i = \frac{2V}{R}$$

Now at $t = \infty$. Current through each capacitor reduce to zero.
So at $t = \infty$ circuit is



94. & 95.

Let the initial temperature, pressure and volume of gas in 'A' be T_0, P_0, V_0 and the area of the position A and B be $a, 2a$. Now gas in chamber 'A' undergoes adiabatic compression whereas gas in chamber 'B' undergoes isothermal expansion.

Now solving for gas in 'A'

$$P_{1A} V_{1A}^\gamma = P_{2A} V_{2A}^\gamma \Rightarrow P_0 V_0^\gamma = P_{2A} (0.25V_0)^\gamma \Rightarrow P_{2A} = 8P_0$$

Solving for gas in chambers 'B'

$$P_{1B} V_{1B} = P_{2B} V_{2B} \Rightarrow P_{1B} V_0 = P_{2B} \times 2.5V_0 \Rightarrow P_{2B} = 0.4 P_{1B}$$

Also

$$(P_{2A} \times a) = (P_{2B} \times 2a) \text{ for gas in 'B'}$$

$$\Rightarrow 8P_0 a = P_{2B} \times 2a \Rightarrow P_{2B} = 4P_0$$

$$\Rightarrow 0.4 P_{1B} = 4P_0 \Rightarrow P_{1B} = 10P_0$$

Now comparing the moles of gas in A and B

$$n_A = \frac{P_0 V_0}{RT_0}$$

$$n_B = \frac{10P_0 V_0}{RT_0} = 10n_A$$

$$m_B = 10 m_A = 10 \times 100\text{gm} = 1\text{kg}.$$

Again for gas in A, $\Delta Q = \Delta U + \Delta W$

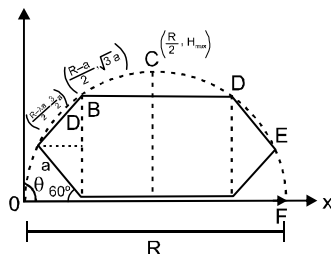
$$\Rightarrow 0 = \Delta U + \Delta W$$

$$\Rightarrow 0 = \Delta U + \left\{ \frac{P_0 V_0 - 8P_0 \times \frac{V_0}{4}}{(\gamma - 1)} \right\}$$

$$\Rightarrow \Delta U = 2P_0 V_0 = 2 \times 10^5 \times 10^{-2} = 2000\text{J}$$

$$F = 8P_0 a = 8 \times 10^5 \times 10^{-2} = 8000\text{N}$$

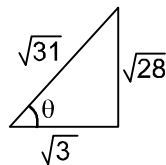
96 & 97.



$$y = x \tan \theta \left(1 - \frac{x}{R} \right)$$

Put co-ordinate of A & B

$$\text{to get } \cos \theta = \sqrt{\frac{3}{31}}$$

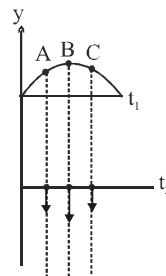


$$\text{and } R = \sqrt{7} a$$

$$\text{Put } x = \frac{R}{2} a, y = H_{\max}$$

$$H_{\max} = \frac{\sqrt{7}}{2} a \times \frac{\sqrt{28}}{\sqrt{3}} \left[\frac{1}{2} \right] = \frac{7}{2\sqrt{3}} a.$$

98.



At t_1 energy A, B, C is zero but at t_2 it is more for B so energy flow from A to B & C to B during t_1 & t_2 , so towards right of A.

99. At B amplitude is maximum so energy is maximum.

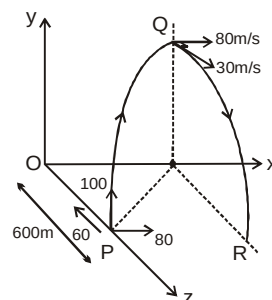
100 & 101.

(a) Since wall is smooth time of flight remains unchanged

$$T = \frac{2u_y}{g} = \frac{2 \times 100}{10} = 20\text{s}$$

The particle reaches the wall

$$\text{at } t = \frac{600}{60} \text{ s} = 10\text{s}$$



At $t = 10\text{s}$, the particle is at its maximum height. Hence, $t = 10\text{s}$ and $\theta = 0^\circ$

(b) Let Q be the point of impact on the wall.

So its X - co-ordinate = $80 \times 10 = 800\text{m}$

$$Y - \text{co-ordinate} = \frac{u_y^2}{2g} = \frac{(100)^2}{2 \times 10} = 500\text{m}$$

$$Z - \text{co-ordinate} = 0$$

(c) Since only velocity component along line of impact changes so, just after collision with wall.

$$\vec{V}_Q = 80\hat{i} + \hat{e} \times 60\hat{k} = (80\hat{i} + 30\hat{k})\text{m/s}$$

And, time to reach the XZ - plane is 10 s from Q, so, for R,

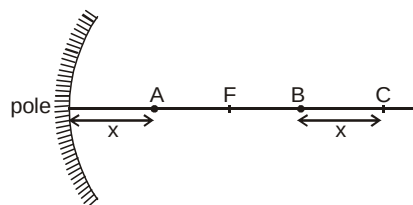
$$X - \text{co-ordinate} = 80 \times 20 = 1600\text{m}$$

$$Y - \text{co-ordinate} = 0$$

$$Z - \text{co-ordinate} = 30 \times 10 = 300\text{m}$$

102. The image distance of A is $v_A = -\frac{(f-x)}{xf}$ ($\because x < f$)

The image distance of B is $v_B = \frac{[(2f-x)-f]}{(2f-x)-x}$



Solving $|v_A| + |v_B| = 4f$ we get $x = \frac{f}{2}$

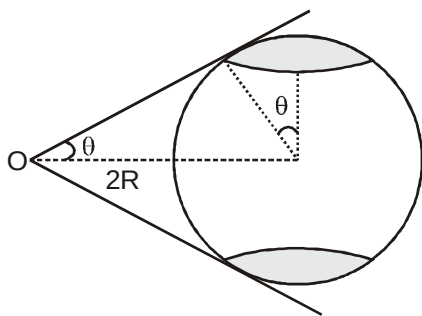
$\therefore$ required time $t = \frac{x}{u} = \frac{f}{2u}$

103. At time $t = 0$, velocity of image of A and B are $-u\hat{i}$ and $+u\hat{i}$ respectively. Therefore magnitude of their relative velocity is $2u$.

104. All area of earth will be covered other than shaded area.

shaded area $= 2[2\pi R^2(1 - \cos\theta)]$

$= 4\pi R^2 \left[1 - \frac{\sqrt{3}}{2} \right] = 2\pi R^2 [2 - \sqrt{3}]$



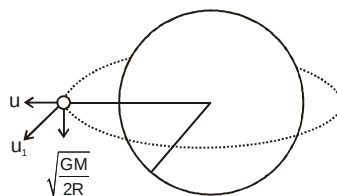
area covered $4\pi R^2 - 2\pi R^2 [2 - \sqrt{3}]$

$= 2\pi R^2 \sqrt{3}$

105. $\frac{1}{2}mu_1^2 - \frac{GMm}{2R} = 0$

$v_1 = \sqrt{\frac{GM}{R}}$

$\frac{GM}{2R} + u^2 = \frac{GM}{R}$



$v = \sqrt{\frac{GM}{2R}}$

$\therefore$ Ans. (B)

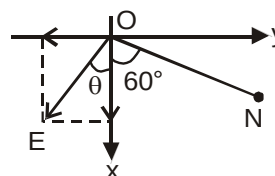
106. $\frac{1}{2}mu^2 - 0 = Q \left[0 - \frac{1}{4\pi\epsilon_0} \frac{P\cos\theta}{r^2} \right]$

$v = \sqrt{\frac{-QP\cos\theta}{2\pi\epsilon_0 mr^2}}$

107. Tangential acceleration of bead at point 'A' is zero

- 108 & 109.

The direction of electric field is in x-y plane as shown in figure



The magnitude of electric field is

$E = \sqrt{E_x^2 + E_y^2} = \sqrt{3+1} = 2V/m$

The direction of electric field is given by

$\theta = \tan^{-1} \frac{E_y}{E_x} = \tan^{-1} \frac{1}{\sqrt{3}} = 30^\circ$

Hence electric field is normal to square frame LMNO as shown in figure.

$\therefore$ electric flux $= \vec{E} \cdot \vec{A} = EA \cos 0 = 2 \times 1 = 2 V/m$

$\therefore$ C is correct option of Q. 15.

Since flux is maximum at $\theta = 60^\circ$, rotation by 30° either way would lead to decrease in flux.

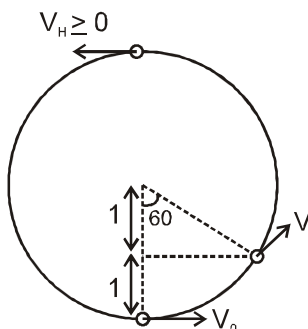
$\therefore$ D is correct option of Q.16.

Lines ON and NM are both normal to uniform electric field $\vec{E}$.

Hence work done by electric field as a point charge $1 \mu C$ is taken from O to M is zero.

110 to 112.

Let mass of bead = m
to complete vertical circle
 $V_H \geq 0$
Apply WE theorem



$$-mg(3) = \frac{1}{2} m (V^2 - V_0^2)$$

$$V^2 = V_H^2 + 6g$$

$$V^2 \geq 6g$$

$$V_{\min} = \sqrt{60}$$

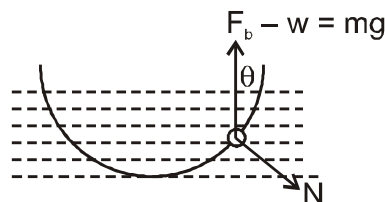
Inside water forces acting are gravity, Buoyant force & Normal reaction.

$$W_a + W_b + W_N = \Delta KE$$

$$-mg(1) + 2mg(1) + 0 = \frac{1}{2} m (V^2 - V_0^2)$$

$$V_0^2 = V^2 - 2g \geq 40$$

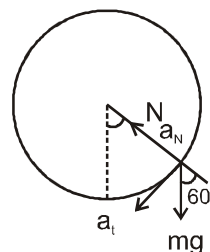
$$V_{0\min} = \sqrt{40} = 2\sqrt{10} \text{ m/s}$$



Inside water $a_t = g \cos \theta$ as it moves up, with time t , θ decreases
 $\therefore a_t$ increases i.e. speed increases and rate of increase of speed (a_t) also increases

$$\text{Just outside water } V = \sqrt{60}$$

$$a_t = g \sin 60 = 5\sqrt{3} \text{ m/s}^2$$



$$a = \sqrt{900 + 75} > 3g$$

113 to 115.

$$\text{Total magnetic flux at any position } \phi = B_z \pi r_0^2 - LI$$

$$\text{Since, } R = 0, \text{ so } \phi = B_0(1 - \alpha z) \pi r_0^2 - LI = \text{constant}$$

From initial condition ($z = 0, I = 0$), the value of constant is

$$\phi = B_0 \pi r_0^2$$

Using the above equation the current in the ring

$$I = \frac{1}{L} B_0 \alpha \pi r_0^2 z$$

The Lorentz force acting on the ring (which can only be vertical, because of the symmetry of the assembly) can be expressed as

$$F_z = -B_0 I(z) 2\pi r_0 = -\frac{2B_0^2 \alpha \beta \pi^2 r_0^4 z}{L} = -kz$$

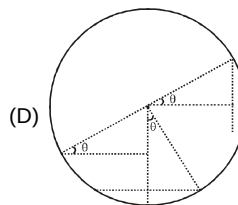
Equation of motion of the ring is $ma_z = F_z - mg = -kz - mg$
Equilibrium position $z_0 = -mg/k$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$117. (A) y = \frac{\omega^2 x^2}{2g} = 1.25, \tan \theta = \frac{1.25}{1} = 1.25$$

$$(B) y = \frac{\omega^2 x^2}{2g} = 1.25, \tan \theta = \frac{1.25}{5} = \frac{1}{4}$$

$$(C) y = 5 \tan \theta = 5 \frac{a}{g} = 2.5 \text{ m}, \quad \tan \theta = \frac{\omega}{g} = \frac{1}{2}$$



Equating the pressure at the junction

$$\Rightarrow \tan \theta = \left(\frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} \right) = \frac{1}{2}$$

118. (Moderate) The electric field due to one dipole at centre of other dipole is parallel to that dipole. Hence torque on all given dipoles is zero.

In case B and C the electric field at second dipole due to first is along the second dipole. hence electrostatic potential energy of second dipole is negative.

In case A and B x-axis is line of zero potential. In case B and C electric field at origin is zero.

119. (B) Phase difference $\phi = \tan^{-1} \left(\frac{X}{R} \right) = \pi/4$ for 1

$$= -\frac{\pi}{4} \text{ for 2}$$

(C) Phase difference $\phi = \tan^{-1} \left(\frac{X}{R} \right) = \pi/4$, for 1

$$= -\pi/4, \text{ for 2}$$

(D) Phase difference $\phi = \tan^{-1} \left(\frac{X}{R} \right) = \pi/4$ for both

120. $\vec{V}_{P,P} = V_2 \hat{i} + 25 \hat{j} + V_1 \hat{k}$

$$\vec{a}_{P,P} = -2 \hat{i} - 12.5 \hat{j}$$

$\vec{V}_{P,P}$ = Velocity of particle relative to platform

$$\text{Time} = \frac{2 \times 25}{12.5} = 4 \text{ sec.}$$

$$8 \leq V_2 \times 4 - \frac{1}{2} \times 2 \times 4^2 \leq 16$$

$$6 \leq V_2 \leq 8$$

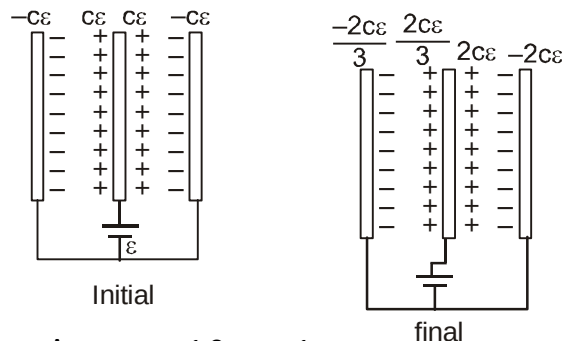
$$16 \leq V_1 \times 4 \leq 24$$

$$4 \leq V_1 \leq 6$$

$$Y = 25 \times 4 - \frac{1}{2} \times 10 \times 4^2$$

$$= 100 - 80 = 20 \text{ m}$$

121. (A)



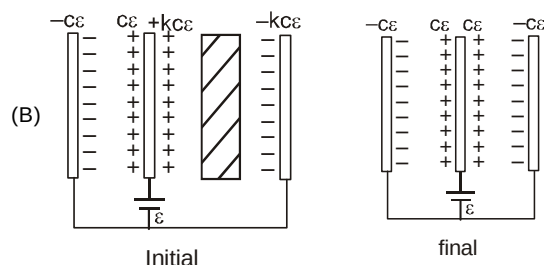
$$C = \frac{A\epsilon_0}{d} \quad U_i = \frac{1}{2} \frac{2C}{3} \epsilon^2 + \frac{1}{2} 2C \epsilon^2$$

$$U_i = \frac{1}{2} C \epsilon^2 + \frac{1}{2} C \epsilon^2 = \frac{4}{3} C \epsilon^2 = C \epsilon^2$$

$$F_{13} = \frac{\frac{2C\epsilon}{3} \cdot 2C\epsilon}{2A\epsilon_0}$$

$$F_{13} = \frac{C^2 \epsilon^2}{2A\epsilon_0} = \frac{2C^2 \epsilon^2}{3A\epsilon_0}$$

$$\text{Work by battery} = \frac{2C\epsilon}{3} \cdot \epsilon = \frac{2}{3} C \epsilon^2$$



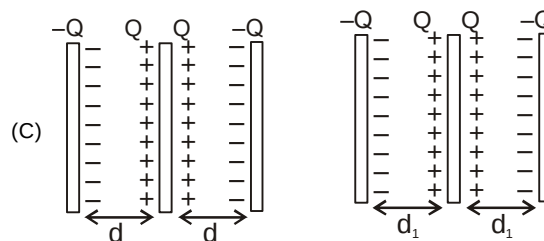
$$C = \frac{A\epsilon_0}{d}$$

$$U_i = \frac{1}{2} C \epsilon^2 + \frac{1}{2} C \epsilon^2$$

$$F_{13} = \frac{kC^2 \epsilon^2}{2A\epsilon_0}$$

$$F_{13} = \frac{C^2 \epsilon^2}{2A\epsilon_0}$$

W by battery is -ve



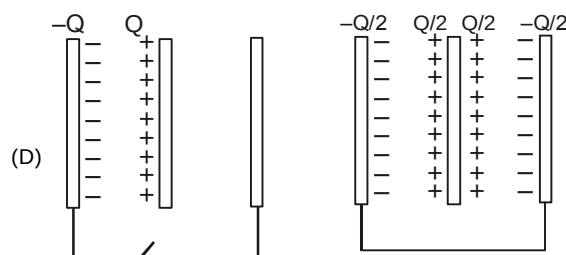
$$U_i = \frac{Q^2}{2C} + \frac{Q^2}{2C}$$

$$U_i = \frac{Q^2}{2C_1} + \frac{Q^2}{2C_1}$$

$$C_1 < C$$

$$F_{13} = \frac{Q^2}{2A\epsilon_0}$$

$$F_{13} = \frac{Q^2}{2A\epsilon_0}$$



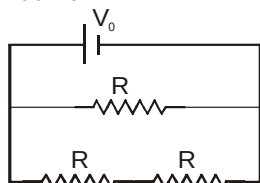
$$U_i = \frac{Q^2}{2C}$$

$$U_i = \frac{Q^2}{8C} + \frac{Q^2}{8C} = \frac{Q^2}{4C}$$

$$F_{13} = 0$$

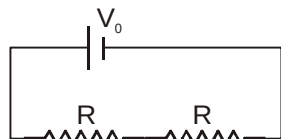
$$F_{13} = \frac{Q^2 / 4}{2A\epsilon_0}$$

122. At $t = 0$



$$P_{\text{consume}} = \frac{V_0^2}{2R/3} = \frac{3}{2} P_0$$

After a long time :



$$P_{\text{consume}} = \frac{V_0^2}{2R} = \frac{P_0}{2}$$

Since current in B_1 decreases with time so its brightness decreases.

Initially brightness of B_2 is less than B_1 but later on B_2 will be brighter.

123. $W_{NA} + W_{NG} + W_G = \frac{1}{2} m_B V^2 - 0$

$$W_{NA} + 0 + 0 = \frac{1}{2} (2) (1)^2 - 0 = 1$$

124. $a_t = \frac{F}{m} \sin \theta$

$$\frac{Rd^2(2\theta)}{dt^2} = \frac{F}{m} \sin \theta$$

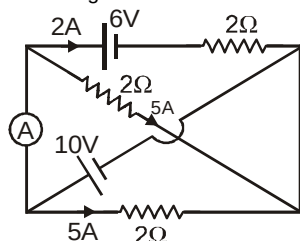
$$\frac{d^2\theta}{dt^2} = \frac{F \sin \theta}{2mR} \quad \dots(1)$$

$$a_c = \frac{F}{m} \cos \theta$$

$$R \left[\frac{d}{dt} (2\theta) \right]^2 = \frac{F}{m} \cos \theta \quad \dots(ii)$$

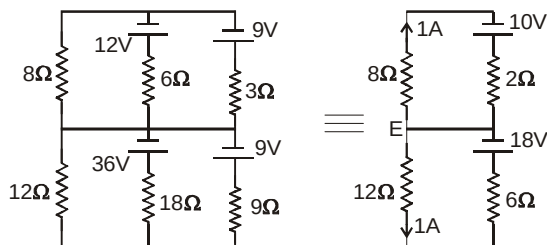
$$\left(\frac{d^2\theta}{dt^2} \right) \left(\frac{d\theta}{dt} \right)^2 = 2 \tan \theta = 2$$

125. From diagram

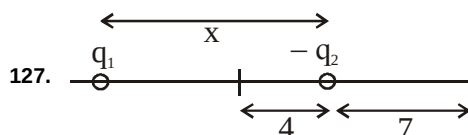


Current through ammeter
= (2 + 5) = 7A

126. Since A and B are at same potential we can redraw the circuit as



Hence current through 8Ω is 1A and
 $V_B - V_D = (V_B - V_E) + (V_E - V_D) = -8 + 12 = 4V$.
 $x + y = 1 + 0 = 1$.



$$\frac{q_1}{x-4} - \frac{q_2}{4} = 0 \quad \frac{q_1}{x+7} - \frac{q_2}{7} = 0$$

$$\frac{q_1}{q_2} = \frac{x-4}{4} \quad \frac{q_1}{q_2} = \frac{x+7}{7}$$

$$\frac{x-4}{4} = \frac{x+7}{7}$$

$$7x - 28 = 4x + 28$$

$$3x = 56$$

$$x = \frac{56}{3} \Rightarrow \frac{q_1}{q_2} = \frac{\frac{56}{3} + 7}{7} = \frac{11}{3}$$

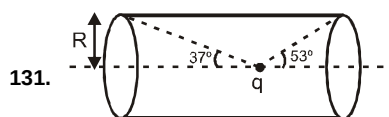
$$|q_2| = + \frac{12}{11} \mu C \Rightarrow q_1 = \frac{12}{11} \times \frac{11}{3} = 4 \mu C$$

129. Only effective pairs will be body diagonals for this interaction

$$\text{electrostatic potential energy is } = \frac{4kq^2}{\sqrt{3} \ell}$$

130. $\frac{nQ}{5\epsilon_0} = \frac{3Q}{5\epsilon_0} - \left[\frac{Q}{\epsilon_0} - \frac{3Q}{5\epsilon_0} \right] = \frac{Q}{5\epsilon_0}$

Ans. $n = 1$

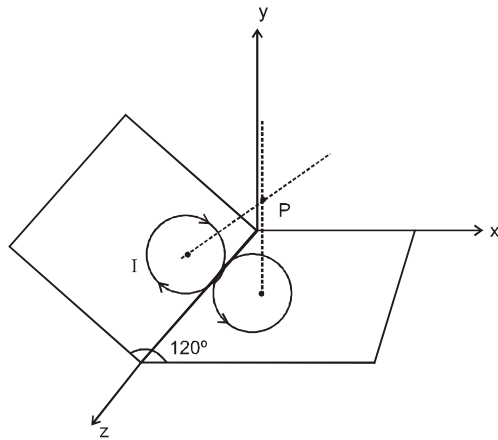


$$\phi = \frac{q}{\epsilon_0} \left(\frac{2\pi(1 - \cos 37^\circ)}{4\pi} \right) - \frac{q}{\epsilon_0} \left(\frac{2\pi(1 - \cos 53^\circ)}{4\pi} \right)$$

$$\phi = \frac{q}{\epsilon_0} \left[1 - \frac{1}{2} \left(1 - \frac{4}{5} \right) - \frac{1}{2} \left(1 - \frac{3}{5} \right) \right]$$

$$\frac{q}{\epsilon_0} \left[1 - \frac{1}{10} - \frac{2}{10} \right] = \frac{7q}{10\epsilon_0}$$

132.

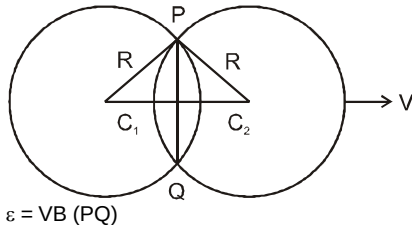


$$B_1 = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + 3R^2)^{3/2}} = \frac{\mu_0 I}{16R}$$

$$B^2 = B_1^2 + B_1^2 + 2B_1 B_1 \cos 120^\circ$$

$$\Rightarrow B = B_1 = \frac{\mu_0 I}{16R} = \frac{3\mu_0 I}{48R}$$

133. $\varepsilon = |(\vec{v} \times \vec{B}) \cdot \vec{\ell}|$



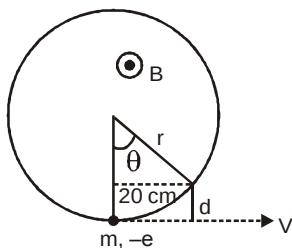
$$\varepsilon = VB(PQ)$$

$$= VB \cdot 2\sqrt{R^2 - \left(\frac{vt}{2}\right)^2} = VB \sqrt{4R^2 - v^2 t^2}$$

$$= 4 \times 0.25 \sqrt{4 \times 25 - 16 \times 4}$$

$$= 6 \text{ volt} \quad (v_Q > v_P)$$

134. Method-I



$$r = \frac{mv}{eB} \cong 3.4 \text{ m}$$

$$r \sin \theta = 20 \text{ cm}$$

$$\theta \cong \frac{1}{17} \text{ radian}$$

$$r - r \cos \theta = d$$

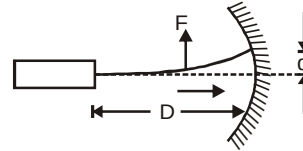
$$d \cong \frac{r\theta^2}{2} \cong 6 \text{ mm}$$

Method-II

The maximum magnetic force is perpendicular to the velocity and has a magnitude

$$F = evB$$

This force produces an acceleration, a , perpendicular to the line from the electron gun to the screen. The force remains in this direction since d is assumed to be small.



$$\text{Thus, } a = \frac{F}{m} = \frac{evB}{m} \text{ and } d = \frac{1}{2}at^2$$

Where $t = \frac{D}{v}$ is the time it takes for the electrons to reach the screen.

$$\text{Hence, } d = \frac{1}{2} \frac{evB}{m} \left(\frac{D}{v}\right)^2 \text{ or } d = \frac{eBD^2}{2mv}$$

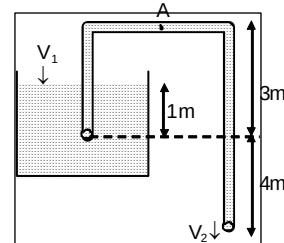
Substituting the numerical values :

$$d = \frac{1.6 \times 10^{-19} \times 5 \times 10^{-5} \times (0.2)^2}{2 \times 9.1 \times 10^{-31} \times 3 \times 10^7}$$

$$\therefore d = 5.9 \times 10^{-3} \text{ m i.e., } d = 6 \text{ mm.}$$

135. $P_0 + \frac{\rho v_1^2}{2} + 0 = P_A + \rho gh + \frac{\rho v_2^2}{2}$

$$(\because v_2 = \sqrt{2g \times 5} = 10 \gg v_1)$$



$$\left(\frac{\rho v_1^2}{2} - \frac{\rho v_2^2}{2}\right) - \rho gh = (P_A - P_0)$$

$$(P_A - P_0) = -\rho gh - \frac{\rho v_2^2}{2}$$

$$= -\frac{10^3}{2} \times 10 \times 2 - \frac{10^3}{2} \times \frac{10 \times 10}{2} = -\frac{7}{2} \times 10^4$$

$$= -\frac{7}{20} \times 10^5 \text{ N/m}^2 = -\frac{7}{20} \text{ atm}$$

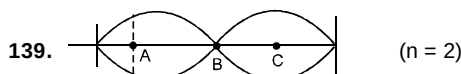
136. The minima will be heard at P when a crest from S_1 and a trough from S_2 reach there at the same time. This will happen if $L_1 - L_2$ is $\lambda/2$ or $\lambda + (\lambda/2)$ or $2\lambda + (\lambda/2)$ and so on. Hence, the increase in L_1 between consecutive minima is λ_1 and from the data we see that $\lambda = 0.40 \text{ m}$. Then from $\lambda = v/f \Rightarrow f = 340/0.40 = 850 \text{ Hz}$.

137. $\frac{P_0^2}{2\rho V} = I = \frac{\text{Power}}{4\pi r^2}$

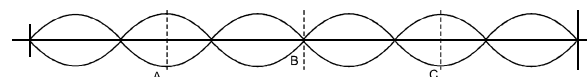
Put the values to get the answer.

138. $(10 \times 11 - 10 \times 6) \times 10^{-4} \times 2T = \Delta w$
 $50 \times 10^{-4} \times 2T = 3 \times 10^{-4}$

$$\Rightarrow T = \frac{3}{100} = 3 \times 10^{-2} \text{ N/m.}$$



$$f_1 = \frac{2v}{2L} \quad \text{--- (i)}$$



$$n = 6$$

$$f_2 = \frac{6v}{2L} \quad \text{--- (i)}$$

$$\frac{f_2}{f_1} = \frac{6v/2L}{2v/2L} = 3$$

$$f_2 = 3f_1 = 3 \times 100 = (300 \text{ Hz})$$

140. $X_{CM} = \frac{\int dm \cdot x}{\int dm} = \frac{2L}{3}$

Mass of rod ;

$$M = \int_0^L \lambda_0 x dx = \frac{10^3}{2}$$

Torque about 'O' ;

$$F_B \frac{L}{2} \cos 37^\circ - Mg \frac{2L}{3} \cos 37^\circ = TL \cos 37^\circ$$

$$10^3 \pi \left(\frac{1}{\pi} \right) \times 10 \frac{1}{2} - \frac{10^4}{2} \times \frac{2}{3} = T$$

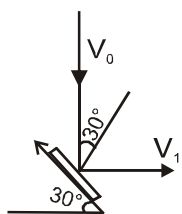
$$T = \frac{10^4}{6} \text{ N}$$

141. $V_0 \sin 30^\circ = V_1 \cos 30^\circ \quad \text{(i)}$

$$eV_0 \cos 30^\circ = V_1 \sin 30^\circ \quad \text{(ii)}$$

Dividing (i) & (ii)

$$\frac{1}{e} \tan 30^\circ = \frac{1}{\tan 30^\circ} \quad \frac{1}{e} = 3$$



142. $\frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R} + \frac{1}{4\pi\epsilon_0} \frac{q}{R} = \epsilon$

$$\frac{q}{2\pi\epsilon_0 R} = \epsilon$$

$$\therefore \text{Ans. } K = 1$$

144. $X_{BF} = \mu \ell_1 + \ell_2$

$$\frac{dx_{BF}}{dt} = \frac{\mu d\ell}{dt} + \frac{d\ell_2}{dt}$$

$$V_{BF} = \frac{4}{3} (-8 + 2) + (4 - 2)$$

$$= \frac{4}{3} \times (-6) + 2 = -8 + 2 = -6 \text{ ms}^{-1}$$

145. at plane surface

$$\text{image of object is } = - \frac{xR}{12} \times 4 \text{ distance}$$

at current surface

$$u = - \left[\frac{xR}{12} 4 + R \right]$$

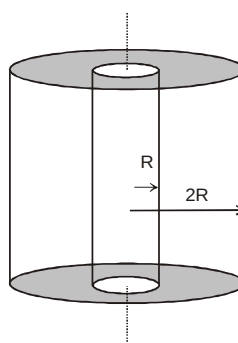
$$v = \infty \quad \text{ROC} = -R$$

$$\frac{\mu_r - \mu_i}{R} = \frac{\mu_r}{v} - \frac{\mu_i}{u}$$

$$\frac{1-4}{-R} = - \left[\frac{xR4}{12} + R \right]$$

$$\Rightarrow x = 1$$

146. $H = -K \cdot 2\pi r l \frac{dT}{dr}$



$$\int_{R_1}^{R_2} \frac{H dr}{2\pi r l} = -K \int_{T_1}^{T_2} dT$$

$$H = \frac{2\pi l k(T_1 - T_2)}{\ln \frac{R_2}{R_1}} \quad H_i = H_f \quad \therefore \text{Ans. } n = 4$$

147. $\Delta Q_{AB} = nC_p \Delta T = \frac{\gamma}{\gamma-1} nR \Delta T = \frac{\gamma}{\gamma-1} [3P_0 V_0 - P_0 V_0]$

$$= 2PV_0 \times \frac{\gamma}{\gamma-1}$$

$$\Delta Q_{AC} = \Delta U + \Delta W$$

$$= \frac{nR}{\gamma-1} \Delta T + \frac{1}{2} \times 3V_0 [P_0 + 4P_0]$$

$$= \frac{[16P_0 V_0 - P_0 V_0]}{\gamma-1} + \frac{15P_0 V_0}{2}$$

$$56 = 2P_0 V_0 \times \frac{\gamma}{\gamma-1}$$

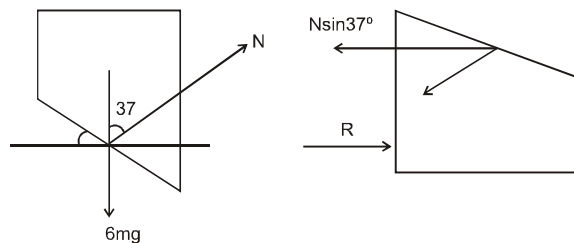
$$360 = 15P_0 V_0 \left[\frac{\gamma+1}{2(\gamma-1)} \right]$$

$$\frac{360}{56} = \frac{15}{4} \frac{(\gamma+1)}{\gamma}$$

$$12\gamma = 7\gamma + 7$$

$$\gamma = \frac{7}{5} = 1 + \frac{2}{f} \Rightarrow f = 5$$

148.

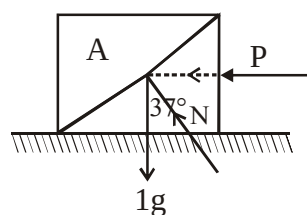


$$\frac{N4}{5} = 6 \text{ mg}$$

$$\Rightarrow 7.5 \text{ mg}$$

$$R = 7.5 \text{ mg} \cdot \frac{3}{5} = 4.5 \text{ mg} = \frac{9}{2} \text{ mg}$$

149.



$$N \cos 37^\circ = 1 \text{ g} \quad \dots (1)$$

$$N \sin 37^\circ = 1a \quad \dots (2)$$

$$\frac{a}{g} = \tan 37^\circ$$

$$a = 10 \times \frac{3}{4} = 7.5 \text{ m/s}^2$$

$$P = 2a$$

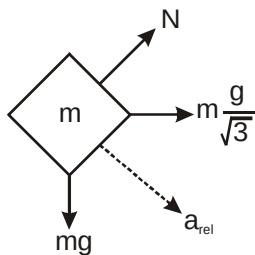
$$P = 15 \text{ N}$$

150. $10\sqrt{3} \cdot \cos 30^\circ = \frac{v}{2}$

$$v = 30.$$

151. Acceleration of wedge equal to $g \tan 30^\circ = \frac{g}{\sqrt{3}}$ towards left.

$$mg \sin 30^\circ + \frac{mg}{\sqrt{3}} \cos 30^\circ = ma_{\text{rel}}$$



w.r. to wedge

$$a_{\text{rel}} = \frac{g}{2} + \frac{g}{2} = g$$

$$3 \frac{g}{\sqrt{3}} = m (a_{\text{rel}} \cos 30^\circ - \frac{g}{\sqrt{3}})$$

$$\therefore m = 6 \text{ Kg.}$$

152. $\frac{dN_A}{dt} = -\lambda_1 N_A, \frac{dN_B}{dt} = 2\lambda_1 N_A - \lambda_2 N_B$

$$N_B = \text{maximum} \Rightarrow \frac{dN_B}{dt} = 0$$

$$\Rightarrow 2\lambda_1 N_A = \lambda_2 N_{B_{\text{max}}} \Rightarrow N_{B_{\text{max}}} = \frac{2\lambda_1}{\lambda_2} N_A$$

$$\Rightarrow N_{B_{\text{max}}} = \frac{2\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t} = 2.$$

153. $8 = 6 \tan \theta \left(1 - \frac{6}{R} \right)$

$$6 = 8 \tan \theta \left(1 - \frac{8}{R} \right)$$

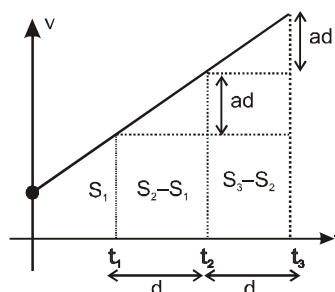
$$\Rightarrow \frac{8}{6} = \frac{6}{8} \left(\frac{R-6}{R-8} \right) \Rightarrow \frac{16}{9} = \frac{R-6}{R-8}$$

$$\Rightarrow 16R - 16 \times 8 = 9R - 9 \times 6$$

$$\Rightarrow 7R = 16 \times 8 - 9 \times 6 = 128 - 54 = 74$$

$$\Rightarrow R = \frac{37 \times 2}{7} \text{ m}$$

154. From figure
 $(S_3 - S_2) - (S_2 - S_1) = ad$
 $S_1 + S_3 - 2S_2 = ad^2$

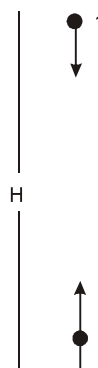


$$a = \frac{S_3 + S_1 - 2\sqrt{S_1 S_3}}{d^2}$$

$$= \frac{(\sqrt{S_3} - \sqrt{S_1})^2}{d^2}$$

$$\therefore \text{Ans. } n = 1$$

155.



$$|S_1| + |S_2| = H$$

$$\frac{1}{2}gt^2 + ut - \frac{1}{2}gt^2 = H$$

$$ut = H \quad \dots (I)$$

$$H = \frac{u^2}{2g} \quad \dots (II)$$

$$t = \frac{u}{2g}$$

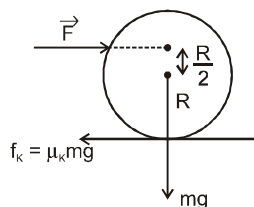
$$S_2 \downarrow ut - \frac{1}{2}gt^2 = 4 \frac{u}{2g} - \frac{1}{2}g \frac{u^2}{4g^2} = \frac{3u^2}{8g} = \frac{3}{4}H$$

156. $\vec{a} \parallel \vec{v}$

$$(2\hat{i} + 3\hat{j} + x\hat{k}) = 3\hat{i} + 4.5\hat{j} - 6\hat{k}$$

$x = 4.$

157.



$$\Delta \vec{p} = \vec{F} \Delta t = m(10 - 0) \quad \dots(i)$$

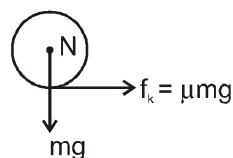
$$\Delta \vec{L}_{\text{centre}} = \vec{F} \Delta t \frac{R}{2} = \frac{2}{5} m R^2 (\omega - 0) \quad \dots(ii)$$

$$\Delta \vec{L}_{\text{centre}} = \vec{F} \Delta t \frac{R}{2} = \frac{2}{5} m R^2 (\omega - 0)$$

Equation (ii) / equation (i)

$$\frac{\frac{2}{5} \omega R^2}{10} = \frac{R}{2}$$

$$\Rightarrow V_{\text{cm}} = 10 \text{ \& } \omega R = 12.5 \text{ kinetic friction acts} \quad \dots(iii)$$



$$a_{\text{cm}} = \frac{\mu mg}{m} = \mu g \quad \dots(iv)$$

$$T_{\text{ac}} = -\mu_k mg R = \frac{2}{5} m R^2 \alpha$$

Let after time t , $V'_{\text{cm}} = \omega' R$

$$10 + \mu_k g t = \left[\frac{12.5}{R} - 2.5 \frac{\mu_k g}{R} \right] R$$

$$\Rightarrow 10 + \mu_k g t = 12.5 - 2.5 \mu_k g$$

$$\Rightarrow 3.5 \mu_k g t = 2.5$$

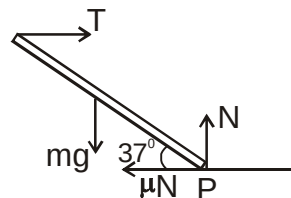
$$\Rightarrow t = \frac{1}{7} \Rightarrow \frac{1}{t} = 7$$

158. The FBD of any one rod is

$$T = \mu N \quad \dots(i)$$

$$mg = N \quad \dots(ii)$$

Taking torque about point 'P'



$$mg \frac{L}{2} \cos 37^\circ = TL \sin 37^\circ$$

$$T = \frac{mg}{2} \frac{4}{3} = \frac{2mg}{3}$$

$$\frac{2mg}{3} = \mu mg \Rightarrow \mu = \frac{2}{3}$$

$$159. \sqrt{\frac{3L}{g \sin \theta}} = \sqrt{\frac{3 \times 15}{10 \times \sin 30^\circ}} = 3s$$

160. The plate is free to rotate about vertical axis yy' .

Let v , v_{cm} and ω be the velocity of particle, velocity of centre of mass of plate and angular velocity of plate just after collision.

$\therefore$ From conservation of angular momentum about vertical axis passing through O is

$$\mu u \frac{a}{2} = mv \frac{a}{2} + \frac{ma^2}{3} \omega \quad \dots(1)$$

since the collision is elastic, the equation of coefficient of restitution is

$$e = \frac{v_{\text{cm}} - v}{u} = 1 \quad \dots(2)$$

$$\text{But } v_{\text{cm}} = \frac{a\omega}{2} \quad \dots(3)$$

solving equation (1), (2) and (3) we get

$$\omega = \frac{12u}{7a} = 5 \text{ rad/s}$$

161. Apply Snell's law on various surfaces one by one :

$$1 \sin 90^\circ = \mu_1 \sin r_1 \Rightarrow \sin r_1 = \frac{1}{\sqrt{2}} \Rightarrow r_1 = 45^\circ$$

$$\mu_1 \cos r_1 = \mu_2 \sin r_2 \Rightarrow \sin r_2 = \frac{1}{\mu_2}$$

$$\mu_2 \cos r_2 = \mu_3 \sin r_3 \Rightarrow \sin r_3 = \frac{\mu_2 \sqrt{1 - \sin^2 r_2}}{\sqrt{3}}$$

$$\mu_3 \cos r_3 = 1 = \frac{\sqrt{\mu_2^2 - 1}}{\sqrt{3}}$$

$$\sin^2 r_3 + \cos^2 r_3 = 1$$

$$\Rightarrow \frac{\mu_2^2 - 1}{3} + \frac{1}{3} = 1 \Rightarrow \mu_2^2 = 3$$

$$\Rightarrow \mu_2 = \sqrt{3}$$

$$162. \frac{1}{f_L} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{20} - \frac{1}{20} \right)$$

$$\Rightarrow F_L = 20 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{-10} - \frac{2}{20} = -\frac{1}{5}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-5} \Rightarrow \frac{1}{v} = \frac{1}{30} - \frac{1}{5}$$

$$\text{for lens } \frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

$$163. f = 25 \Rightarrow \frac{1}{25} = \frac{1}{v} + \frac{1}{-25} \Rightarrow v = \frac{25}{2}$$

$$\Rightarrow m = \frac{-v}{u} = \frac{1}{2}$$

$$164. T \propto \sqrt{\text{length}} \text{ free length becomes } \frac{\ell}{4} \text{ So time period}$$

$$\text{becomes } T_1 = \frac{T}{2}$$

Half oscillation is with free length ℓ and half with free length

$$\frac{\ell}{4}$$

$$\text{Total time} = \frac{T}{2} + \frac{T_1}{2} = \frac{3T}{4}$$

$$165. f = 2N$$

$$a_{1kg} = 2 \text{ m/s}^2$$

$$S = \frac{1}{2} \times 2 \times 4 = 4 \text{ m}$$

$$W_{\text{friction}} = -2 \times 4 = -8J$$

$$166. \text{Energy entering in the windmill}$$

$$= \frac{1}{2} mv^2$$

$$P_{\text{in}} = \frac{dE}{dt} = \left(\frac{1}{2} v^2 \right) \left(\frac{dm}{dt} \right)$$

$$P_{\text{in}} = \left(\frac{1}{2} v^2 \right) (\rho AV) = \frac{1}{2} \rho AV^3$$

Electrical power output

$$P_{\text{out}} = \frac{1}{3} \left(\frac{1}{2} \rho AV^3 \right)$$

$$P_{\text{out}} = \frac{1}{6} \rho AV^3 = \frac{1}{6} \times 1.2 \times 10 \times (20)^3$$

$$P_{\text{out}} = 16 \text{ kW.}$$

$$167. a_{\text{cart}} = \frac{50 - \mu_k (20)}{5 + 3}$$

$$a_{\text{brick}} = \mu_k 10$$

$$a_{\text{rel}} = \frac{50 - 20\mu_k}{8} - \mu_k (10)$$

$$\Rightarrow 0.4 = \frac{1}{2} \left(\frac{50 - 100\mu_k}{8} \right) (0.8)^2$$

$$\Rightarrow 1 = 5 - 10\mu_k$$

$$\mu_k = 0.4$$

$$168. P = \left(\frac{E}{R} \right)^2 R_x$$

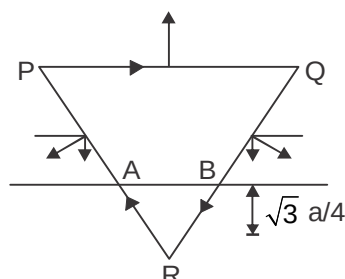
$$\Rightarrow \frac{dp}{dx} = \frac{E^2}{R^2} \frac{dR}{dx}$$

$$= \frac{E^2}{\rho_0^2 L^4} \frac{\rho_0 x}{4A^2 L^2}$$

$$169. \text{From lenz's law the current } i \text{ in the loop is clockwise}$$

For equilibrium

$$Mg = \frac{Bla}{2}$$



$$i = \frac{2mg}{Ba} = \frac{2 \times 2 \times 10}{10 \times 1} = 4$$

$$170. \vec{v} \parallel \vec{B}$$

$$\text{So, } F = 0$$

$$171. \text{If the block has moved by distance } x.$$

$$Ma = 2\mu Mg - \left[\frac{\mu Mg}{3} + \frac{2\mu Mg}{3} + \frac{3\mu Mg}{3\ell} (\ell - x) \right]$$

$$a = 2\mu g - \left(\frac{\mu g}{3} + \frac{2\mu g}{3} + \frac{3\mu g}{3\ell} (\ell - x) \right)$$

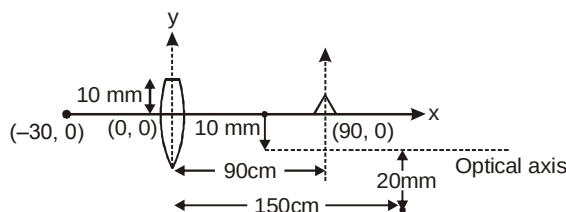
$$\frac{da_1}{dx} = \frac{3\mu g}{3\ell} \text{ [for 2nd and 3rd cells similarly]}$$

$$\tan \theta_1 : \tan \theta_2 : \tan \theta_3 = 3 : 2 : 1$$

$$172. \text{For first lens}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$$



$$\Rightarrow v = 60 \text{ cm}$$

For second lens

$$u = 30 \text{ cm, } f = +20 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30}$$

$$\Rightarrow \frac{1}{v} = \frac{3-2}{60} = \frac{1}{60}$$

$$\therefore v_2 = 60 \text{ cm}$$

$$\Rightarrow m = -2$$

So coordinates are (150 cm, -30 mm)

$$173. \quad h\nu = 13.6(3)^2 \left[\frac{1}{4^2} - \frac{1}{5^2} \right] = 2.75 \text{ eV}$$

for $n = 4$ to $n = 3$

$$h\nu = (13.6) \times (3)^2 \left[\frac{1}{3^2} - \frac{1}{4^2} \right]$$

$$= 5.95 \text{ eV}$$

for shorter wavelength

$$3.95 = 5.95 - \phi \Rightarrow \phi = 2 \text{ eV}$$

for longer wavelength

$$\text{eV}_s = 2.75 - 2 = 0.75 \text{ eV}$$

$$174. \quad \text{Pressure} = \frac{(1+r)I\cos^2\theta}{C}$$

$$\text{Where, } \theta \text{ is the angle with normal} = \frac{3I}{8C}$$

$$175. \quad \overline{PQ} = \sqrt{15^2 + 20^2} = 25 \text{ m}$$

$$25 = \frac{V_p^2 \sin 2(45)}{g} \Rightarrow V_p = 5\sqrt{10} \text{ m/s}$$

$$\therefore V_A^2 \sin^2 \theta - 2 \times 10 \times 12.5 = (5\sqrt{5})^2 \Rightarrow V_A \sin \theta = 5\sqrt{15} \text{ m/s}$$

$$V_A \cos \theta = V_p \cos 45^\circ = 5\sqrt{5} \text{ m/s and } \theta = 60^\circ$$

$$\therefore \overline{AB} = \frac{V_A^2 \sin 2\theta}{g} = 25\sqrt{3}$$

176. Consider equation of torque about C

$$\left[mg \frac{4R}{3\pi} \right] = \frac{mR^2}{2} \alpha$$

$$\alpha = \frac{8g}{3\pi R} = \frac{8\pi g}{3\pi^2 R} = \frac{8\pi g}{30R}$$

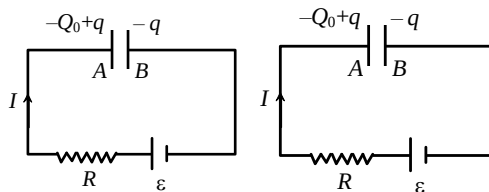
$$177. \quad f = \frac{3v}{4[L + 0.6r]}$$

$$\Rightarrow \frac{df}{dt} = \frac{3v[-0.6 \frac{dr}{dt}]}{4(L + 0.6r)^2}$$

$$\frac{dr}{dt} = \frac{1}{72} \text{ m/s}$$

$$178. \quad \eta = \frac{\left(\frac{F}{A} \right)}{\left(\frac{dv}{dx} \right)} = \frac{\frac{0.02}{2}}{\frac{1}{1}} = 10^{-3} \text{ kg/ms} = 10^{-2} \text{ poise}$$

179. Let at any time t charge flow through the plate B to plate A is q and instantaneous current is I .



From loop theorem

$$\left(\frac{2q - Q_0}{2C} \right) + IR - \varepsilon = 0$$

$$\Rightarrow R \frac{dq}{dt} = \frac{-2q + 2\varepsilon C + Q_0}{2C}$$

$$\Rightarrow \frac{dq}{2\varepsilon C + Q_0 - 2q} = \frac{dt}{2RC}$$

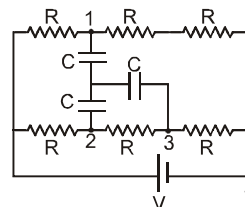
Now for charge on plate A to be zero $q = Q_0$.

$$\text{Integrating } \int_0^{Q_0} \frac{dq}{2\varepsilon C + Q_0 - 2q} = \int_0^t \frac{dt}{2RC}$$

$$= t = RC \ln \left[\frac{2\varepsilon C + Q_0}{2\varepsilon C - Q_0} \right]$$

Putting the value of C , Q_0 , ε and R We get $t = 2$ seconds.

180. No current passes through capacitors in steady state. Assume potential at point '4' to be zero.



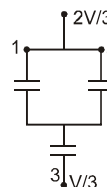
Then points '1' and '2' are at same potential $\frac{2V}{3}$.

Hence C_1 and C_2 can be taken in parallel.

The potential at point 3 is $\frac{V}{3}$.

$\therefore$ Equivalent circuit of all three capacitors is shown

Hence potential difference across capacitor C_3 is



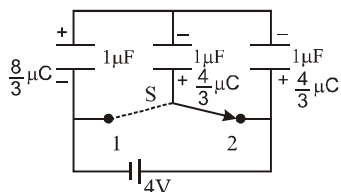
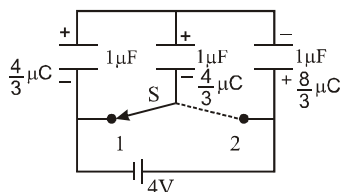
$$= \frac{2C}{2C + C} \times \left(\frac{2V}{3} - \frac{V}{3} \right) = \frac{2V}{9}$$

181. Applying work energy theorem we can write

$$\frac{\rho}{2} \left(\frac{\pi R}{2} \right) v^2 - 0 = \frac{\rho \pi R}{2} g \left(\frac{2R}{\pi} + \frac{\pi R}{4} \right)$$

$$v^2 = \frac{gR}{2} \left(\frac{8+10}{\pi} \right) = 900$$

182.



$$W_b = \Delta U + \Delta H$$

$$4 \times \frac{4}{3} = 0 + \Delta H$$

$$\therefore \Delta H = \frac{16}{3} \mu\text{J}$$

$$\frac{16}{3} = \frac{x}{2} \Rightarrow x = \frac{16 \times 12}{3}$$

$$= 64$$

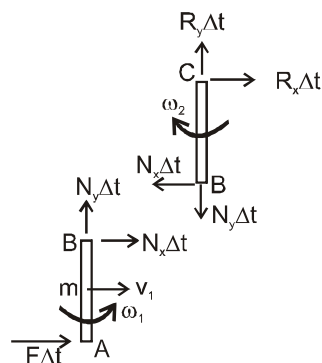
183. $(\pi r^2) \sqrt{2gy} = \pi x^2 \left(-\frac{dy}{dt} \right)$

$$\Rightarrow (r^2) \sqrt{2gy} = x^2 \lambda$$

$$\Rightarrow y \propto x^4$$

Hence अतः, $n = 4$.

184.



$$AB : F\Delta t + N_x \Delta t = mv_1 \quad \dots(1)$$

$$F\Delta t(\ell/2) - N_x \Delta t (\ell/2)$$

$$= (m\ell^2/12)\omega_1 \quad \dots(2)$$

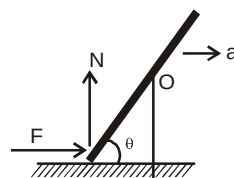
$$BC : (N_x \Delta t)t = (m\ell^2/3)\omega_2 \quad \dots(3)$$

$$\omega_2 \ell = (\ell/2) \omega_1 - v_1 \quad \dots(4)$$

$$\text{Solving we get } \omega_2 = 6(F\Delta t)/7ml$$

$$= 6 \text{ rad/s.}$$

185. Since the rod is in translation so, $\alpha = 0$
 $\therefore$ about O,



$$F \frac{\ell}{2} \sin \theta - N \frac{\ell}{2} \cos \theta = 0$$

$$\Rightarrow F \sin \theta = N \cos \theta$$

$$\Rightarrow \tan \theta = \frac{N}{F} \quad \dots\dots (1)$$

$$\text{Also, } N = Mg \quad \dots\dots (2)$$

$$\text{So, } \Rightarrow \tan \theta = \frac{Mg}{F} = \frac{40}{60} = \frac{2}{3}$$

$$\therefore \theta = \tan^{-1} \left(\frac{2}{3} \right)$$

$$X = 3$$

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